

Informality and financial frictions

David Morales ^{*} Daniela Puggioni [†]

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Abstract

This paper shows that expanding bank credit in developing economies stimulates firm activity but tilts its composition toward informality. Using matched administrative records on 1.7 million Mexican firms and 127 banks over 2004–2024, we identify bank-specific credit-supply shocks within firms across banks and construct a leave-one-market-out shift-share instrument for local credit growth. A one-percent credit expansion raises the local stock of firms with an elasticity of 0.18, and the response is statistically indistinguishable across the formal and informal sectors (0.20 vs. 0.17). Entry, exit, and informal-to-formal transitions move together; informal employment responds more than formal employment; and average value added per firm *falls* by 0.07%. A dynamic model of firm entry, exit, and sector choice under sector-specific financial frictions reproduces these patterns through a single mechanism: relaxing credit constraints disproportionately admits low-productivity entrants into the informal sector, expanding aggregate activity while lowering average productivity. Financial development in emerging economies therefore confronts a fundamental trade-off, better credit access stimulates firm activity but reshapes it toward informality.

Keywords: Financial frictions, firm dynamics, informality, shift-share IV.

^{*}Toulouse School of economics, david.morales@tse-fr

[†]Banco de México, dpuggionih@banxico.org.mx

1 Introduction

Access to credit is widely viewed as a key determinant of firm growth, employment creation, and structural transformation. Yet in many developing economies, financial markets remain shallow and a large share of firms operate informally, outside the reach of formal institutions. Whether and how relaxing credit constraints reallocates activity across the formal and informal margins is, therefore, central to assessing the aggregate consequences of financial frictions in contexts where informality is pervasive. This paper provides new causal evidence on these questions using administrative credit, payroll, and firm-level census data from Mexico, a country where 88% of firms are informal and bank lending is a concentrated source of finance for both formal and informal firms.

I combine matched bank-firm-worker administrative records with a *leave-one-market-out* shift-share identification strategy to isolate plausibly exogenous variation in local credit supply. In a first step, bank-specific credit-supply shocks are recovered from within-firm, across-bank variation in the universe of Mexican commercial credit relationships, following [Khawaja and Mian \(2008\)](#) and [?.](#) In a second step, the bank-specific shocks are aggregated to the local labor market level using predetermined exposure shares; the leave-one-market-out correction ([Borusyak et al., 2021](#)) eliminates the mechanical bias that would arise from constructing each market’s instrument with that market’s own loans. Inference accounts for the cross-market correlation induced by shared bank shocks following [?.](#) The rich administrative data allow me to track formal and informal firm dynamics, wages, employment, and financial conditions within finely defined local labor markets, providing a comprehensive view of how credit-supply shocks propagate through the local economy.

Four main empirical findings emerge. *First*, credit-supply expansions raise the local stock of firms with an elasticity of 0.18, an effect that is precise under the AKM inference appropriate for shift-share IV, that is robust across nine combinations of formality status (formal vs. informal) and margin (existing, entering, exiting), and that survives a battery of robustness checks documented in Section 6.3. Evaluated at the average local labor market, a one-percent increase in five-year credit growth adds approximately 13 firms (2 formal and 11 informal), or about 7,200 firms at the national level.

Second, the response operates predominantly through entry and exit: gross flow margins move symmetrically (entry elasticity of 0.19, exit elasticity of 0.20), consistent with credit shocks raising market churn rather than the net firm count alone. The five additional entrants per market translate into a 0.07 percentage point increase in the five-year startup rate, roughly one tenth of the U.S. startup deficit documented by [Pugsley and Sahin \(2019\)](#) over the same horizon, an informative benchmark given the limited evidence on informal firm dynamics in middle-income economies. On the transition margin, credit shocks raise informal-to-formal upgrading (elasticity of 0.20) and the formal-to-informal downgrading, indicating that credit access increase the porbability of formalization at the

intensive margin.

Third, credit-supply shocks pass-through to the broader local labor market. A one-percent increase in credit raises total local employment by 0.15% (about 56 workers at the average market), with informal employment (0.19%) responding more elastically than formal employment (0.12%); the aggregate wagebill rises by 0.20%, an economically small magnitude. Total value added rises by 0.10%, while *average* value added per firm falls by 0.07%. This composition, rising aggregate output and employment alongside a fall in per-firm productivity, is the empirical signature of credit-supply shocks operating through the extensive margin of low-productivity entry.

Fourth, the headline results are not artifacts of any single methodological choice. The shift-share instrument is uncorrelated with predetermined market controls and pre-period firm-dynamics trajectories; the estimated elasticities survive the exclusion of the three largest metropolitan areas, the removal of credit weighting, variation in the coverage threshold from zero to 0.8, the renormalization of the instrument by coverage, and the conventional (non-leave-one-out) shift-share construction. Most compellingly, a structurally distinct identification strategy at the bank level—the cell-fixed-effects design of [Degryse et al. \(2019\)](#), which identifies bank shocks from within-cell variation across banks rather than across markets and permits identification from both multi-bank *and* single-bank firms, recovers point estimates within 0.03 of the market-level results across every firm-margin outcome. The intermediate financial channels (collateral and interest rates) are less consistently identified across specifications, and I accordingly treat the firm-margin and aggregate labor-market findings as the paper’s central contributions.

To interpret these empirical patterns I develop a dynamic model of firm entry, exit, and sector choice under sector-specific financial frictions. Firms differ in productivity and choose whether to operate formally or informally; informal firms face lower regulatory burdens but a strictly steeper marginal cost of external finance. A counterfactual that relaxes the informal-sector borrowing constraint in proportion to the empirical bank-supply shock reproduces all seven qualitative patterns documented in the data, including the observation that credit expansions *lower* average productivity while raising aggregate output. The mechanism is a marginal-firm story: relaxing credit disproportionately admits low-productivity entrants into the informal sector, lowering average productivity through composition while expanding the firm distribution. The quantitative calibration of the model is ongoing work; the qualitative pattern of the counterfactual is robust across the parameter neighborhoods I have explored.

This paper contributes to three related literatures. First, it adds to a growing body of work on how credit-supply shocks affect firm real outcomes ([Chodorow-Reich, 2014](#); [Greenstone et al., 2020](#); [Khwaja and Mian, 2008](#); [Gutierrez et al., 2023](#)), by providing causal estimates of how credit shocks reallocate firm activity across the formal-informal margin in

a context where informality is the dominant mode of operation. Second, it extends the literature on informality and firm dynamics (Ulyssea, 2018; Dix-Carneiro et al., 2021; Imbert and Ulyssea, 2023) by documenting that, in Mexico, the empirically dominant transition is from formality *into* informality rather than the reverse, a qualitative pattern that has received less attention than the extensive literature on informal-to-formal upgrading. Third, by combining a granular shift-share IV design with a model of sector choice under financial frictions, the paper provides a unified framework linking credit-supply shocks to the joint determination of firm entry, exit, and formality, an integration that has been largely absent from the literature on financial development in middle-income economies.

The remainder of the paper is organized as follows. Section 2 describes the related literature. Section 3 describes the administrative datasets and presents descriptive evidence on credit concentration, formality, and formal-informal transitions. Section 5 develops the leave-one-market-out shift-share identification strategy. Section 6 presents the main empirical findings, including the headline elasticities, the formality decomposition, and the battery of robustness checks. Section ?? develops the theoretical framework. Section ?? parametrizes the model and Section 10 reports the counterfactual. Section 11 concludes.

2 Literature review

I contribute to three broad strands of the literature. First, this paper adds to the growing evidence on how credit supply shocks affect firms’ real outcomes, particularly employment, in different contexts. A large body of work shows that credit market disruptions can have significant employment effects at the firm level. For example, Chodorow-Reich (2014) showed that in the 2008–09 crisis, firms reliant on distressed banks were less able to borrow and consequently cut more jobs; indeed, the withdrawal of credit accounted for roughly one-third to one-half of the employment decline at affected small and mid-sized firms. In contrast, other studies find more muted aggregate effects: Greenstone et al. (2020) use a county-level shift-share design and conclude that small business lending shocks had an economically insignificant impact on overall employment in the United States. Evidence from developing economies, however, suggests credit frictions play a more binding role. Khwaja and Mian (2008) first documented in Pakistan that idiosyncratic bank liquidity shocks can depress borrowing firms’ output and employment. More recently, Gutierrez et al. (2023) find that in Mexico a positive bank credit shock leads to substantial gains in formal jobs – a one standard deviation increase in local credit supply raises annual employment growth by about 1.4 percentage points. I build on this literature by providing new causal evidence that connects credit shocks to both employment and wages in a middle-income country, leveraging rich matched firm-bank-worker data. My analysis highlights that credit constraints can meaningfully influence not only how many workers firms hire, but also how

those workers are paid.

Second, I contribute to the literature on rent-sharing and the role of firm heterogeneity in wage determination. A seminal contribution by [Abowd et al. \(1999\)](#) introduced a framework (AKM) to separately identify worker and firm pay components, revealing that firms pay systematically different wages even to similar workers. Subsequent studies across advanced economies confirm that firm effects explain a significant share of wage dispersion. For instance, AKM find an important role for firm-specific wage premiums in France, and similar conclusions have been reached for Germany, Italy, Portugal, Brazil ([Alvarez et al. \(2018\)](#)), and other countries. Beyond wage dispersion, many authors have examined the link between firm performance and worker pay. In previous work, [Blanchflower et al. \(1996\)](#) suggest a direct test of rent-sharing in US manufacturing, finding that wages co-move with firm profits. [Van Reenen \(1996\)](#) shows that firm innovations translate into higher worker wages in the UK, and [Guiso et al. \(2005\)](#) find that firm-specific productivity shocks have a positive effect on employees' wages in Italy. Consistent with these patterns, studies using matched employer-employee data often report modest but significant rent-sharing elasticities (e.g. a 3–7% wage increase per 100% rise in firm value-added, [Card et al. \(2014\)](#)). This paper extends this literature by examining whether credit-driven changes in firm performance are passed through to workers. In doing so, I leverage a three-way fixed-effects framework (worker–firm–bank) to decompose wages in the presence of financial shocks. This approach allows us to assess how much firm–worker pay premiums are influenced by access to credit and whether credit constraints at the firm level translate into changes in wages for employees. By providing novel evidence of rent-sharing in a credit-constrained environment, I shed light on the channels through which financial frictions can affect labor earnings in developing economies.

Third, I add to the literature on general equilibrium and local labor market effects of financial shocks in developing countries. A number of studies have analyzed how regional labor markets respond to shocks in credit availability or other financial interventions. For instance, [Greenstone et al. \(2020\)](#) implement a shift-share strategy to isolate local credit supply shocks in US counties and find little overall impact on employment, suggesting that alternative financing or labor mobility can dampen localized effects. In middle-income economies with less developed financial markets and larger informal sectors, credit shocks may induce different adjustments. Recent research in Mexico indicates that credit expansions disproportionately benefit smaller, younger, and previously credit-starved firms [Gutierrez et al. \(2023\)](#), which can lead to greater job creation at the local level. We contribute to this strand by examining credit impacts at the level of Local Labor Markets (LLMs), linking firm-level shocks to broader community outcomes. This approach allows us to capture spillovers and equilibrium effects – for example, whether increased formal lending in a region draws workers out of informality or raises prevailing wages even for

firms not directly receiving new loans. My work is related to studies of local labor demand shocks in developing countries (e.g. the impact of trade liberalization or infrastructure investments on regional employment and informal work). By focusing on credit supply, I provide evidence on a relatively understudied driver of local labor market fluctuations in developing economies. In summary, my analysis bridges the finance and labor literatures by documenting not only firm-level rent-sharing of credit benefits but also the aggregate implications when credit shocks propagate through local economies.

3 Data

I use five databases in this paper. The main database is a matched employer-employee data that comes from administrative records of formal employment at the Social Security Institution (**IMSS**, by its Spanish acronym). This data contains information on wages in the formal sector and some characteristics of firms (e.g., industry code and geographical location) and employees (e.g., age and sex).

The second database is matched bank-employer from commercial loan administrative records that is provided by the Mexican National Security Commission (**CNBV**, by its Spanish acronym). This data provides information on the number of credits, the quantity borrowed, and the conditions of the credits (e.g, interest rate). Importantly, it contains banking credits to both formal and informal firms.

I have access to the individual level non-public version of these databases through the **EconLab** environment of the Bank of Mexico (Banxico). I merge the IMSS and the CNBV data at the firm level, creating a unique monthly bank-firm-worker matched dataset spanning 20 years that includes the universe of formal firms (holding a bank loan or not) and the universe of firms with bank loans (both formal and informal). In other words, I do not observe the subset of informal firms without banking loans.

However, big entities (bank or firms) could have a general equilibrium effect on the credit conditions and labor markets of smaller players. To account for this issue, I study the implications at the Local Labor Market (LLM) level. Therefore, I use the final database, which is the definition of the 777 local labor markets (LLMs) in the country to collapse the data at this LLM level. This data comes from Banxico’s **Database for Local Labor Markets Analysis in Mexico**.

To overcome the limitation that informal firms without banking loans are not observed in matched bank-employer-employee data, I rely on the quinquennial **Economic Census** (EC) dataset published by the National Statistic Institution (**INEGI**, by its acronym in Spanish), which contains the informal sector and is matched with the previous data at the Local Labor Market (LLM) level.

3.1 Matched employer-employee data (IMSS)

The data from IMSS includes, by definition, only formal workers who must be register by their employer in order to comply with the Mexican labor regulations. The wages are daily earnings that include cash payments, gratuities (such as profit sharing and year-end bonuses), wages, food, housing, vacation bonuses, commissions, in-kind payments, and any other remuneration for work—excluding food vouchers and savings fund contributions. The data on wages is bottom-coded at the minimum wage¹ and top-coded at 25 times the general minimum wage in Mexico City. The data is monthly and it is available from 2004 to date. However, I use the monthly observations during a three years window, from 2017 to 2019, of workers between 25 and 54 years old. Is worth to notice that the IMSS information excludes the sizeable informal sector in Mexico. In the fourth quarter of 2024, the the National Survey of Occupation and Employment ([ENOE](#)) estimated the rate of informal employment at 54.5 percent.

3.2 Matched bank-employer data (CNBV)

The report "RC04" contains administrative data on credits, which by law is reported by banks to the Mexican Security and Exchange Commission (CNBV by its acronym in Spanish)². This report includes detailed information on each credit that each bank grants. It includes granular information such as the amount of the credit, the interest rate, the firm to which it was granted, the number of workers employed in the firm, and the age of the firm that received the credit, among other details. The data is available on a monthly basis starting from 2004 to date; however I focus on the monthly observations during a three years window, from 2017 to 2019.

3.3 Local Labor Market data (LLM)

I use the official definition of Local Labor Markets (LLM) in Mexico constructed by the National Institute of Statistics and Geography (INEGI) and available through the [Database for Local Labor Markets Analysis in Mexico](#). INEGI uses commuting flows between house and work locations to build a dissimilarity index. Then an iteratively algorithm creates geographic areas that optimize the mean dissimilarity index across clusters. This procedure creates the 777 distinct LLM that divide the country.

¹It is possible to observe wages below the minimum wage due to the late implementation in July of the approved changes of minimum wage in January.

²A public aggregate version of this data can be found in the [CNBV's](#) website.

3.4 Economic Census data (EC)

The EC is a firm level survey representative of all non-agricultural sectors at all locations with more than 2,500 inhabitants. The periodicity is quinquennial and it captures entrepreneurial activity taking place at physical locations. Thus, economic activity such as street sellers without a physical facility is not taken into account. Importantly, it captures the formal and the informal sectors of the economy.

4 Descriptive statistics

The CNBV data spans over the period 2004–2024. The data include detailed information on 1,678,422 firms and their credit relationships with 127 banks. In 2019, for example, 726,542 establishments received at least one loan—corresponding to 11.4 percent of the 6,373,169 establishments reported in the Economic Census, highlighting the broad reach of bank credit in the corporate sector. The resulting panel comprises 3,001,320 firm–bank–year observations, allowing us to track the credit variations over two decades with comprehensive coverage and substantial cross-sectional variation.

Figure 1: Distribution of the number of banks per firm

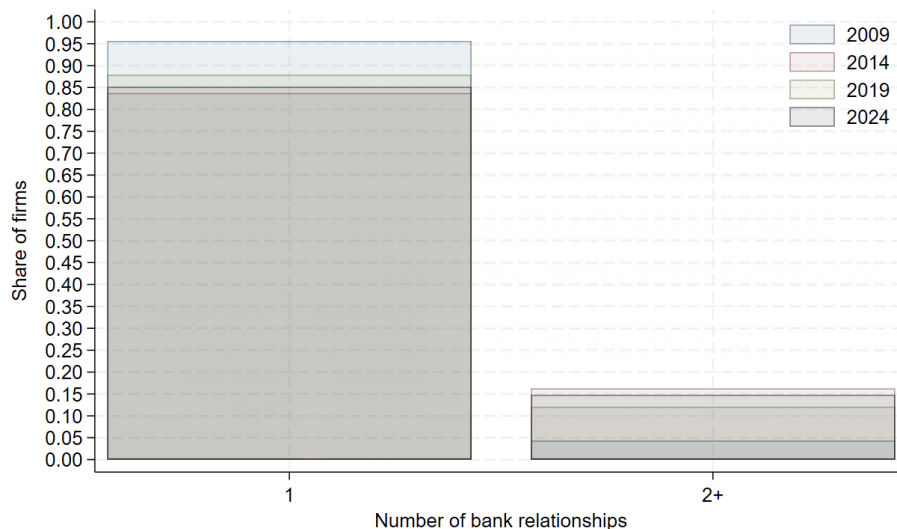
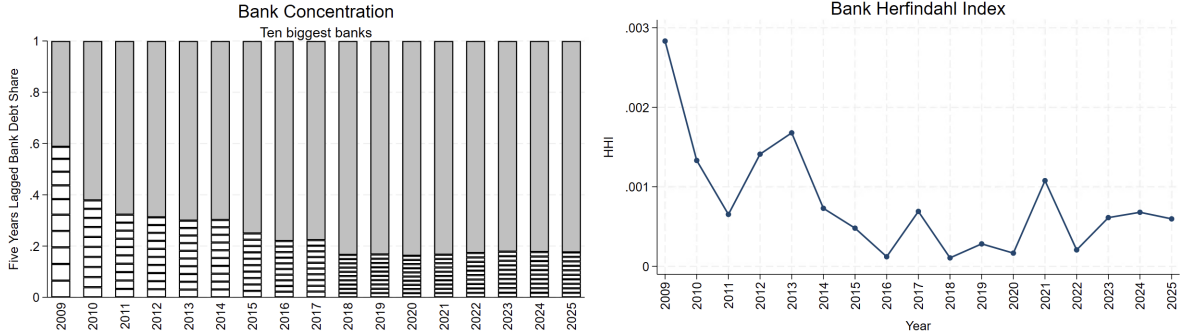


Figure 1 shows the distribution of the number of banking relationships per firm in 2009, 2014, 2019, and 2024. Most firms borrow from a single bank, although multi-bank relationships become modestly more common over time. The figure therefore points to a gradual but limited increase in bank diversification, with single-bank borrowing remaining the dominant mode of credit access throughout the period. This distinction is relevant because our baseline bank-shock estimation is identified from multi-bank firms. As a robustness check,

we also estimate shocks using specifications that allow single-bank firms to contribute as in Degryse et al. (2019); the results are similar, indicating that the findings are not driven solely by the multi-bank sample.

Figure 2: Bank concentration and Herfindahl Index



In terms of credit concentration, the left panel of Figure 2 reports the annual distribution of five-year lagged bank credit shares for the ten largest banks. These are the same predetermined exposure shares used to construct the shift-share instrument. The figure shows that credit exposure is not heavily concentrated and becomes less so over time. The right panel reports the corresponding Herfindahl–Hirschman Index. These evidence implies that our latter empirical strategy identifies the bank supply shocks from multiple independent bank institutions, rather than leaning on the idiosyncratic behavior of any single institution.

Figure 3: Distribution of firms by formality and banking credit holding

	Formal	Informal	Total
Credit	87,788 (1.6%)	249,588 (4.6%)	337,376 (6.2%)
No Credit	570,277 (10.4%)	4,560,527 (83.4%)	5,130,804 (93.8%)
Total	658,065 (12.0%)	4,810,115 (88.0%)	5,468,180 (100%)

Source: Economic Census 2024.

Table 3 documents the joint distribution of firms by formality status and access to bank credit. Informality is pervasive: under our definition of informality, firms with positive income but zero social security contributions, 88% of firms are informal. Access to bank

credit is limited: only 6.2% of firms report holding credit from a bank. Although most firms with bank credit are informal (74%), this mainly reflects the large mass of informal firms in the economy. Conditional on formality status, formal firms are substantially more likely to borrow from banks: 13.3% of formal firms have bank credit, compared with 5.2% of informal firms. The table therefore confirms two well-known facts: informality is widespread and formal firms have greater access to credit. At the same time, it highlights a less emphasized fact: informal firms also borrow from formal financial intermediaries. As a result, changes in bank credit conditions can affect informal firms directly, not only through spillovers from the formal sector.

Figure 4: Transition matrix 2014-2019 and 2019-2024

		Destination		
		Formal	Informal	Exit
2014 Origin	Formal (%)	122,784 70.5	23,526 13.5	56,994 32.7
	Informal (%)	263,565 6.5	2,232,216 55.4	1,531,660 38.0
2019 Origin	Formal (%)	324,052 85.0	138,041 36.2	170,946 44.9
	Informal (%)	146,071 3.5	2,547,652 61.1	1,473,395 35.4

Note: Formal if Nb workers with SS greather than zero, Informal if Nb of workers with SS is zero.

Figure 4 documents the dynamics of establishment status across the formal and informal sectors using the 2014, 2019, and 2024 Economic Censuses. The patterns reveal that between 55 to 60 percent of establishments remain informal throughout the period, while between 70 to 85 percent remain formal. Transitions out of formality are substantially more frequent than transitions into it: between 13 and 36 percent of firms move from formal to informal status, compared with only between 3.5 to 6.5 percent moving in the opposite direction.

These empirical facts underscore that, in the Mexican context, the dominant and economically relevant transition is *formal-to-informal*, not informal-to-formal. This is at stark contrast with much of the recent literature that has focused on the latter—examining barriers to formalization and the determinants of informal-to-formal upgrading (e.g., [Dix-Carneiro et al. \(2021\)](#); [Imbert and Ulyssea \(2023\)](#))—but I present new evidence here suggesting that the opposite direction of movement is quantitatively far more prevalent. Understanding the forces driving firms out of formality, rather than only into it, is therefore central for interpreting the effects of credit supply shocks and the broader functioning of labor and product markets.

5 Empirical strategy

5.1 Shift–share identification with exogenous shocks

We estimate the impact of credit supply shocks on firm-level transition outcomes (formality status, entry, and exit). OLS estimates are contaminated by reverse causality and omitted-variable bias: markets with more informal-to-formal transitions may demand more credit, and an unobserved technology shock may simultaneously raise firm entry and credit volumes.

A shift-share design with an exogenous shifter resolves this endogeneity. The ideal experiment randomly allocates additional funding across banks, raising their credit supply independently of firms’ demand. Because the allocation is random, it is uncorrelated with firms’ demand for credit and, hence, with firm transitions. The identification requires the lottery to affect outcomes solely through equilibrium credit.

We implement an empirical analogue with a Leave-One-Out Shift-Share Instrumental Variable (LOO-SSIV) following [Amiti and Weinstein \(2018\)](#) and [Greenstone et al. \(2020\)](#). In the first step, for each market-year pair (m, t) , we decompose bank-firm credit growth into demand and supply components on the sample that excludes market m . Specifically, we estimate

$$\Delta \ln L_{bft}^{-m} = \alpha_{ft}^{-m} + \gamma_{bt}^{-m} + \epsilon_{bft}, \quad (1)$$

where α_{ft}^{-m} is a firm-year fixed effect and γ_{bt}^{-m} is a bank-year fixed effect. The firm-year fixed effect absorbs credit demand; the bank-year fixed effect recovers the bank-specific supply shock. Because identification of γ_{bt}^{-m} exploits within-firm, across-bank variation, only multi-bank firms contribute.³ We maintain the standard assumption that the supply shock identified on multi-bank firms also applies to single-bank firms. For robustness, we replace firm-year fixed effects with market-sector-year fixed effects as in [Degryse et al. \(2019\)](#), which allows single-bank firms to contribute to identification at the cost of potentially absorbing demand shifters into the estimated supply shocks; results are similar.

In the second step, we construct bank shares in municipality m at baseline,

$$s_{bmt-5} = \frac{L_{bmt-5}}{\sum_{b'} L_{b'mt-5}}, \quad (2)$$

and define the shift-share instrument as

$$\hat{Z}_{mt}^{LOO} = \sum_b s_{bmt-5} \hat{\gamma}_{bt}^{-m}. \quad (3)$$

³Multi-bank firms account for 100, 69, and 73% of banks, markets, and bank-market observations in 2025, respectively.

Following the exogenous-shifter framework of [Borusyak et al. \(2021\)](#), identification requires that the shifters $\hat{\gamma}_{bt}^{-m}$ be as-good-as-randomly assigned across banks conditional on the included controls. The assumption would fail if banks more exposed to credit shocks systematically lend in markets with atypical transition dynamics. For example, if an oil-sector boom raises both lending by sector-specialized banks and firm entry in oil-intensive markets, the estimated coefficient would partly capture sectoral demand rather than credit supply. We address these concerns by including controls that may correlate with both the shifter and the outcome, and by conducting a battery of robustness checks (Section 6.3). The baseline controls include year fixed effects, market-level sector shares from both credit volumes and firm transitions, and lagged market-level interest rates, number of loans, and number of banks.

The leave-one-out construction eliminates the mechanical bias that would arise from estimating $\hat{\gamma}_{bt}^{-m}$ on data that include market m 's own observations, but it does not substitute for shifter exogeneity. For valid inference, we follow [Adão et al. \(2019\)](#).

The main estimating equation relates local credit growth to firm dynamics at the market level:

$$y_{mt} = \beta \Delta_5 \ln(C_{mt}) + X'_{mt-5} \Gamma + \lambda_t + \varepsilon_{mt}, \quad (4)$$

where y_{mt} is a firm-dynamics outcome (such as the log number of active firms, entrants, exits, or transitions across formality states), and $\Delta_5 \ln(C_{mt})$ is the five-year log-change in credit in market m , instrumented by $\hat{Z}_{mt}^{LOO}$. The vector X_{mt-5} contains baseline market and bank controls (market-level sector shares from both credit volumes and firm transitions, and lagged market-level interest rates, number of loans, and number of banks) all measured five years prior to the outcome window. Year fixed effects are captured by λ_t .

The corresponding first stage is

$$\Delta_5 \ln(C_{mt}) = \pi \hat{Z}_{mt}^{LOO} + X'_{mt-5} \Pi + \lambda_t + u_{mt}. \quad (5)$$

The coefficient β is the causal elasticity of firm dynamics with respect to credit at the market level, identified by variation in credit induced by exogenous bank-level supply shocks: a one-percent increase in credit in market m implies a $\hat{\beta}$ -percent change in the outcome. All regressions are weighted by baseline market credit $C_{m,t-5}$. Standard errors follow the closed-form shift-share correction of [Adão et al. \(2019\)](#), which accounts for correlation across markets sharing common bank shocks.

6 Results

6.1 Descriptive statistics of the shocks

Table 5 reports the [Borusyak et al. \(2021\)](#) diagnostic of our shift-share design at the bank-year level, the unit at which identifying variation lives, with all moments weighted by each

bank’s average market exposure share. Pooled across waves, the design draws on a diffuse set of shocks: the effective sample size, $ESS = 1/\sum_b w_{bt}^2$ computed over normalized exposure shares, reaches 63.99 across the 60 banks observed at least once during 2009–2024, and the largest single bank-year contributes only 4% of the share-weighted variation. Identification therefore does not lean on the idiosyncratic behavior of any single institution. The within-year decomposition tracks the gradual deepening of the Mexican banking system: the number of banks with estimable shocks rises from 10 in 2009 to 39 in 2024, and the within-year ESS from 2.17 to 5.80. The 2009 wave displays markedly narrower shock dispersion than subsequent waves, a feature visible in both share-weighted and unweighted moments (Appendix A, Table 12), consistent with the relatively homogeneous behavior of Mexican incumbent banks at the start of our sample period, prior to the entry of mid-sized and specialized institutions in the 2010s. Although 2009 contributes proportionally less to identification than later waves, the pooled diagnostic combines variation from all four Economic Census waves and is not driven by any single one.

Figure 5: Descriptive statistics of the standardized bank shocks

Year	Mean	Std. deviation	IQR	No. banks	ESS	Largest w
2009	-0.08	0.02	0.00	10	2.17	0.66
2014	0.04	0.77	1.58	17	5.75	0.25
2019	-0.45	1.61	3.26	29	2.72	0.57
2024	-0.21	0.43	0.78	39	5.80	0.36
All	0.00	1.00	1.05	60	63.99	0.04

This table summarizes the distribution of standardized (overall the years) bank-level credit shocks (bank fix-effects), $\hat{\gamma}_{bt}^{std}$, across banks b and years t .

No. banks is the number of banks with non missing estimated shocks.

The effective sample size (ESS) is computed as $1/\sum w_{bt}^2$, and the largest weight is the maximum exposure share.

The Inter-Quartile Range (IQR) is defined as the difference between the 25 and the 75 percentiles.

6.2 Main estimates

This section report our main IV estimates. We follow the recent shift-share literature in reporting standard errors following ?, which account for the cross-market correlation induced by common shocks to bank credit supply; significance stars are based on AKM p -values. We additionally report wild cluster bootstrap p -values in brackets (?) as a nonparametric robustness check.

Table 6 reports the estimates of the main specification. The first-stage results in panel B show that bank shocks, by construction independent of the demand side, are a relevant predictor of the amount of credit observed in equilibrium. One standard deviation increase in the credit supply shocks yields a increase on the amount of loans in a market of around 130% on average. The relevance test, using the fact that the first-stage Kleibergen-Paap exceeds 85 in every column, well above conventional weak-instrument thresholds, gives evidence that the instrument is relevant.

The second-stage estimates in Table 6 shows that credit-supply shocks robustly expand the local stock of formal and informal firms. This effect is mainly driven by an entry margin, there are more firms entering than existing, both as formal and informal. As a result, the total number of firms increases; however, the composition shifts toward a higher informal-to-formal ratio. To grasp the magnitudes, consider that a one percent increase in instrumented five-year market credit growth raises the existing-firm count by 0.18 percent overall, with formal and informal stocks responding by 0.20 and 0.17 percent, respectively. The point estimates are tightly clustered between 0.17 and 0.22 across the nine combinations of formality and margin (existing, entering, exiting), indicating that credit-supply variation propagates through the local economy without disproportionately favoring either the formal or the informal segment, an empirically important fact for a context in which informal firms account for the majority of local establishments.

Evaluated at the average market, the shock corresponds to approximately two additional formal firms and eleven more informal firms. Evaluated at the national level, i.e. multiplied by the number of markets this effect is estimated, this implies 1,106 additional formal firms and 6,116 more informal firms.

To put the magnitude of entry in perspective, this corresponds to an increase of approximately 0.07 percentage points in the startup rate over a five-year period⁴. This magnitude is around one tenth of what Pugsley and Şahin (2019) document as the U.S. start up deficit, a decline of about 5 percentage points in the U.S. startup rate over 32 years (approximately a quinquennial decrease of 0.78 percentage points). While this comparison is imperfect given the inclusion of informal firms it provides a useful benchmark given the limited evidence on informal firm dynamics.

Entry and exit respond symmetrically, both at roughly 0.20 percent, consistent with credit shocks raising market churn rather than only the net firm count. At the average market the shock increases by five (2,780 nationally) the number of firms entering and by 4 the number of firms exiting (2,224 nationally). This firm replacement can be seen through the lenses of the creative-destruction of (Schumpeter, 1942), where the creation of a new more productive firm destroys and replaces old unproductive firms.

⁴This number comes from the 2,574 new entering firms (2,569 plus the 5 new entrants with the shock) divided by the original 7,152 existing firms (35.99%) which we rest from the original 2,569 entrants divided by 7,152 existing firms (35.92%).

Wild-cluster bootstrap p -values confirm significance at conventional levels for all nine specifications, with the most conservative inference at $p < 0.10$ for the headline existing-firms total ($p_{\text{boot}} = 0.078$).

Figure 6: Shift-share IV estimates of the effect of bank shocks on the number of firms

Dependent variable: Log number of firms									
	Existing			Entering			Exiting		
	Total	Formal	Informal	Total	Formal	Informal	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.176*** (0.040) [0.078]	0.197*** (0.039) [0.018]	0.173*** (0.045) [0.090]	0.192*** (0.036) [0.029]	0.219*** (0.042) [0.001]	0.190*** (0.038) [0.043]	0.203*** (0.039) [0.036]	0.183*** (0.042) [0.062]	0.205*** (0.041) [0.039]
FS Coeff.	1.309 (0.144)	1.355 (0.156)	1.355 (0.156)	1.374 (0.161)	1.374 (0.161)	1.374 (0.161)	1.374 (0.161)	1.374 (0.161)	1.374 (0.161)
FS F	88.96	88.97	88.96	86.40	86.43	86.40	86.40	86.39	86.40
Mean	7,152	868	6,284	2,569	269	2,301	1,887	198	1,689
Obs.	1,679	1,675	1,679	1,390	1,378	1,390	1,390	1,369	1,390
Markets	556	553	556	556	549	556	556	547	556

Notes: Two-stage least squares estimates at the market (LLM) $\times$ Economic Census wave level (2009, 2014, 2019, 2024). The endogenous regressor is the five-year log change in market credit, instrumented by the leave-one-market-out shift-share bank credit-supply shock. The mean row is in levels. All columns include year fixed effects and predetermined controls for market sectoral composition (employment- and credit-weighted), lagged interest rate, lagged collateral coverage, log number of banks, and log number of credits. Sample restricted to market-years with instrument coverage ≥ 0.5 . Inference: AKM standard errors following [Adão et al. \(2019\)](#) in parentheses, with stars from the corresponding p -values; wild cluster bootstrap p -values ([Cameron et al., 2008](#)), clustered at the market level, in brackets. “FS F” is the Kleibergen-Paap Wald statistic. The average new loan amount in a market is 96.4 billion MXN (≈ 6 billion USD at 16 MXN/USD). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Nevertheless, the additional number of existing formal and informal firms can not be explained only by entry or exit. To understand better where this change in the existing firms is coming from, Table 7 present the second stage of number of firms transitioning between formality status. The results show that the strongest and most precisely estimated response is an increase of formalization (informal firms becoming formal, 0.195, $p_{\text{boot}} = 0.006$), which is consistent with credit access increasing the likelihood of formalization, and an unexpected increase of formal firms transitioning to informality (0.113, $p_{\text{boot}} = 0.037$), although short of conventional significance under the more conservative wild bootstrap. We therefore interpret the positive informal-to-formal margin as the design’s most robust transition finding.

Figure 7: Shift-share IV estimates of the effect of bank shocks on firm transitions

	Dependent variable: Log number of firm transitions			
	Formal–Formal	Informal–Formal	Formal–Informal	Informal–Informal
$\widehat{\Delta_5 \ln(C)}$	0.172*** (0.042) [0.089]	0.195*** (0.030) [0.006]	0.113*** (0.037) [0.135]	0.165*** (0.053) [0.107]
FS F	86.43	85.62	86.42	86.40
Mean	260	166	151	2,548
Obs.	1,372	1,366	1,372	1,390
Markets	545	541	549	556

Notes: See table 6 for specification and inference details. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To understand better what kind of firms are more affected by the credit supply shock, table 8 shows the effect on the number of firms with credit by formality status. The results show two important aspects. First, credit shocks affect essentially identical both formal and informal firms (0.179), indicating that credit access constraints slack in similar magnitudes at both sectors. Second, the positive effects suggests the response operates predominantly on the extensive margin of credit access rather than through deepening existing relationships. Taken together with the previous results, they shows that informal firms are relevantly affected by banking credits and that financial constraints are a cornerstone of informality.

Figure 8: Shift-share IV estimates of effect of bank shocks on number of firms with credit

	Dep. var.: Log No. firms w/credit		
	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.179*** (0.040) [0.070]	0.174*** (0.038) [0.037]	0.190*** (0.047) [0.111]
FS F	88.96	88.96	88.96
Mean	454	110	344
Obs.	1,679	1,679	1,679
Markets	556	556	556

Notes: See Table 6 for specification and inference details. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Now, we turn to analyze what are the impact of these credit supply shocks and its firm dynamics in terms of number of workers, wages, and value added. Table 9 shows that increasing the quantity of banking loans increases the total number of workers (0.15 percent or 56 workers in the average market), but this is mainly through additional informal employment (0.19 percent, or 26 informal workers in the average market), rather than formal one (0.12 percent, or 7 formal workers in the average market).

Regarding formal wages, they are positively correlated (0.20 percent) with additional banking loans, although the magnitude is economically small. Finally, these results give a clear picture regarding the value added. In the presence of additional loans, the average value added decreases (0.07 percent), while the total value added in a market increases (10 percent). Combining this last result with the results found in terms of firm dynamics, the picture becomes clear. Additional loan resources in market allow more informal, low productive, firms to enter the market, lowering the average value added; however, the total value added in a market increases, driven by all the additional firms that entered into the market.

Both value-added results are precisely estimated under AKM but not robust to the wild bootstrap ($p_{\text{boot}} = 0.139$ and 0.226 , respectively), and we therefore interpret them with caution: the design appears to identify margin-of-entry and labor-reallocation effects with substantially greater precision than per-firm productivity effects.

Figure 9: Shift-share IV estimates of the effect of bank shocks on workers, wagebill, and value added

	Log number of workers			Log total wagebill	Log avg. VA	Log total VA
	Total	Formal	Informal			
$\widehat{\Delta_5 \ln(C)}$	0.150*** (0.028) [0.047]	0.122*** (0.039) [0.184]	0.185*** (0.028) [0.025]	0.200*** (0.039) [0.022]	-0.073*** (0.020) [0.226]	0.102*** (0.034) [0.139]
FS F	84.66	84.65	84.66	88.96	88.96	88.96
Mean	37,304	5,566	13,955	10.9	48.8	0.9
Obs.	992	968	992	1,671	1,679	1,679
Markets	546	534	546	552	556	556

Notes: See Table 6 for specification and inference details. “Log total wagebill” is the log of total formal-sector wagebill; “Log avg. VA” is the log of average per-firm value added; “Log total VA” is the log of aggregate value added. Means in levels: daily wage 174 MXN (≈ 10.9 USD), annual average value added 780 MXN (≈ 48.8 USD), annual total value added 14.5 million MXN (≈ 0.9 million USD) at 16 MXN/USD. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10 examines two financial-market channels through which the credit-supply shock transmits to firms. Credit growth raises the credit-weighted market interest rate by 0.46 percentage points, bust to both inference procedures ($p_{\text{boot}} = 0.004$), and reduces the credit-weighted log of collateral by 0.73 percent ($p < 0.10$); however, the bootstrap p -value of 0.175 indicates we cannot reject the null of no effect under the more conservative inference. We therefore highlight the positive interest-rate response as the cleaner channel finding, while noting that the sign on collateral is qualitatively consistent with banks easing pledgeability requirements in response to favorable funding conditions. The positive interest-rate response, while at first counterintuitive for a supply-driven expansion, is consistent with a composition effect in which new lending extends to riskier borrowers who command higher pricing. Together, the two estimates suggest that the marginal credit relationship created by a positive bank shock is one in which collateral requirements fall but average interest rates rise—a pattern compatible with the extensive-margin entry of previously credit-rationed (and observably riskier) firms.

Figure 10: Shift-share IV estimates of effect of bank shocks on collateral and interest rate

Dependent variable	Log collateral	Interest rate
$\widehat{\Delta_5 \ln(C)}$	−0.733*** (0.225) [0.175]	0.460*** (0.073) [0.004]
FS F	86.50	88.96
Mean	5.2	14.97
Obs.	1,287	1,679
Markets	525	556

Notes: See Table 6 for specification and inference details. Both outcomes are credit-weighted market-level aggregates of the underlying bank-loan-level variables. The interest rate is expressed in percentage points. Average market collateral is 82.3 (one-year) and 82.9 (five-year) billion MXN (≈ 5.1 and 5.2 billion USD at 16 MXN/USD). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Overall, these results support the conclusion that bank credit supply is a quantitatively important driver of firm dynamics in Mexican local economies. We found an important pass-through of loans to firm dynamics and formality composition. Additional financial resources in a market finances both formal and *informal* firms, increasing firm entry, and allowing more existing informal firms to become formal.

6.3 Robustness checks

A central identifying assumption of our design is that the bank credit-supply shocks aggregated by the shift-share instrument are as-good-as-randomly assigned conditional on the controls in our main specification. Following the balance-test prescription in [Borusyak et al. \(2021\)](#), we test this assumption by regressing each of the predetermined market controls included in our second-stage specification on the leave-one-market-out shift-share instrument. Table B in Appendix B reports the results for the seven controls, each at lags of five and ten years before the outcome wave. None of the fourteen coefficients is statistically distinguishable from zero at conventional levels (cluster-robust p -values exceed 0.10 throughout) and the point estimates are economically small. The pattern is consistent with the orthogonality condition required by the design: conditional on year fixed effects and the controls included in our main specification, the residual variation in the instrument is uncorrelated with the level or trajectory of predetermined market characteristics. We

therefore interpret the identifying variation in our shift-share as exogenous with respect to the pre-existing economic conditions of Mexican local labor markets.

As a model-free check on our identification, Table 14 in Appendix C reports reduced-form estimates of the relationship between the leave-one-market-out shift-share instrument and each outcome. Each row presents the coefficient from an OLS regression of the outcome on the standardized instrument, with the same controls and sample restrictions as our main IV specification. By construction, the reduced-form coefficient equals the product of the corresponding first-stage and second-stage IV coefficients. The pattern of signs and significance mirrors the IV estimates throughout: a one-standard-deviation increase in the shift-share exposure to bank credit-supply shocks is associated with statistically significant increases in the stock, entry, and exit margins of firms, the share of firms with credit access, employment, and the aggregate wagebill, alongside a reduction in collateral and a rise in the credit-weighted interest rate. We report the IV estimates as our preferred specification because they recover the elasticity of each outcome with respect to credit growth, the economically interpretable parameter, rather than the response to a standardized exposure measure; the reduced form serves to demonstrate that the underlying correlation is present in the data without invoking the IV decomposition.

The credit-weighted regression places disproportionate weight on the three largest Mexican metropolitan areas (Mexico City, Monterrey, and Guadalajara) which together account for nearly half of the base-period credit in our sample. To address the concern that our main estimates may reflect dynamics specific to these urban banking centers rather than the typical Mexican local labor market, Table 15 in Appendix D re-estimates the main specification excluding these three metropolitan areas. The estimated elasticities of the stock, entry, exit, and credit-access margins of firms, of total and sectoral employment, and of the aggregate wagebill remain positive, precisely estimated, and statistically significant at the 1% level under AKM inference (Adão et al., 2019) across virtually every outcome. The point estimates are uniformly *larger* than in the full sample, the headline elasticity of existing firms, for instance, doubles from 0.176 to 0.357, consistent with the credit-weighted average masking the more elastic response of the non-metropolitan markets in which the typical Mexican firm is located. The interest-rate response is similarly robust ($0.46 \rightarrow 0.52$). The one substantive change is the sign on collateral, which reverses outside the metropolitan areas; we interpret this as evidence that the credit-supply-induced relaxation of collateral requirements documented in our main results is a metropolitan-banking phenomenon and does not extend to smaller markets, where credit expansion is associated with somewhat stricter collateralization. Taken together, these results indicate that our baseline findings are not artifacts of the concentrated weight of the three largest markets; if anything, they may understate the elasticity of local labor market outcomes to bank credit-supply shocks in the typical Mexican market.

The credit weighting in our main specification (amount of credit in the market at baseline) implies that the elasticity reported in Tables 6–10 is a credit-weighted average of market-specific responses, in which markets with larger base-period credit (notably the three metropolitan areas) carry proportionally more weight. To examine the importance of this weighting choice, Table 16 in Appendix E re-estimates our main specification with each market-year contributing equally. The estimated elasticities are uniformly larger than in our main specification, the existing-firm stock elasticity rises from 0.176 to 0.320, firm entry from 0.192 to 0.418, and informal firms from 0.173 to 0.322, consistent with the excluding the three largest metropolitan result in Table 15 and with the broader interpretation that the typical Mexican local labor market is more responsive to bank credit-supply shocks than the credit-weighted average implies. Under cluster-robust inference at the market level, all firm-margin outcomes remain statistically significant at the 1% level. Under AKM inference (reported in Table 16), the standard errors widen substantially, a feature of unweighted shift-share estimation in which each market contributes equally to the identification, and several stock-margin outcomes are no longer significant at conventional levels. Notably, however, the entry, exit, and formality-transition margins survive AKM inference at the 5% level, as do the informal-firm and informal-worker outcomes. Together with the exclusion of the three biggest metropolitan result, this evidence indicates that our headline conclusions are not artifacts of the credit weighting; if anything, the typical-market elasticities suggested by the unweighted specification are larger than those reported in our credit-weighted headline.

As a direct test of the identifying assumption that our shift-share instrument is uncorrelated with pre-existing dynamics in local labor market outcomes, Table 17 in Appendix F reports placebo IV estimates for the main firm transition variables in which we replace the contemporaneous outcome with its pre-period five-year change, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$. The exercise asks directly whether the bank credit-supply shocks predict trajectories in firm dynamics that materialized in the five years *before* the shock window. The stock of firms, the share with credit access, and the gross flow margins, the placebo coefficients are an order of magnitude smaller than the corresponding main IV estimates (e.g., 0.009 for the stock of existing firms, against a main effect of 0.176). The estimates on entry, exit, and formality transitions are uninformative because the pre-period sample shrinks substantially when the further lag is required, producing first-stage F -statistics below conventional weak-instrument thresholds; we report them for transparency rather than as evidence in either direction. Overall, the absence of consistent predictive power for lagged outcomes provides reassurance that the estimated effects are not driven by pre-trends.

A natural question is whether our headline findings depend on the leave-one-market-out construction of the shift-share instrument, or whether the substantive conclusions would survive under the conventional (non-leave-one-out) design used in much of the shift-share

literature. Table 18 in Appendix G reports IV estimates using this conventional construction, in which bank credit-supply shocks $\hat{\gamma}_{bt}$ are estimated on the full sample including each market’s own loans, so that the instrument for market m at time t is $\hat{Z}_{m,t} = \sum_b s_{b,m,t-5} \cdot \hat{\gamma}_{b,t}$ with every bank b contributing the same shock value regardless of which market is being instrumented. As emphasized by Borusyak et al. (2021), this design is vulnerable to a mechanical own-overlap bias: the instrument for market m partially reflects m ’s own contemporaneous credit dynamics, which inflates the apparent IV estimate in the direction of the underlying causal effect. The data display exactly this pattern. The conventional shift-share elasticities are uniformly larger in magnitude than those in our leave-one-market-out specification, the headline elasticity of the stock of existing firms rises from 0.176 to 0.273, the effect on informal firms from 0.173 to 0.294, and the magnitude of the collateral response from -0.73 to -1.18 . The qualitative pattern of our results is preserved nonetheless: every firm-margin response, every formality transition, and every aggregate labor and credit outcome retains its sign, its statistical significance, and its order of magnitude under both AKM inference and the wild cluster bootstrap. We adopt the leave-one-market-out instrument as our preferred specification because it eliminates the overlap bias by construction; the robustness reported in Table 18 confirms that our findings are not artifacts of the methodological refinement, but rather are conservative relative to what the conventional shift-share design would deliver.

As a stand-alone identification check using a structurally distinct research design, Table 19 in Appendix H reports IV estimates at the bank-year level. We identify bank credit-supply shocks following the cell-fixed-effects approach of Degryse et al. (2019), an extension of the within-firm framework of Khwaja and Mian (2008) in which credit demand is absorbed at the market-sector (LLM $\times$ industry) cell level rather than at the individual firm level. This refinement is consequential in our setting because it permits identification from *both* multi-bank and single-bank firms, substantially expanding the borrower sample used to recover the shocks, a particularly important feature for Mexico, where single-bank relationships dominate the local credit market. The bank-level specification aggregates each bank’s outcome contribution across the markets it serves and instruments the bank’s five-year credit growth with this cell-FE pre-identified credit-supply shock. The two designs we report, market-level shift-share and bank-level cell-FE shift-share, identify the elasticity of local outcomes to credit-supply variation from *structurally different* sources: within-market-across-banks at the market level, within-(LLM $\times$ sector)-cell-across-banks at the bank level. They operate on different unit populations, deploy different identification arguments, and rely on different assumptions about the orthogonality structure of the shocks; yet both should recover the same economic parameter if our story about credit-supply pass-through is correct.

The data deliver this convergence in striking detail. The bank-level elasticity of the

existing-firm stock is 0.188, indistinguishable from the market-level estimate of 0.176. The entry margin yields 0.181 at the bank level versus 0.192 at the market level; informal firms yield 0.197 versus 0.173; informal-to-formal transitions yield 0.187 versus 0.195. Across every margin of firm response, every formality transition, and every aggregate labor outcome, the two designs deliver point estimates within 0.03 of each other and the same pattern of statistical significance under wild cluster bootstrap inference. The first-stage Kleibergen-Paap statistic at the bank level ($F \approx 34$) is naturally lower than at the market level ($F \approx 89$) given the substantially smaller cross-section of banks, but remains well above conventional weak-instrument thresholds. We interpret the convergence of two structurally distinct IV designs on essentially the same elasticities as among the strongest pieces of evidence for the substantive conclusions of this paper. The collateral and interest-rate channels are too imprecisely estimated at the bank level, owing to the small bank-year cross-section, to draw inference from there; for these intermediate channels the market-level results in Table 10 remain our primary evidence.

Our main analysis restricts the estimation sample to market-years for which the leave-one-market-out shift-share instrument is constructed from at least half of the local base-period banking exposure. Table I in Appendix 20 re-estimates our headline specification at the two extreme alternatives: a threshold of zero, which includes every market-year for which an instrument can be constructed regardless of how thinly the underlying banking exposure is sampled, and a threshold of 0.8, which restricts identification to market-years where the instrument pools the great majority of local banking activity. The point estimates rise monotonically with the threshold across nearly every outcome, for the headline existing-firm stock, the elasticity moves from 0.160 ($\text{cov} \geq 0$) to 0.176 (main spec) to 0.201 ($\text{cov} \geq 0.8$); the corresponding patterns for entry, exit, formality transitions, and the aggregate labor outcomes are quantitatively similar. We interpret this regularity as consistent with classical attenuation bias from measurement error in the LOO instrument: as the threshold tightens, the bank-shock exposure underlying the instrument is more precisely measured and the resulting IV estimate approaches more closely the underlying causal parameter. Crucially, all elasticities remain precisely estimated and statistically significant under AKM inference at *both* extreme thresholds, with the first-stage Kleibergen-Paap statistic exceeding 67 in every column. Our central findings are therefore not sensitive to the specific coverage threshold chosen and are, if anything, conservative relative to what tightening the threshold would deliver.

A potential concern with our leave-one-market-out shift-share instrument is that its magnitude scales mechanically with coverage: in a market where only half of the base-period banks have an identifiable shock, the instrument is a coverage-weighted sum that is mechanically half as large as it would otherwise be. As a robustness check on this scale concern, Table 21 in Appendix J re estimates our main specification using the *renormal-*

ized instrument $\tilde{z}_{m,t} = z_{m,t}/\text{cov_z_5}_{m,t}$, which converts the coverage-weighted sum into a coverage-weighted average. The estimated elasticities are virtually identical to those of the main specification—the headline existing-firm elasticity moves from 0.176 to 0.174, the entry margin from 0.192 to 0.192, the informal-to-formal transition from 0.195 to 0.196, the interest-rate response from 0.46 to 0.46—and statistical significance is preserved at the 1% level under AKM inference for every firm-margin outcome. We conclude that our results are not artifacts of the instrument’s coverage-weighted scaling.

The robustness checks reported in this section converge on a single conclusion: the headline elasticity of approximately 0.18 for the effect of bank credit-supply shocks on the extensive margin of firm activity in Mexican local labor markets is not driven by any single methodological choice. The shift-share instrument is orthogonal to predetermined market controls and to pre-period outcome changes; the estimated elasticities survive the removal of the leave-one-market-out correction, the exclusion of the three largest metropolitan areas, the removal of credit weighting, variation in the coverage threshold from zero to 0.8, and the renormalization of the instrument by coverage. Most compellingly, a structurally distinct identification strategy at the bank level, the cell-fixed-effects design of [Degryse et al. \(2019\)](#) that identifies from *within*-cell variation across banks rather than across markets, recovers point estimates within 0.03 of our market-level results across every firm-margin outcome. The intermediate financial channels of collateral and interest rates are less consistently identified across specifications, and we accordingly treat the firm-margin and aggregate labor-market findings as our central substantive contributions. The weight of the accumulated evidence supports the broader conclusion that bank credit supply is a quantitatively important driver of firm dynamics in Mexican local economies.

7 A two-period model of formality and finance

This section develops a two-period version of the full model of Section 8. Its purpose is threefold. First, it delivers *closed-form* decision rules and transition thresholds for every margin observed in the data: entry into the informal or the formal sector, exit from either sector, and switching in both directions. Second, it signs, in closed form, the response of every threshold and of every transition *probability* to the two financial primitives that the credit-supply shocks of Section ?? move: the marginal cost of external finance and the collateral capacity of each sector. Third, it produces the composition results that discipline the interpretation of the shift-share estimates: a credit expansion operates through *marginal* firms, so it simultaneously raises entry, raises the subsequent churn of the entering cohort, and lowers average productivity per firm. The general model of Section 8 adds sector-specific productivity processes, a size-dependent cost of informality on the labor margin, a fixed cost of borrowing, and an infinite horizon; every mechanism there is already visible

here.

7.1 Environment

Agents, horizons, preferences. Time lasts two periods, $t \in \{1, 2\}$. A firm is run by a risk-neutral entrepreneur who consumes her dividends and discounts period 2 at factor $\beta \in (0, 1)$. The economy is populated by (i) *incumbents*, firms that begin period 1 attached to a sector $s \in \{i, f\}$ (informal or formal) with installed capital $k_1 > 0$, and (ii) *potential entrants*, agents endowed with financial wealth $k_0 \geq 0$ who may create a firm that operates in period 2. The output good is the numéraire and the wage w is exogenous to the firm (in the quantitative model w is the local-labor-market wage, normalized to 1; the partial-equilibrium treatment mirrors the within-market identification of the empirical design).

Technology and productivity. An active firm in sector s with capital k and labor ℓ produces

$$y = \theta k^\alpha \ell^\nu, \quad \alpha, \nu > 0, \quad \alpha + \nu < 1. \quad (6)$$

Log productivity combines a permanent component v , drawn at birth from $\ln v \sim N(0, \sigma_v^2)$ and known to the firm, and transitory shocks:

$$\ln \theta_1 = \ln v + \varepsilon_1, \quad \ln \theta_2 = \rho \ln \theta_1 + (1 - \rho) \ln v + \varepsilon_2, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2), \quad (7)$$

with $\rho \in (0, 1)$. In the general model $(\rho, \sigma_\varepsilon)$ are sector-specific; here they are common so that sector choice is driven by the frictions rather than by the productivity process.

Taxes and the cost of informality. A formal firm pays a proportional tax $\tau_y \in [0, 1]$ on revenue and a payroll tax $\tau_\ell \geq 0$ on its wage bill. An informal firm evades both. To isolate the financial mechanism, the two-period model shuts down the size-dependent audit technology of the general model (formally, $\eta_i \rightarrow \infty$ in the notation of Section 8), so the *only* advantages of formality are financial and the *only* advantages of informality are fiscal and its lower fixed costs. Operating in sector s in period 2 also requires a per-period fixed cost $c_s \geq 0$, with $c_f > c_i$ (registration upkeep, accounting, visibility).

Financial frictions. All within-period payments (wages, taxes) are settled out of contemporaneous revenue. Investment, in contrast, must be paid one period ahead of the revenue it generates, so a firm whose desired capital exceeds its internal funds must raise *external funds* $a \geq 0$. External finance is intermediated at a sector-specific cost: raising a costs the firm a fee

$$\Phi_s(a) = \phi_s a, \quad \phi_i > \phi_f \geq 0, \quad (8)$$

paid on top of repayment of principal within the period. The parameter ϕ_s is the *spread* between the marginal cost of external funds and the risk-free rate in sector s ; informality

raises it because informal firms cannot pledge audited accounts, registered assets, or formal payroll as information collateral (Section ?? documents the corresponding gap in loan pricing). In addition, external funds are limited by a collateral constraint on the capital being installed,

$$a \leq b_s k', \quad 0 \leq b_i < b_f < 1, \quad (9)$$

where k' is next-period capital: lenders finance at most a fraction b_s of the installed capital of a sector- s borrower, and formal status raises pledgeability. The pair (ϕ_s, b_s) summarizes the financial conditions faced by sector s ; the empirical counterpart of a bank credit-supply shock is a change in this pair, and all comparative statics below are taken with respect to it. The general model adds a fixed cost of borrowing, $\Phi_s(a) = \phi_{0s}\mathbf{1}\{a > 0\} + \phi_s a$ (Remark 7.4); the linear case makes every formula below explicit.

Entry, switching, exit, and death. Creating a firm in sector s costs c_s^e , with $c_f^e > c_i^e$ (registration, licenses, documentation). An informal incumbent that formalizes pays $c_{if} \geq 0$; a formal incumbent that de-formalizes pays $c_{fi} \geq 0$. A firm operating in sector s in period 2 is destroyed with exogenous probability $\lambda_s \in [0, 1)$ *before* producing, in which case the owner recovers only the undepreciated capital; the ranking $\lambda_i \geq \lambda_f$ (formal status insures against expropriation, eviction, and enforcement shocks) is allowed but not imposed. Capital depreciates at rate $\delta \in (0, 1)$ and is resold at its undepreciated value.

Timing. Figure 11 summarizes the timing, which mirrors the recursive model: *sector choices for period $t+1$ are made at the end of period t , before the period- $(t+1)$ innovation ε_{t+1} is observed.*

1. **Period 1 (production).** Incumbents observe (θ_1, v) , hire labor at wage w , produce, settle taxes, and pay the fixed cost c_s .
2. **Period 1 (transition stage).** Each incumbent chooses $d \in \{\text{exit}, i, f\}$ and, if continuing in sector s' , next-period capital k_2 , borrowing $a = \max\{0, k_2 - n\}$ at fee $\Phi_{s'}(a)$ subject to $a \leq b_{s'} k_2$, where n denotes *internal funds* at this stage (defined in (10) below). Potential entrants, knowing (v, k_0) but *not* ε_2 , choose $d \in \{\text{stay out}, i, f\}$, pay $c_{s'}^e$ if entering, and choose k_2 under the same financial technology. Dividends (possibly negative, interpreted as the owner's outside funds absorbed at the financing fee's shadow price) are consumed.
3. **Period 2.** Death shocks realize (probability $\lambda_{s'}$): a destroyed firm's owner keeps $(1 - \delta)k_2$. Surviving firms observe θ_2 , hire labor, produce, pay taxes and the fixed cost $c_{s'}$, then sell undepreciated capital $(1 - \delta)k_2$. The world ends.

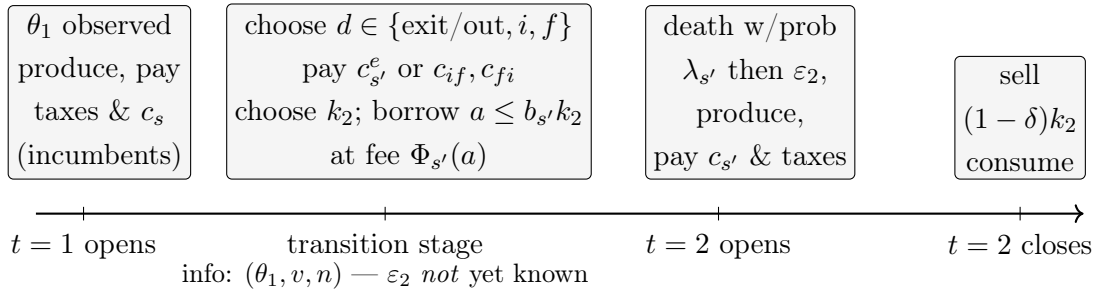


Figure 11: Timing of the two-period model. Sector choices for period 2 are made before the period-2 innovation is observed, exactly as in the recursive model of Section 8.

Internal funds. At the transition stage the resources available inside the firm are the period-1 operating profit plus the resale value of undepreciated capital, net of the fixed cost and of any switching or entry fee:

$$n = \begin{cases} \pi_s(\theta_1, k_1) - c_s + (1 - \delta)k_1 - c_{ss'} \mathbf{1}\{s' \neq s\}, & \text{incumbent of sector } s, \\ k_0 - c_{s'}^e, & \text{entrant into } s', \end{cases} \quad (10)$$

with $c_{ss} \equiv 0$. An exiting incumbent simply consumes $n^{\text{exit}} = \pi_s(\theta_1, k_1) - c_s + (1 - \delta)k_1$; a non-entrant consumes k_0 . Throughout, “financial conditions” means the four-vector $(\phi_i, \phi_f, b_i, b_f)$.

7.2 Static block: labor demand and the profit function

Let τ_y^s and τ_ℓ^s denote the taxes effectively paid in sector s : $\tau_y^f = \tau_y$, $\tau_\ell^f = \tau_\ell$, $\tau_y^i = \tau_\ell^i = 0$. Conditional on operating with state (θ, k) , the firm solves $\max_{\ell \geq 0} (1 - \tau_y^s) \theta k^\alpha \ell^\nu - (1 + \tau_\ell^s) w \ell$; substituting the optimal ℓ back into the objective yields the operating-profit function

$$\pi_s(\theta, k) = (1 - \nu) (1 - \tau_y^s) \theta k^\alpha \left[\frac{\nu (1 - \tau_y^s) \theta k^\alpha}{(1 + \tau_\ell^s) w} \right]^{\frac{\nu}{1-\nu}},$$

which the next lemma rewrites in the separable form used throughout.

Lemma 7.1 (Labor demand and profits). *Optimal labor and the operating-profit function are*

$$\ell_s^*(\theta, k) = \left[\frac{\nu (1 - \tau_y^s) \theta k^\alpha}{(1 + \tau_\ell^s) w} \right]^{\frac{1}{1-\nu}}, \quad \pi_s(\theta, k) = \Omega_s x k^\gamma, \quad (11)$$

where

$$x \equiv \theta^{\frac{1}{1-\nu}}, \quad \gamma \equiv \frac{\alpha}{1-\nu} \in (0, 1), \quad \Omega_s \equiv (1 - \nu) (1 - \tau_y^s)^{\frac{1}{1-\nu}} \left[\frac{\nu}{(1 + \tau_\ell^s) w} \right]^{\frac{\nu}{1-\nu}}. \quad (12)$$

Labor demand is strictly increasing in θ and k ($\partial \ell^* / \partial \theta > 0$, $\partial \ell^* / \partial k > 0$) and strictly decreasing in the wage and in own-sector taxes ($\partial \ell^* / \partial w < 0$, $\partial \ell^* / \partial \tau_y < 0$, $\partial \ell^* / \partial \tau_\ell < 0$).

The profit function is strictly increasing and strictly concave in k ($\gamma < 1$), and the tax wedge

$$T \equiv \frac{\Omega_f}{\Omega_i} = (1 - \tau_y)^{\frac{1}{1-\nu}} (1 + \tau_\ell)^{-\frac{\nu}{1-\nu}} < 1 \quad (13)$$

summarizes the fiscal cost of formality per unit of the profitability index x .

All proofs are collected in Section K, an appendix to this section. The index $x = \theta^{1/(1-\nu)}$ is the natural unit of firm heterogeneity: profits are *linear* in x , so all expectations below reduce to means of a log-normal random variable. Conditional on the transition-stage information set, define the *expected profitability index*

$$\hat{x} \equiv \mathbb{E}[x_2 \mid \mathcal{I}_1] = \exp \left\{ \frac{\rho \ln \theta_1 + (1 - \rho) \ln v}{1 - \nu} + \frac{\sigma_\varepsilon^2}{2(1 - \nu)^2} \right\} \quad (\text{incumbents}), \quad (14)$$

and $\hat{x}_e(v) = \exp\{\ln v/(1 - \nu) + \sigma_\varepsilon^2/(2(1 - \nu)^2)\}$ for entrants (whose period-2 productivity is $\ln \theta_2 = \ln v + \varepsilon_2$). $\hat{x}$ is strictly increasing in θ_1 and in v ; it is the single sufficient statistic for expected productivity in every decision below.

7.3 Investment and financing: a four-region policy

Consider a firm that has chosen to operate in sector s in period 2 (the sector subscript is momentarily suppressed) with internal funds $n \geq 0$ and expected index $\hat{x}$. Its investment problem at the transition stage is

$$G_s(\hat{x}, n) \equiv \max_{k \geq 0} -k - \Phi_s(a) + \beta \left[(1 - \lambda_s) \Omega_s \hat{x} k^\gamma + (1 - \delta)k \right], \quad a = \max\{0, k - n\}, \quad a \leq b_s k, \quad (15)$$

so that the value of operating in s is $U_s(\hat{x}, n) = n + G_s(\hat{x}, n) - \beta(1 - \lambda_s)c_s$: the firm consumes n minus the resources it puts into the firm, and expects profit net of the fixed cost if it survives, plus the capital resale value in either survival state. Define the internal and external *user costs of capital* and the two notional targets

$$R^0 \equiv 1 - \beta(1 - \delta), \quad R_s \equiv R^0 + \phi_s, \quad \bar{k}_s(\hat{x}) = \left[\frac{\gamma M_s \hat{x}}{R^0} \right]^{\frac{1}{1-\gamma}}, \quad \tilde{k}_s(\hat{x}) = \left[\frac{\gamma M_s \hat{x}}{R_s} \right]^{\frac{1}{1-\gamma}}, \quad (16)$$

with $M_s \equiv \beta(1 - \lambda_s)\Omega_s$. Since $\phi_s > 0$, $\tilde{k}_s < \bar{k}_s$: a borrower's optimal scale is compressed relative to a self-financed firm's.

Lemma 7.2 (Capital and borrowing policy). *The solution to (15) is*

$$k_s^*(\hat{x}, n) = \begin{cases} \bar{k}_s(\hat{x}), & n \geq \bar{k}_s(\hat{x}) & (R1: \text{unconstrained}) \\ n, & \tilde{k}_s(\hat{x}) \leq n < \bar{k}_s(\hat{x}) & (R2: \text{self-financed at the kink}) \\ \tilde{k}_s(\hat{x}), & (1 - b_s)\tilde{k}_s(\hat{x}) \leq n < \tilde{k}_s(\hat{x}) & (R3: \text{interior borrowing}) \\ \frac{n}{1 - b_s}, & 0 \leq n < (1 - b_s)\tilde{k}_s(\hat{x}) & (R4: \text{collateral-constrained}) \end{cases} \quad (17)$$

with borrowing $a_s^* = \max\{0, k_s^* - n\}$, strictly positive exactly in R3–R4. The policy is continuous, weakly increasing in $\hat{x}$ and in n , and satisfies the signed derivatives

$$\begin{aligned} \frac{\partial k_s^*}{\partial \phi_s} &= -\frac{\tilde{k}_s}{(1-\gamma)R_s} < 0 \text{ in } R3, & 0 \text{ in } R1, R2, R4; & \quad \frac{\partial k_s^*}{\partial b_s} = \frac{n}{(1-b_s)^2} > 0 \text{ in } R4, & \quad 0 \text{ otherwise;} \\ \frac{\partial k_s^*}{\partial n} &= 1 \text{ in } R2, & \frac{1}{1-b_s} > 1 \text{ in } R4, & \quad 0 \text{ in } R1, R3; & \quad \frac{\partial k_s^*}{\partial \hat{x}} > 0 \text{ in } R1, R3, & \quad 0 \text{ in } R2, R4. \end{aligned} \quad (18)$$

Three remarks connect Lemma 7.2 to the data. First, investment–cash-flow sensitivity ($\partial k^*/\partial n > 0$) is positive *only* for constrained firms (R2, R4), and is largest exactly where the collateral constraint binds; this is the firm-level footprint of the credit channel that the shift-share design picks up at the market level. Second, the marginal cost of external funds ϕ_s acts on the *intensive* margin of constrained borrowers (R3), while collateral capacity b_s acts on the most severely constrained firms (R4) — typically young, low- n entrants; the two instruments of a credit expansion therefore reach different parts of the firm distribution. Third, borrowing incidence $\mathbb{P}(a^* > 0)$ rises with expected productivity and falls with internal funds; Remark 7.4 below turns this margin into the identification argument for the fixed cost of borrowing in Section 9.

Lemma 7.3 (Value of operating, closed form). *Define $\Psi_s(\hat{x}; R) \equiv (1-\gamma) \gamma^{\frac{\gamma}{1-\gamma}} (M_s \hat{x})^{\frac{1}{1-\gamma}} R^{-\frac{\gamma}{1-\gamma}}$, which is strictly increasing and strictly convex in $\hat{x}$ and strictly decreasing in R . Then*

$$G_s(\hat{x}, n) = \begin{cases} \Psi_s(\hat{x}; R^0), & R1 \\ M_s \hat{x} n^\gamma - R^0 n, & R2 \\ \phi_s n + \Psi_s(\hat{x}; R_s), & R3 \\ \phi_s n + M_s \hat{x} \left(\frac{n}{1-b_s}\right)^\gamma - R_s \frac{n}{1-b_s}, & R4, \end{cases} \quad (19)$$

and G_s is continuous, strictly increasing and strictly convex in $\hat{x}$, and concave and nondecreasing in n . The marginal value of internal funds, $q_s \equiv 1 + \partial G_s / \partial n$, satisfies

$$q_s = 1 \text{ (R1);} \quad q_s = 1 + \gamma M_s \hat{x} n^{\gamma-1} - R^0 \in (1, 1 + \phi_s] \text{ (R2);} \quad q_s = 1 + \phi_s \text{ (R3);} \quad (20)$$

$$q_s = 1 + \phi_s + \frac{\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s}{1-b_s} > 1 + \phi_s \text{ (R4).} \quad (21)$$

The step function q_s is the shadow price of a peso inside the firm. It exceeds one exactly for constrained firms and jumps above $1 + \phi_s$ when collateral binds; it is the model object behind the role of entrant wealth $(\mu_{k_0}, \sigma_{k_0})$ in the quantitative model.

Remark 7.4 (Borrowing participation and the fixed cost of borrowing). *In the general model the financing fee has a fixed component, $\Phi_s(a) = \phi_{0s} \mathbf{1}\{a > 0\} + \phi_s a$. All results in this section extend directly:*

conditional on borrowing the fixed cost is sunk, so all policies and envelope derivatives are unchanged, and thresholds are unchanged up to the bookkeeping $m \rightarrow m + \phi_{0s}$ (and $C_o \rightarrow C_o + \phi_{0f} - \phi_{0i}$) on branches whose continuation borrows. Moreover, decision to borrow at all acquires its own margin. Define the participation premium as the surplus gain from external expansion over staying at the kink,

$$\pi_s^B(\hat{x}, n) \equiv G_s^{R3/R4}(\hat{x}, n) - [M_s \hat{x} n^\gamma - R^0 n],$$

which equals zero at the R2/R3 boundary and is strictly increasing in $\hat{x}$ beyond it (its $\hat{x}$ -derivative is $M_s(k_s^*{}^\gamma - n^\gamma) > 0$ whenever $k_s^* > n$). With $\phi_{0s} > 0$ the firm borrows iff $\pi_s^B \geq \phi_{0s}$, so the R2/R3 boundary shifts to the unique root of $\pi_s^B = \phi_{0s}$: a borrowing-participation cutoff in $\hat{x}$ given n . The population mass beyond the cutoff — borrowing incidence — is strictly decreasing in ϕ_{0s} given (ϕ_s, b_s) , which is the moment that identifies the fixed cost in Section 9; the intensive-margin objects (a^* , the thresholds, their derivatives) are unchanged conditional on participation.

The next lemma is the analytical workhorse of the paper: by the envelope theorem, the value of a sector responds to its financial conditions in proportion to *equilibrium borrowing*, and to collateral capacity in proportion to the *shadow value of pledgeability*.

Lemma 7.5 (Value of financial conditions). G_s (hence U_s) is differentiable in (ϕ_s, b_s) except at region boundaries, with

$$\frac{\partial U_s}{\partial \phi_s} = -a_s^*(\hat{x}, n) \leq 0, \quad \frac{\partial U_s}{\partial b_s} = \mu_s k_s^* = \mathbf{1}\{R4\} \left[\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s \right] \frac{n}{(1-b_s)^2} \geq 0, \quad (22)$$

where $\mu_s \geq 0$ is the multiplier on (9). Both inequalities are strict exactly when the firm borrows (respectively, when the collateral constraint binds). Cross-derivatives are zero: $\partial U_s / \partial \phi_{s'} = \partial U_s / \partial b_{s'} = 0$ for $s' \neq s$.

7.4 Sector choice: a single-crossing formality margin

Fix an origin state $o \in \{\text{entrant}, i\text{-incumbent}, f\text{-incumbent}\}$ with internal funds n_o available before paying the origin-specific fee of each destination: destination s leaves $n_{o,s} = n_o - \kappa_{os}$ inside the firm, where $\kappa_{os} \in \{c_s^e, c_{if}, c_{fi}, 0\}$ per (10). The formality margin compares

$$\Delta_o(\hat{x}, n_o) \equiv U_f(\hat{x}, n_o - \kappa_{of}) - U_i(\hat{x}, n_o - \kappa_{oi}). \quad (23)$$

The empirically relevant region — growing firms whose desired capital exceeds internal funds — is R3 for both destinations; the statement below is for that region, and Section K verifies that the threshold structure extends piecewise to R2/R4 (and characterizes the cash-rich case in Remark 7.8). On R3×R3,

$$\Delta_o(\hat{x}, n_o) = A \hat{x}^{\frac{1}{1-\gamma}} - B n_o - C_o, \quad (24)$$

with

$$A \equiv (1 - \gamma) \gamma^{\frac{\gamma}{1-\gamma}} \left[M_f^{\frac{1}{1-\gamma}} R_f^{-\frac{\gamma}{1-\gamma}} - M_i^{\frac{1}{1-\gamma}} R_i^{-\frac{\gamma}{1-\gamma}} \right], \quad B \equiv \phi_i - \phi_f > 0, \quad (25)$$

$$C_o \equiv (1 + \phi_f) \kappa_{of} - (1 + \phi_i) \kappa_{oi} + \beta[(1 - \lambda_f)c_f - (1 - \lambda_i)c_i]. \quad (26)$$

Assumption 1 (Finance-driven formality). *Let $T_\lambda \equiv T \cdot \frac{1 - \lambda_f}{1 - \lambda_i}$ be the tax wedge adjusted by the survival advantage of formality. Then*

$$\left(\frac{R_f}{R_i} \right)^\gamma < T_\lambda. \quad (27)$$

Assumption 1 states that, for a firm that must borrow, the financing advantage of formality (right-hand side smaller than one because $R_f < R_i$, raised to the profit curvature γ) outweighs its fiscal cost net of the survival advantage (T_λ , which may be below or above one). It is equivalent to $A > 0$, and it is the exact sense in which formality is, in this model, *bought access to capital*: absent a financing or survival advantage ($R_f = R_i$, $\lambda_f = \lambda_i$), $T < 1$ implies $A < 0$ and no borrower would ever formalize.

Assumption 2 (Interior thresholds and cost ordering). *(a) $C_o + B n_o > 0$ for the origin states considered, i.e. the fixed and transition costs of formality are not dominated by its switching subsidies. (b) The switching and entry fees satisfy $c_{if}, c_{fi} > 0$, $0 \leq c_i^e \leq c_f^e$, and*

$$(1 + \phi_f) c_f^e - (1 + \phi_i) c_i^e < (1 + \phi_f) c_{if}, \quad (\text{A2-i})$$

$$(1 + \phi_f) c_f^e + (1 + \phi_i) c_{fi} > (1 + \phi_i) c_i^e. \quad (\text{A2-ii})$$

Conditions (A2-i)–(A2-ii) are exactly equivalent (Lemma 7.6) to the cost ordering $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ that places the entrant formality margin between the two incumbent margins. (A2-i) says that formalizing *de novo* — the financing-adjusted entry-cost gap — is cheaper than converting an existing informal firm; (A2-ii) says that entering informally is not so expensive as to undo the formal incumbent's lock-in (the fee c_{fi} acts as a subsidy to remaining formal). (A2-ii) holds automatically whenever $c_{fi} \geq c_i^e$, and in particular whenever informal entry is (nearly) free — the empirically relevant case.

Lemma 7.6 (Cost ordering). *Evaluate (26) at each origin's fee pair $(\kappa_{of}, \kappa_{oi}) \rightarrow (c_{if}, 0)$ for informal incumbents, $(0, c_{fi})$ for formal incumbents, (c_f^e, c_i^e) for entrants — and let $K \equiv \beta[(1 - \lambda_f)c_f - (1 - \lambda_i)c_i]$. Then*

$$C_{i\text{-inc}} = (1 + \phi_f)c_{if} + K, \quad C_{f\text{-inc}} = -(1 + \phi_i)c_{fi} + K, \quad C_e = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + K, \quad (28)$$

and $C_e < C_{i\text{-inc}} \iff (\text{A2-i})$ while $C_{f\text{-inc}} < C_e \iff (\text{A2-ii})$. Under $c_{if}, c_{fi} > 0$ alone, $C_{f\text{-inc}} < C_{i\text{-inc}}$ always holds (the hysteresis band exists); (A2-i)–(A2-ii) are what places the entrant margin strictly inside it.

Proposition 7.7 (Formality threshold). *Under Assumptions 1–2, for each origin o and internal funds n_o , $\Delta_o(\cdot, n_o)$ is strictly increasing and strictly convex in $\hat{x}$ and crosses zero exactly once, at*

$$\hat{x}_o^F(n_o) = \left[\frac{B n_o + C_o}{A} \right]^{1-\gamma}. \quad (29)$$

The firm chooses the formal sector if and only if $\hat{x} \geq \hat{x}_o^F(n_o)$. The threshold is strictly increasing in internal funds,

$$\frac{\partial \hat{x}_o^F}{\partial n_o} = (1-\gamma) \frac{B}{A} \left[\frac{B n_o + C_o}{A} \right]^{-\gamma} > 0 : \quad (30)$$

conditional on expected productivity, firms with more internal funds are less likely to formalize, because self-finance substitutes for the financial advantage of formality.

Proposition 7.7 formalizes the sorting logic: the firms that select into formality are those whose *growth needs are large relative to their internal resources* — high $\hat{x}$, low n . These are precisely the firms for which a bank credit-supply shock is decisive, which is why the shift-share elasticities of Section ?? are informative about (ϕ_s, b_s) .

Remark 7.8 (Cash-rich firms and the role of the audit margin). *For unconstrained firms ($R1 \times R1$) the finance advantage is irrelevant and $\text{sgn } \Delta_o = \text{sgn}\{\Psi_f(\hat{x}; R^0) - \Psi_i(\hat{x}; R^0) - \text{costs}\}$, which under $T_\lambda < 1$ is negative for large $\hat{x}$: with taxes as the only non-financial difference, a sufficiently wealthy firm prefers informality. Two observations discipline this artifact. First, with $\lambda_i > \lambda_f$ large enough, $T_\lambda \geq 1$ and formality remains absorbing even for the cash-rich. Second, the general model restores a size-dependent cost of informality on the labor margin (the audit function $\tau_i(\ell)$ of Section 8), which caps informal scale and makes formality optimal for large firms independently of finance. The empirical prevalence of formal-to-informal transitions documented in Section ?? — the dominant switching direction in Mexico — suggests that for a sizable mass of firms the net non-financial advantage of formality is indeed weak, i.e. T_λ close to or below one; the quantitative model lets the data settle this through $(\tau_y, \tau_\ell, \lambda_s, \eta_i)$.*

7.5 The full transition map and its comparative statics

Combining the exit option with Proposition 7.7 yields the complete policy of each origin state, mirroring the decision rules s_n, s_i, s_f of the recursive model.

Exit and entry margins. Every exit and entry margin reduces to one comparison: continuation in the informal sector (the relevant alternative at low $\hat{x}$, by the ordering of Proposition 7.9) against consuming the outside value. Because exit and stay-out payoffs are consumed, the transition fee involved is valued at par in the constant of the comparison,

while inside the continuation value it is priced at the destination's shadow price of funds — both appear automatically below. Write

$$A_s \equiv (1 - \gamma) \left(\frac{\gamma}{R_s} \right)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad A_s^0 \equiv (1 - \gamma) \left(\frac{\gamma}{R^0} \right)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad \mathcal{K}_s \equiv \beta(1 - \lambda_s)c_s, \quad (31)$$

so that $\Psi_s(\hat{x}; R_s) = A_s \hat{x}^{1/(1-\gamma)}$, and note the identities $A = A_f - A_i$ for the formality-margin slope in (25) and $A^0 = A_f^0 - A_i^0$ for the cash-rich slope of Remark 7.8. Each margin is characterized by a condition of the form $G_s(\hat{x}, \tilde{n}) = m$, where $\tilde{n}$ are the funds carried into the continuation and m collects the operating and transition costs. Solving branch by branch of (19) defines the *master threshold*

$$\mathcal{X}_s(\tilde{n}; m) = \begin{cases} (m/A_s^0)^{1-\gamma}, & \text{R1} \\ \frac{m + R^0 \tilde{n}}{M_s \tilde{n}^\gamma}, & \text{R2} \\ \left[\frac{m - \phi_s \tilde{n}}{A_s} \right]^{1-\gamma}, & \text{R3, if } m > \phi_s \tilde{n} \\ (1 - b_s)^\gamma \frac{m + \tilde{n} \frac{R^0 + \phi_s b_s}{1 - b_s}}{M_s \tilde{n}^\gamma}, & \text{R4,} \end{cases} \quad (32)$$

where the applicable branch is the one containing the crossing (the region boundaries depend on $\hat{x}$ through $\tilde{k}_s, \bar{k}_s$, so the branch self-selects; at low $\hat{x}$ every firm is unconstrained relative to the vanishing targets and R1 anchors the threshold). $\mathcal{X}_s$ is continuous across the R1/R2, R2/R3 and R3/R4 boundaries, weakly decreasing in $\tilde{n}$, independent of financial conditions on R1, and satisfies $\mathcal{X}_s \rightarrow \infty$ as $\tilde{n} \rightarrow 0^+$ on R4: firms without internal funds exit — or fail to enter — at *any* productivity (financial exit). On R2, G_s is linear in $\hat{x}$, hence the non-power form. If $m \leq \phi_s \tilde{n}$ on R3 (or the analogous condition on R4), the constrained firm never exits: the quasi-rents $\phi_s \tilde{n}$ earned by internal funds keep it alive at any productivity.

The four margins are instances of (32):

$$\text{exit as informal: } \hat{x}_i^X(n) = \mathcal{X}_i(n; \mathcal{K}_i), \quad n = n^{\text{exit}}, \quad (33)$$

$$\text{exit as formal: } \hat{x}_f^X(n) = \mathcal{X}_i(n - c_{fi}; \mathcal{K}_i + c_{fi}), \quad n = n^{\text{exit}}, \quad (34)$$

$$\text{entry as informal: } \hat{x}_i^E(k_0) = \mathcal{X}_i(k_0 - c_i^e; \mathcal{K}_i + c_i^e), \quad (35)$$

$$\text{entry as formal: } \hat{x}_f^E(k_0) = \hat{x}_e^F(k_0) = \left[\frac{B k_0 + C_e}{A} \right]^{1-\gamma} \quad (\text{Prop. 7.7}). \quad (36)$$

Three reading notes. (i) In (34) the formal incumbent's relevant alternative to exiting is switching down (below $\hat{x}_f^F$ staying formal is dominated by Proposition 7.9), so the comparison is $U_i(\hat{x}, n - c_{fi}) \geq n$; on R3 the numerator of (32) becomes $\mathcal{K}_i + (1 + \phi_i)c_{fi} - \phi_i n$ — the switch fee priced at the informal shadow price of funds, exactly as in Lemma 7.6.

(ii) Likewise on R3 the entry margin (35) has numerator $\mathcal{K}_i + (1 + \phi_i)c_i^e - \phi_i k_0$, and on R4 it inherits $\hat{x}_i^E \rightarrow \infty$ as $k_0 \downarrow c_i^e$: wealth-constrained non-entry in the spirit of Evans–Jovanovic. (iii) Under Assumption 2 the ordering $\hat{x}_i^E(k_0) < \hat{x}_e^F(k_0)$ makes the formality margin the binding condition for formal entry in (36); the direct enter-formal-versus-out threshold, relevant only outside that ordering, is $\mathcal{X}_f(k_0 - c_f^e; \mathcal{K}_f + c_f^e)$. In primitive units, entrant thresholds convert through $\bar{v}(k_0) = [\hat{x}^\bullet(k_0)]^{1-\nu} \exp\{-\sigma_\varepsilon^2/(2(1-\nu))\}$ and incumbent thresholds through (38). A complete taxonomy of these thresholds is developed in Section M.

Proposition 7.9 (Ordered thresholds and hysteresis). *Under Assumptions 1–2 and $c_i > 0$, for each level of internal funds the policy of every origin state is a partition of the $\hat{x}$ line:*

$$\begin{aligned} \text{entrant:} \quad & \text{out on } [0, \hat{x}_i^E), \quad \text{enter } i \text{ on } [\hat{x}_i^E, \hat{x}_e^F), \quad \text{enter } f \text{ on } [\hat{x}_e^F, \infty); \\ i\text{-incumbent:} \quad & \text{exit on } [0, \hat{x}_i^X), \quad \text{stay } i \text{ on } [\hat{x}_i^X, \hat{x}_i^F), \quad \text{switch to } f \text{ on } [\hat{x}_i^F, \infty); \\ f\text{-incumbent:} \quad & \text{exit on } [0, \hat{x}_f^X), \quad \text{switch to } i \text{ on } [\hat{x}_f^X, \hat{x}_f^F), \quad \text{stay } f \text{ on } [\hat{x}_f^F, \infty). \end{aligned}$$

Moreover $\hat{x}_f^F(n) < \hat{x}_e^F(n) < \hat{x}_i^F(n)$: switching costs open a band $[\hat{x}_f^F, \hat{x}_i^F)$ of expected productivity on which informal incumbents remain informal and formal incumbents remain formal (hysteresis in formality status), while entrants sort in between.

Proposition 7.10 (Comparative statics of all thresholds). *On the borrowing regions, with $D_o \equiv Bn_o + C_o > 0$ and a_s^* evaluated at the relevant margin, the thresholds respond to financial conditions as follows (“0” means zero except through inactive regions):*

	$\phi_f \uparrow$	$\phi_i \uparrow$	$b_f \uparrow$	$b_i \uparrow$
Formality margins $\hat{x}_e^F, \hat{x}_i^F, \hat{x}_f^F$	$\frac{(1-\gamma)a_f^*}{D_o} > 0$	$-\frac{(1-\gamma)a_i^*}{D_o} < 0$	≤ 0	≥ 0
Informal entry/exit $\hat{x}_i^E, \hat{x}_i^X$	0	$\frac{a_i^*}{\Psi_i'} > 0$	0	≤ 0
Formal exit $\hat{x}_f^X$	0	$\frac{a_i^*}{\Psi_i'} > 0$	0	≤ 0

Note: entries assume the threshold ordering of Proposition 7.9, so the formal incumbent’s relevant alternative at its exit margin is the switch-down option; off that ordering the formal-exit row picks up ≥ 0 (ϕ_f) and ≤ 0 (b_f) terms through the value of staying formal. The first row reports $\partial \ln(\text{threshold})/\partial(\text{parameter})$; the second and third rows report level derivatives, with a_i^* evaluated at the margin’s funds (n for informal exit/entry, $n - c_{fi}$ for formal exit).

In words: (i) cheaper or more abundant formal credit ($\phi_f \downarrow$ or $b_f \uparrow$) lowers every formality threshold — more formal entry, more informal-to-formal switching, less formal-to-informal switching — and leaves the formal exit margin unchanged, because the exiting formal firm’s relevant alternative is the switch-down option, which formal-side terms do not enter; (ii)

cheaper or more abundant informal credit ($\phi_i \downarrow$ or $b_i \uparrow$) lowers the informal entry and exit thresholds and the formal exit threshold (through the switch-down option) — more informal entry, less exit from either sector — and raises every formality threshold — less informal-to-formal switching, more formal-to-informal switching; (iii) all effects are proportional to equilibrium borrowing at the margin, hence concentrated on firms with high growth needs relative to internal funds, and are zero for unconstrained firms.

A remark on what survives the passage to the general model. The exact zeros in the table are *exclusion restrictions* delivered by the two-period structure: with an infinite horizon, continuation values embed the option to switch sectors again later, so a formal-side parameter leaks into an informal continuation value at second order (and conversely). The general model therefore preserves the table's sign pattern and its near-diagonal loading — each margin responds first and foremost to its own sector's terms, in proportion to marginal borrowing — but not the literal zeros. Section 9 verifies exactly this at the estimated parameters, by reporting the full Jacobian of moments with respect to parameters: the quantitative counterpart of this table.

7.6 From thresholds to transition rates

The thresholds translate into transition *probabilities*, the objects the shift-share design estimates, by integrating over the period-1 population. Consider informal incumbents with permanent component v and internal funds n (the same algebra applies to each origin state). Since $\ln \theta_1 = \ln v + \varepsilon_1$ and $\hat{x}$ is monotone in θ_1 , the probability of an informal-to-formal switch is

$$\mathbb{P}(i \rightarrow f \mid v, n) = 1 - \Phi(z_{if}), \quad z_{if} \equiv \frac{\ln \bar{\theta}_{if}(v, n) - \ln v}{\sigma_\varepsilon}, \quad (37)$$

where the productivity threshold inverts (14):

$$\ln \bar{\theta}_{if}(v, n) = \frac{1}{\rho} \left[(1 - \nu) \ln \hat{x}_i^F(n) - (1 - \rho) \ln v \right] - \frac{\sigma_\varepsilon^2}{2\rho(1 - \nu)}. \quad (38)$$

Expression (38) is *region-free*: its derivation uses only the log-linear map (14) and the existence of a cutoff in $\hat{x}$ -units, so it converts any threshold — whichever branch of (32) delivers it, closed-form or implicit — into a cutoff for θ_1 . The financing region enters only through the *value* of the threshold and through its derivatives in the chain rule below. Analogous expressions hold for $\mathbb{P}(\text{exit} \mid i) = \Phi(z_{iX})$, $\mathbb{P}(f \rightarrow i) = \Phi(z_{fi}) - \Phi(z_{fX})$, entry probabilities (with $\hat{x}_e(v)$ and thresholds in v), etc.

Proposition 7.11 (Transition semi-elasticities). *Let $p = \mathbb{P}(i \rightarrow f \mid v, n)$ from (37). Then, when both destinations borrow at an interior optimum (R3) at the margin,*

$$\frac{\partial p}{\partial \phi_f} = -\phi(z_{if}) \frac{(1 - \nu)(1 - \gamma)}{\rho \sigma_\varepsilon} \frac{a_f^*}{D_i} < 0, \quad \frac{\partial p}{\partial \phi_i} = +\phi(z_{if}) \frac{(1 - \nu)(1 - \gamma)}{\rho \sigma_\varepsilon} \frac{a_i^*}{D_i} > 0, \quad (39)$$

while, when the relevant destination is collateral-constrained (R4) at the margin, the collateral responses are strict:

$$\frac{\partial p}{\partial b_f} = +\phi(z_{if}) \frac{1-\nu}{\rho\sigma_\varepsilon} \frac{\mu_f k_f^*}{N_i} \geq 0, \quad \frac{\partial p}{\partial b_i} = -\phi(z_{if}) \frac{1-\nu}{\rho\sigma_\varepsilon} \frac{\mu_i k_i^*}{N_i} \leq 0, \quad (40)$$

where N_i is the numerator of the operative branch of $\hat{x}_i^F$ at the crossing (equal to $D_i \hat{x}^*$ on $R4 \times R4$; Section M) and $\mu_s k_s^*$ is the shadow value of collateral of Lemma 7.5. The same structure, with the appropriate threshold and sign from Proposition 7.10, applies to every other transition probability. The factor $(1-\gamma)$ is specific to the power branch (R3) and drops on the linear branches; all responses are scaled by $\phi(z)/\sigma_\varepsilon$, the density of firms at the margin, so aggregate responses are largest where the firm distribution is thick near thresholds.

Two aggregation notes. First, because (38) is region-free, the chain rule composes with whichever region's log-derivative is operative at the margin for the given (v, n) : the ϕ -formulas displayed are the R3 instances, the collateral formulas the R4 instances, and R1/R2-financed firms contribute exactly zero. Second, aggregate transition responses are therefore *incidence-weighted integrals* — $\phi(z)$ times the region-specific term, integrated over the joint distribution of (v, n) — which concentrates predicted responses among small, young, low-net-worth firms: the sample counterpart of the heterogeneity documented in Section ??.

Corollary 7.12 (A common credit expansion moves both switching margins up). *Consider a credit-supply expansion that improves both sectors' borrowing terms: spreads fall by the absolute amounts $\varepsilon_s \equiv \chi_s \phi_s \geq 0$, i.e. $d\phi_s = -\varepsilon_s$, and/or collateral capacities rise, $db_s \geq 0$. The conditions below are stated for the spread components; a collateral improvement adds $\mu_s k_s^* db_s$ to its own sector's side of each inequality. Then:*

(i) Entry and exit. Under Proposition 7.9, the binding entry margin and both exit margins, $\hat{x}_i^E$, $\hat{x}_i^X$, $\hat{x}_f^X$, are informal-continuation comparisons, so they respond only to informal-side terms: all three fall when $\varepsilon_i > 0$ or $db_i > 0$, and are unchanged under a purely formal-side expansion. Hence total entry rises and exit falls at both incumbent margins whenever the expansion reaches the informal side; the formal/informal composition of entry is governed by $\hat{x}_e^F$, which obeys the same condition as the switching margins of (ii).

(ii) Switching up. $\mathbb{P}(i \rightarrow f)$ rises if and only if

$$\varepsilon_f a_f^* > \varepsilon_i a_i^* \quad \text{at the } i \rightarrow f \text{ margin.} \quad (41)$$

Because the switch fee travels with the formal option, $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_{if} > 0$ under Assumption 1; hence equal absolute improvements ($\varepsilon_f = \varepsilon_i$) always raise

informal-to-formal switching. Under equal proportional improvements ($\chi_f = \chi_i$), (41) reads $\phi_f a_f^* > \phi_i a_i^*$: the borrowing-scale gap must outweigh the spread gap. The same condition, with the a^* 's evaluated at the entrant's funds ($a_s^* = \tilde{k}_s - (k_0 - c_s^e)$), governs the entrant margin $\hat{x}_e^F$: there $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e > 0$, so equal absolute improvements also raise the formal share of entrants.

(iii) Switching down. $\mathbb{P}(f \rightarrow i)$ rises if and only if

$$\varepsilon_i a_i^* > \varepsilon_f a_f^* \quad \text{at the } f \rightarrow i \text{ margin,} \quad (42)$$

where now the fee travels with the informal option, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$. Both switching margins can therefore rise together only if the expansion has a sufficiently informal-side component (ε_i , or $b_i \uparrow$).

Hence the joint empirical finding of Section ?? — credit expansions raise entry, exit, and both switching directions — is informative about the incidence of bank credit-supply shocks across sectors: both formality margins can rise together only if the marginal borrowing terms of both sectors improve, with a meaningful informal-side component. This is the restriction taken to the data in Section 9: the price-side responses of a credit expansion (interest-rate pass-through and collateral use) pin (χ_f, κ_f) , the formal-to-informal switching response — the margin that can rise only through informal-side incidence — pins χ_i , and the remaining transition and output responses are reserved as out-of-sample validation.

Corollary 7.12 is the toy model's central payoff for the empirical strategy: it converts the seemingly puzzling pattern that formalization *and* de-formalization both accelerate after a credit expansion into a statement about which sector's borrowing terms the shock reached — consistent with the direct evidence that informal firms hold bank credit (Section ??) and that the marginal loan carries higher rates and lower collateral.

7.7 Composition: marginal entrants, churn, and average productivity

The last set of results concerns the *composition* of the response, which is where the model meets the negative average value-added result of Section ?. Allow period-2 firms a shut-down option: after observing θ_2 , an active firm operates iff variable profit covers the fixed cost, $\Omega_s x_2 k_2^\gamma \geq c_s$; otherwise it stays idle for the period (it still resells its capital). Shut-down is the two-period analogue of endogenous exit for a young cohort. For an entering cohort into sector i (the margin a credit expansion moves most), entry occurs iff $v \geq \bar{v}(k_0)$, where $\bar{v}$ solves $\hat{x}_e(\bar{v}) = \hat{x}_i^E(k_0)$, and:

shutdown probability of an entrant with (v, k_2) :
$$\pi^X(v) = \Phi\left(\frac{(1 - \nu) \ln(c_i/(\Omega_i k_2^\gamma)) - \ln v}{\sigma_\varepsilon}\right), \quad (43)$$

which is strictly decreasing in v (directly, and indirectly through k_2).

Proposition 7.13 (Composition of a credit expansion). *Let financial conditions improve for the informal sector ($\phi_i \downarrow$ or $b_i \uparrow$), so that $\bar{v}$ falls for every constrained k_0 . Then, for the entering cohort:*

- (i) Entry rises: *the mass of entrants $\int \mathbb{P}(v \geq \bar{v}(k_0)) dF(k_0)$ increases.*
- (ii) Average entrant productivity falls: *for $\ln v \sim N(0, \sigma_v^2)$, $\mathbb{E}[v \mid v \geq \bar{v}]$ is strictly increasing in $\bar{v}$; hence a lower threshold mechanically lowers the mean (and every quantile) of entrant productivity, and with it average value added per entrant.*
- (iii) Cohort churn rises: *the cohort shutdown rate $\mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ is strictly decreasing in $\bar{v}$; the marginal entrants admitted by the expansion fail at higher rates than infra-marginal ones, so measured exit rises with a lag even as incumbent exit thresholds fall.*

Part (iii) isolates the composition channel: cheaper credit also raises some entrants' capital, which lowers their shutdown risk through k_2 in (43). For the collateral-constrained entrants where the margin lives (R4), however, $k_2 = (k_0 - c_i^e)/(1 - b_i)$ is pinned by wealth and collateral (Lemma 7.2), so a pure spread reduction leaves their capital unchanged and the composition effect is the entire response; a collateral relaxation ($b_i \uparrow$) does deepen their capital and partially offsets it.

Proposition 7.13 reproduces, inside the two-period model, the empirical signature documented in Section ??: a credit expansion raises the stock of firms and gross flows in both directions (entry *and* exit), while *lowering* average value added per firm — growth through the extensive margin of low-productivity, credit-dependent entrants, not through upgrading of infra-marginal firms.

7.8 The model-implied shift-share estimand

Finally, the toy model makes precise what the leave-one-out shift-share coefficient identifies through the lens of the model. Appendix L shows as an example the detailed case of the effect on the mass of informal firms. Given the exact computation in the quantitative model is done numerically, the generic case presented here is shown in a schematic manner to draw general conclusions. Let the bank-supply shock in market m shift financial conditions by $d\Xi_m = (-\chi_i \phi_i, -\chi_f \phi_f, \kappa_i b_i, \kappa_f b_f)' \zeta_m$, where ζ_m is the (exogenous) shock intensity.

Aggregate borrowing is $C = \int a^* d\mu$ over active firms, and each firm-count outcome Y (stocks, entry, exit, transitions by sector) aggregates the threshold rules. To first order, the model counterpart of the IV estimand is the ratio of two equilibrium responses to the *same* financial-conditions shift:

$$\beta_Y^{\text{model}} = \frac{d \ln Y / d \zeta}{d \ln C / d \zeta} = \frac{\sum_{j \in \text{margins of } Y} \frac{\phi(z_j)}{Y} \frac{(1-\nu)(1-\gamma)}{\rho \sigma_\varepsilon} \frac{\phi a^* \chi}{D_j}}{\frac{1}{C} \left[\underbrace{\int_{\text{borrowers}} \frac{\chi_{s'} \phi_{s'} \tilde{k}_{s'}(\hat{x})}{(1-\gamma) R_{s'}} d\mu}_{\text{intensive: loan deepening}} - \underbrace{\sum_j \int w_M s_M \Delta a_j^* d\mu}_{\text{extensive: Created/swapped loans}} \right]}, \quad (44)$$

where w_M and s_M are as defined in Appendix L. Three lessons that carry to the quantitative model. First, the *numerator* is a density-weighted sum of threshold shifts: the IV coefficient is large where many firms sit near their thresholds, young, small, constrained firms, which rationalizes both the magnitude ordering of the empirical elasticities (entry $\approx$ exit $>$ stocks) and their similarity across sectors. Second, the *denominator* mixes intensive responses (continuing borrowers deepen loans by $\chi_s \phi_s \tilde{k}_s / [(1-\gamma) R_s]$ each) with extensive ones (loans created and swapped at the margins), and both are built from parameters estimated off cross-sectional moments, spread levels, participation costs, collateral caps. Matching the observed first stage jointly with the loan-rate pass-through therefore tests that financing block: the pass-through pins the price movement $\chi_f \phi_f$, and the quantity response then has no free parameter left. Third, because ζ enters numerator and denominator symmetrically, β_Y^{model} is invariant to the scale of the shock for small ζ : the model counterpart can be computed from any small perturbation, and only the *incidence* (χ_i, χ_f, κ_f), not the size, of the credit-supply shock needs to be estimated ($\kappa_i = 0$ in the baseline: bank credit-supply shocks do not shift informal collateral technology); Section 9 pins it from three responses and reserves the rest for validation.

8 The quantitative model

This section embeds the mechanisms of Section 7 in a stationary firm-dynamics model in the tradition of ?, with two operating regimes as in ? and costly external finance as in ?. Relative to the two-period model, it adds (i) an infinite horizon with persistent productivity, (ii) a size-dependent cost of informality on the labor margin, (iii) sector-specific productivity dynamics, and (iv) a fixed cost of borrowing. These additions matter quantitatively — they discipline the size distribution, the age profiles, and the relative churn of the two sectors — but every comparative static that drives the counterfactual is inherited from the closed forms of Section 7.

8.1 Environment

Demographics. Time is discrete and infinite, $t = 0, 1, 2, \dots$; a period is a year. The economy of a local labor market contains a unit continuum of production sites. At each date a measure m_0 of *potential entrants* is born. A potential entrant draws a permanent productivity type v , with $\ln v \sim N(0, \sigma_v^2)$, and initial wealth k_0 , with $\ln k_0 \sim N(\mu_{k_0}, \sigma_{k_0}^2)$, independent of v . Incumbent firms are indexed by their sector $s \in \{i, f\}$, their permanent type v , their lagged productivity θ_{-1} , and their installed capital k . An incumbent operating in sector s dies exogenously at the beginning of each period with probability $\lambda_s \in [0, 1)$, in which case the owner recovers the undepreciated capital $(1 - \delta)k$.

Preferences and prices. Owners are risk neutral, consume dividends, and discount at $\beta \in (0, 1)$. The output good is the numéraire. The wage w is common to both sectors and exogenous to the firm; in the baseline quantitative exercise it is normalized to $w = 1$ and held fixed, consistent with the within-market, within-period identification of the empirical design (Section ??) and with local wage stickiness over the outcome window; general-equilibrium adjustment of w at the local-labor-market level is discussed in Section ??.

Technology and productivity. Production is $y = \theta k^\alpha \ell^\nu$ with $\alpha + \nu < 1$, where ℓ is *total* labor input, the sum of the owner’s inelastically supplied unit and hired workers: $\ell = 1 + \ell^h$, $\ell^h \geq 0$. Productivity follows the sector-specific process

$$\ln \theta_t = \ln v + z_t + u_t, \quad z_t = \rho_s z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_{\varepsilon s}^2), \quad u_t \sim N\left(-\frac{1}{2}\sigma_{u,s}^2, \sigma_{u,s}^2\right) \text{ i.i.d.}, \quad (45)$$

where s is the sector in which the firm operates at t ; an entrant’s first draws are $z_1 = \varepsilon_1$ (the special case $z_0 = 0$, i.e. θ centered on v) and a fresh u_1 . Productivity thus has three layers with distinct jobs. The permanent component v generates selection at entry and the long tails of the age–size profiles. The persistent component z carries the slow mean-reversion the census autocovariances identify. The transitory-real component u — one-off contracts, demand swings, disruptions — is observed by the firm *before* its exit, switching, input, and borrowing decisions, and is what generates realistic annual churn; it is mean-preserving ($\mathbb{E}e^u = 1$) and, being i.i.d., adds no state variable beyond the composite current shock. Measured log-TFP adds classical measurement error on top of (45); the error affects the data, never decisions, and Section 9 shows how the census autocovariances separate the three layers and bound the error. The sector-specific $(\rho_s, \sigma_{\varepsilon s}, \sigma_{u,s})$ are set there. The dispersion σ_v of the permanent type is common across sectors — v is drawn at birth, before any sector is chosen — and the observed by-sector dispersion of firm-mean productivity is an equilibrium outcome of selection, used as an untargeted check in Section 9.

Formality and the labor margin. A *formal* firm pays the revenue tax τ_y and hires all its workers on registered contracts at the payroll-inclusive wage, so its labor cost is

$$\mathcal{C}_f(\ell^h) = (1 + \tau_\ell) w \ell^h. \quad (46)$$

An *informal* firm evades τ_y and τ_ℓ entirely but its whole payroll is exposed to audit: hiring ℓ^h costs

$$\mathcal{C}_i(\ell^h) = w \ell^h \left(1 + \frac{\ell^h}{\eta_i}\right), \quad \eta_i > 0, \quad (47)$$

an expected-penalty schedule increasing in size in the spirit of ? and ?: informal firms find it progressively harder to remain undetected as they grow. The labor saving of informality is therefore τ_ℓ per worker at small scale, eroded at rate ℓ^h/η_i as the firm grows. The baseline abstracts from formal firms' off-the-books hiring — the *intensive* margin of informality in ? — so that formality is a firm-level status aligned exactly with the census classification used throughout the empirical work; the underreporting of registered firms' payrolls is an interesting margin in its own right, but it is not the one the credit-supply variation moves. Static profits are

$$\pi_s(\theta, k) = \max_{\ell^h \geq 0} (1 - \tau_y^s) \theta k^\alpha (1 + \ell^h)^\nu - \mathcal{C}_s(\ell^h), \quad \tau_y^f = \tau_y, \tau_y^i = 0. \quad (48)$$

The objective is strictly concave in ℓ^h , so labor demand is unique; the corner $\ell^h = 0$ is the self-employment margin that accounts for the mass of owner-only firms in the Census. As $\eta_i \rightarrow \infty$ and $\ell^h \gg 1$, (48) converges to the constant-wedge profit function of Section 7; conversely, for finite η_i informal profits are eventually *less* elastic to θ than formal profits, which restores formality as the optimal regime for large firms even absent financial motives (Remark 7.8).

Financial frictions. Investment must be financed one period ahead of the revenue it generates. A firm entering period $t + 1$ in sector s' with target capital k' and internal funds n (defined below) raises external funds $a = \max\{0, k' - n\}$ at total cost

$$\Phi_{s'}(a) = \phi_{0s'} \mathbf{1}\{a > 0\} + \phi_{s'} a, \quad (49)$$

subject to the collateral constraint

$$a \leq b_{s'} k'. \quad (50)$$

The two cost parameters have distinct empirical counterparts: the *level* ϕ_s is the spread of marginal loan pricing over the risk-free rate, and the *fixed component* ϕ_{0s} captures application, documentation, and relationship costs that keep incidence $\mathbb{P}(a > 0)$ far below one even when spreads are moderate (Remark 7.4). A convex term $\frac{1}{2}\phi_{1s}a^2$ is a straightforward extension; it is omitted from the baseline because the exactly identified design of Section 9 assigns each financial parameter one dedicated moment, and no moment in the extraction

isolates loan-size-dependent pricing. The sector ranking $(\phi_{0i}, \phi_i) \geq (\phi_{0f}, \phi_f)$ and $b_i \leq b_f$ is not imposed but estimated; Section 9 maps each parameter to its identifying moment. A bank credit-supply shock is a shift of (ϕ_s, b_s) ; the empirical incidence across sectors is estimated, not assumed (Corollary 7.12).

Entry, switching, exit. Entry into sector s costs c_s^e ; switching costs are c_{if} ($i \rightarrow f$) and c_{fi} ($f \rightarrow i$); exit is free and yields the firm's internal funds (including the resale value of undepreciated capital). All costs are denominated in units of output.

8.2 Timing and recursive formulation

The within-period timing replicates Figure 11: (1) death shocks; (2) θ_t is drawn and observed; (3) production, taxes, fixed cost; (4) the *transition stage*: the firm chooses next period's status $d \in \{\text{exit}, i, f\}$ and, if continuing, (k', a) ; entrants make their entry decision; (5) dividends are consumed. Sector status for period $t + 1$ is therefore chosen under the information set $\{v, \theta_t, k\}$, before ε_{t+1} is observed — the timing assumption that makes lagged productivity the relevant state for transitions, as in the policy maps of Figure ??.

Let $W_s(v, \theta_{-1}, k)$ denote the value of beginning a period committed to sector s with capital k :

$$W_s(v, \theta_{-1}, k) = \lambda_s (1 - \delta)k + (1 - \lambda_s) \mathbb{E}_{\theta|\theta_{-1}, v}^s \left[V_s(v, \theta, k) \right], \quad (51)$$

where $\mathbb{E}^s$ integrates (45) with the sector- s parameters. The post-production value solves

$$V_s(v, \theta, k) = \max \left\{ \underbrace{n_s^{\text{exit}}}_{\text{exit}}, \max_{s' \in \{i, f\}} \underbrace{\max_{k' \geq 0} n_{ss'} - k' - \Phi_{s'}(a) + \beta W_{s'}(v, \theta, k')}_{\text{continue in } s'} \right\}, \quad (52)$$

subject to $a = \max\{0, k' - n_{ss'}\}$ and $a \leq b_{s'}k'$, with internal funds

$$n_s^{\text{exit}} = \pi_s(\theta, k) - c_s + (1 - \delta)k, \quad n_{ss'} = n_s^{\text{exit}} - c_{ss'} \mathbf{1}\{s' \neq s\}. \quad (53)$$

A potential entrant with (v, k_0) solves

$$V^e(v, k_0) = \max \left\{ k_0, \max_{s \in \{i, f\}} \max_{k' \geq 0} (k_0 - c_s^e) - k' - \Phi_s(a) + \beta W_s(v, v, k') \right\}, \quad (54)$$

subject to $a = \max\{0, k' - (k_0 - c_s^e)\}$ and $a \leq b_s k'$; the second state argument of W_s is v because an entrant's first-period productivity is drawn from (45) at $\theta_{-1} = v$. Standard arguments (bounded per-period payoffs after truncating the capital grid, monotone mixing of (45), and Blackwell's conditions for the operator defined by (51)–(52)) deliver existence and uniqueness of (W_s, V_s) and measurable policies

$$d_s(v, \theta, k) \in \{\text{exit}, i, f\}, \quad k'_s(v, \theta, k), \quad a_s(v, \theta, k), \quad \ell_s^h(\theta, k). \quad (55)$$

Decision rules and thresholds. Because $W_{s'}$ is increasing in θ (monotonicity of (45) in the FOSD order) and the exit value is flat in future productivity, the policies inherit the threshold structure of Section 7 conditional on (v, k) :

$$d_i(v, \theta, k) = \begin{cases} \text{exit,} & \theta < \theta_i^X(v, k) \\ i, & \theta_i^X(v, k) \leq \theta < \theta_{if}(v, k) \\ f, & \theta \geq \theta_{if}(v, k), \end{cases} \quad d_f(v, \theta, k) = \begin{cases} \text{exit,} & \theta < \theta_f^X(v, k) \\ i, & \theta_f^X(v, k) \leq \theta < \theta_{fi}(v, k) \\ f, & \theta \geq \theta_{fi}(v, k), \end{cases} \quad (56)$$

and entrants stay out below $\theta_i^E(v, k_0)$ (expressed through v), enter informally on an intermediate interval, and enter formally at the top. Two forces shape the boundaries in (k, θ) space: the *finance* force of Section 7, which makes formality attractive for firms whose growth needs are large relative to internal funds, and the *audit* force in (47), which makes formality attractive at large scale regardless of finance. Both are estimated; the policy maps in Figure ?? display their combined effect at the estimated parameters. In the exactly identified estimation of Section 9, the sign pattern of the comparative statics of these thresholds — Proposition 7.10, exact in the two-period model, near-diagonal here — is the object verified by the Jacobian exhibit.

8.3 Stationary equilibrium

Let $\mathcal{S} = \{i, f\} \times \mathbb{R}_+^3$ collect (s, v, θ, k) . The policies induce a transition kernel Q on $\mathcal{S}$: incumbents die with probability λ_s , draw θ' from (45), produce, and transit according to d_s ; entrants add mass $m_0 \times (\text{entry policy})$ at states (s', v, θ_1, k') . A *stationary equilibrium* given w is a measure μ on $\mathcal{S}$ and value/policy functions such that:

1. (W_s, V_s, V^e) solve (51)–(54) and the policies are optimal;
2. μ is invariant: $\mu = Q(\mu) + m_0 \mathcal{E}$, where $\mathcal{E}$ is the entrant inflow measure implied by V^e ;
3. aggregate flows balance: the mass of entrants equals the mass of exits (endogenous plus exogenous) sector by sector net of switching flows.

Aggregates are integrals against μ : the formal share $N_f/(N_i + N_f)$ with $N_s = \mu(\{s\} \times \mathbb{R}_+^3)$, aggregate credit $C = \int a \, d\mu$, employment $L = \int \ell \, d\mu$, value added $Y = \int \theta k^\alpha \ell^\nu \, d\mu$, and the flow masses of entry E_s , exit X_s , and switching S_{if}, S_{fi} used in estimation. The flow identity

$$N_{s,t+1} = \underbrace{(1 - \lambda_s) \text{Stay}_{s,t}}_{\text{survivors that stay}} + \underbrace{(1 - \lambda_{s'}) S_{s',t}}_{\text{switchers in}} + \underbrace{E_{s,t}}_{\text{entrants}}, \quad (57)$$

is the model counterpart of the margin decomposition (existing/entering/exiting $\times$ formal/informal and the four transition cells) estimated in Section ???. Two implications of

stationarity matter for the estimation design. First, the entrant share of the firm stock is pinned by the exit flows through the balance condition, so it is a *consistency check* rather than a free moment. Second, the stationary distribution $\mu(\theta)$ is the population over which every targeted moment is an integral — the object updated at each sweep of the estimation loop in Section 9.

8.4 The model counterpart of the shift-share experiment

The empirical design identifies the semi-elasticity of firm-margin outcomes to *equilibrium credit* induced by bank credit-supply shocks within a local labor market, using leave-one-out exposure across markets. Its model counterpart replicates the design, not just the estimand: the model is simulated as a collection of markets with heterogeneous exposure, and the empirical estimator is run on the simulated data.

Step 1: markets and the shock. The economy consists of M ex-ante identical local labor markets, each in the stationary equilibrium of the estimated model, sharing the technology and cost parameters and differing only in their exposure ω_m to a common credit-supply shifter g (the empirical counterpart is the leave-one-out bank-exposure design of Section ??; ω_m is calibrated to the cross-market dispersion of the observed exposure instrument). At $t = 0$ an unanticipated, permanent credit-supply shock of intensity $\zeta_m = \omega_m g$ shifts market m 's financing schedules to

$$\phi_s(\zeta_m) = (1 - \chi_s \zeta_m) \phi_s, \quad b_s(\zeta_m) = (1 + \kappa_s \zeta_m) b_s, \quad (58)$$

where $(\chi_i, \chi_f, \kappa_f)$ is the *incidence vector*: how much of a marginal peso of bank funding reaches each sector's price and quantity margins. In the baseline $\kappa_i = 0$ — bank credit-supply shocks do not shift the collateral technology available to informal firms — and this restriction is relaxed in robustness. The wage is held at w in every market, mirroring the within-market identification (bank shocks are small relative to the local labor market, and local wages are sticky over the five-year outcome window).

Step 2: the response. For each market, solve the new policies and iterate the distribution forward five years from μ_0 — the horizon of the empirical outcome window — obtaining $\{\mu_t^m\}_{t=1}^5$ and the implied paths of every outcome Y_t^m (firm stocks by sector, entry, exit, the four transition cells, employment by sector, wagebill, value added, average value added, credit C_t^m , borrowing incidence, the credit-weighted average marginal loan rate $\bar{r}_t^m = r_f + \mathbb{E}_{\mu_t}^{a\text{-weighted}}[\phi_s]$, and collateral coverage). Measurement noise calibrated to the empirical first-stage residual is added at the market level.

Step 3: the estimator, not just the estimand. The *actual* leave-one-out shift-share estimator of Section ?? — same exposure construction, same five-year differences, same inference — is then run on the simulated market panel, producing coefficients $\hat{\beta}_Y^{\text{model}}$ directly comparable to the empirical $\hat{\beta}_Y$. For small g each coefficient approaches the population estimand

$$\beta_Y^{\text{model}} = \frac{\Delta_5 \ln Y}{\Delta_5 \ln C} = \frac{\ln Y_5(\zeta) - \ln Y_5(0)}{\ln C_5(\zeta) - \ln C_5(0)}, \quad (59)$$

which is invariant to the scale of the shock (numerator and denominator are both proportional to ζ to first order, as in (44)); in practice g is set so that the model first stage matches the observed one-standard-deviation bank-shock first stage, and invariance is verified. Three properties of this construction organize its use in Section 9: (i) each coefficient is a density-weighted aggregate of the threshold responses of Section 7, concentrated on constrained, marginal firms; (ii) three responses — the loan-rate pass-through, the collateral-use response, and the formal-to-informal switching response — pin the incidence vector one-for-one (Corollary 7.12); (iii) every remaining coefficient (firm stocks, entry, exit, informal-to-formal switching, employment, average value added) is produced by the model with *zero* remaining free parameters, and the comparison with its empirical counterpart is the paper’s validation exercise.

8.5 Discussion of assumptions

Partial equilibrium in w . Identification in Section ?? is within market $\times$ year, purged of aggregate shocks; the estimated elasticities are relative responses to relative credit shifts. Holding w fixed in the model experiment is the exact analogue, and is additionally supported by local wage stickiness at the five-year horizon of the outcome window. The counterfactuals that change *aggregate* credit conditions (Section ??) additionally report a labor-market-clearing version in which w adjusts to keep local labor supply unchanged; the composition results survive because they are driven by selection at entry, not by scale.

One-period external finance. Debt is within-period: firms borrow to install capital, repay principal plus the fee out of the following period’s resources, and carry no debt stock across transition-stage decisions. This keeps net worth out of the state vector — the firm’s internal funds are a *function* of (θ, k) — at the cost of abstracting from long-term leverage dynamics. The measurement counterpart is that model borrowing a maps to gross new credit within a year (flows), and the leverage cap b_s to the ratio of credit lines to capital among borrowers; Section 9 constructs both analogues explicitly. Extending the state with net worth is conceptually straightforward and quantitatively secondary for the margins studied here, because the empirical variation exploited is five-year and cross-market rather than high-frequency.

No strategic default. Loans are safe (the collateral cap guarantees repayment), so ϕ_s is an intermediation cost rather than a default premium. The estimated informal–formal spread gap then bounds the total of screening and enforcement costs that formality saves; interpreting part of it as a default premium would not change the firm’s problem, only the welfare accounting of Section ??.

Exogenous financial parameters. The triple (ϕ_{0s}, ϕ_s, b_s) is measured, not derived. A large literature derives such objects from primitives: limited enforcement and dynamic lending contracts generate endogenous borrowing limits (Albuquerque and Hopenhayn, 2004; Clementi and Hopenhayn, 2006), default pricing and costly intermediation generate endogenous spreads (Cooley and Quadrini, 2001; Arellano et al., 2012), and D’Erasmus and Boedo (2012) embed endogenous credit in a formal–informal economy. Each of these theories delivers exactly the two objects assumed here — a quantity limit and a price wedge — and predicts their sector ranking: informal assets are harder to seize (lower b_i) and informal borrowers are more opaque to lenders (higher ϕ_i). Treating the triple as measured primitives, as in the quantitative misallocation literature (e.g., Midrigan and Xu, 2014; Moll, 2014), is what lets each parameter be read off its own data moment (Section 9) and keeps the mapping from the shift-share evidence to the model transparent. The cost is a caveat on the counterfactuals: under a large formalization, ϕ_i and b_i would themselves improve as information and enforceability improve, so experiments holding them fixed likely *understate* the financial channel — Brooks and DAVIS (2020) show that endogenizing borrowing constraints changes counterfactual magnitudes in exactly this direction. We sign this bias but do not quantify it.

Sector-specific processes. Allowing $(\rho_s, \sigma_{\varepsilon s})$ to differ by sector is essential: the Census autocovariances imply materially higher persistence for formal firms and higher innovation variance for informal firms (Section 9), and imposing a common process would force the switching costs to absorb the difference, contaminating the counterfactual.

Scope: the intensive margin of informality. Formality here is a firm-level status, matching the census classification that underlies every empirical moment. The model is therefore silent on registered firms that underreport part of their payroll; that margin responds to enforcement and labor-cost policy rather than to the credit-supply variation exploited here, and its omission buys a one-to-one mapping between the model’s sector indicator and the data’s.

9 Estimation

The model is estimated *exactly identified*: every parameter is assigned one named moment, every moment enters through one equation, and there is no joint objective function and no weighting matrix anywhere in the procedure. The parameters are organized in four recursive blocks. Block 0 (fourteen parameters) is preset by direct, closed-form estimation on data outside the model. Block 1 (five parameters) is read one-for-one off financial and factor-share moments, again with no model solve. Block 2 (ten parameters) is the core: each parameter solves the equation “model moment = data moment” for *its* moment, with the ten equations made mutually consistent by a damped fixed-point loop over the model’s stationary distribution. Block 3 (three parameters) pins the incidence of the empirical credit-supply shock from three quasi-experimental responses, conditional on all earlier blocks. Identification arguments for every assignment come from the closed-form comparative statics of Section 7; their validity in the general model is verified ex post by a Jacobian exhibit. Validation then uses *external variation* rather than held-out cross-sectional moments: the model is simulated as a collection of local markets, the actual leave-one-out shift-share estimator of Section ?? is run on the simulated data, and the resulting coefficients are compared with their empirical counterparts for the six responses that no parameter was allowed to touch. Every moment in the extraction has exactly one job — preset input, assigned target, variance input, robustness alternate, or validation outcome — so all available information is used, none of it twice.

9.1 Measurement: mapping model objects to the data

Because the three data sources live on separated secure servers, every moment is defined once, in the microdata language of its source, and the model reproduces the *measurement process* rather than an idealized object. Table 1 collects the conventions; four deserve emphasis.

First, *census timing*. The EC is quinquennial, so exit and switching are observed as five-year outcomes contaminated by death and by round-tripping. Rather than annualizing with a constant-hazard formula (reported only as a diagnostic), the estimation computes `exit5`, `switch5` and the entrant shares from the model’s five-year transition kernel with the same definitions that the Stata scripts apply to the EC. Second, *who counts as labor*. EC employment includes owners and unpaid family workers; the model’s owner unit and self-employment corner ($\ell^h = 0$) exist precisely so that the model size distribution is comparable to the census one. Third, *financing is measured as flows first*. ENAFIN reports credit raised during the reference year and the approved line of the main credit; the model’s within-period external funds a and cap $b_s k'$ are their exact analogues, so no debt-stock imputation is needed; the stock-based (P24) leverage variant brackets the flow-based reads.

Fourth, *units*. All monetary moments are converted to multiples of the EC median annual wage per paid worker, the model’s numéraire; loan-pricing moments are spreads over the 28-day CETES rate, matching the model convention that $\Phi'_s(a)$ prices external funds *over* the risk-free rate.

9.2 Block 0: presets (fourteen parameters, no model solve)

Table 2 lists the preset parameters with their sources; three groups require discussion.

Production function. The baseline (α, ν) come from cost shares in the EC 2024: ν is the labor share of value added with unpaid labor imputed at local median wages (medians: $\nu = 0.59$), and $\alpha = (1 - \nu) \times (\text{gross-operating-surplus share})$, yielding $\alpha = 0.17$ and decreasing returns $\alpha + \nu \approx 0.76$. The within-industry OLS panel estimate of the capital coefficient (0.14) corroborates α ; the OLS labor coefficient (1.26) exceeds one — the classic transmission bias when inputs respond to productivity — and is reported only to justify preferring cost shares. Because the profit curvature $\gamma = \alpha/(1 - \nu)$ governs how strongly financial conditions move thresholds (Section 7), the paper reports the counterfactuals also under the literature values $(\alpha, \nu) = (0.50, 0.40)$ used in earlier drafts.

Productivity process. The process (45) implies, for the cross-section of firms observed in waves $h \in \{0, 5, 10\}$ years apart, autocovariances of demeaned log-TFP $a_h = \text{Cov}(\ln \tilde{\theta}_t, \ln \tilde{\theta}_{t-h})$ of the form

$$a_h = \sigma_v^2 + \rho_s^h S_s, \quad S_s \equiv \frac{\sigma_{\varepsilon s}^2}{1 - \rho_s^2}, \quad (60)$$

with additive i.i.d. measurement error in value added contributing a further σ_{me}^2 at $h = 0$ only: $a_0 = \sigma_v^2 + S_s + \sigma_{\text{me},s}^2$. This is why the system must not lean on a_0 . The seemingly natural identification $\rho_s^5 = (a_5 - a_{10})/(a_0 - a_5)$ differences the permanent component out of the numerator but leaves σ_{me}^2 in the denominator, biasing persistence *down* and the innovation variance *up* — manufacturing spurious annual churn. The baseline therefore identifies the process from the ME-free covariances alone, with σ_v preset:

$$\rho_s^5 = \frac{a_{10} - \sigma_v^2}{a_5 - \sigma_v^2}, \quad S_s = \frac{a_5 - \sigma_v^2}{\rho_s^5}, \quad \sigma_{\varepsilon s}^2 = S_s (1 - \rho_s^2), \quad (61)$$

giving $\rho_i = 0.953$, $\rho_f = 0.952$, $\sigma_{\varepsilon i} = 0.136$, $\sigma_{\varepsilon f} = 0.209$, and — reported transparently as the addendum extraction intends — an $h=0$ excess $a_0 - \sigma_v^2 - S_s$ of 1.38 (informal) and 0.56 (formal), 73% and 42% of a_0 . This excess is a *mixture* of two observationally identical objects that the model treats very differently: classical measurement error, which lives in the data and never in decisions, and the transitory-real shock u of eq. (45) — genuine one-year variation the firm observes and acts on. Autocovariances cannot split the mixture (both components load only on a_0), but the economy can: only the real part generates

exit, switching, and borrowing flows. The baseline assigns 50% of the excess variance to the real component, $\sigma_{u,s}^2 = \frac{1}{2}(a_0 - \sigma_v^2 - S_s)$, i.e. $\sigma_{u,i} = 0.83$ and $\sigma_{u,f} = 0.53$, with real shares of 25% and 75% re-run as robustness. The logic of the split is familiar from earnings dynamics: [Blundell et al. \(2008\)](#) observe that transitory shocks and measurement error are not separately identified from income autocovariances alone, and separate them with consumption responses — that is, with behavior. Firm behavior plays that role here: only the real component moves exit, switching, and borrowing, so the churn margins targeted by the core equations discipline the share, and the estimated fees absorb what the chosen value over- or under-delivers. Three disclosures. First, the system conditions on the preset $\sigma_v = 0.56$; over $\sigma_v \in [0.50, 0.60]$ the implied annual persistence stays inside $[0.93, 0.97]$, and the naive a_0 -based system ($\rho = 0.41, 0.60$; $\sigma_\varepsilon = 1.11, 0.72$) is re-run as robustness. Second, survivor selection in the balanced three-wave panel raises measured persistence; it is disclosed and bounded by re-estimating on the 2019–2024 two-wave panel. Third, because the permanent type is drawn at birth, before any sector is chosen, σ_v is a single structural parameter: the by-sector dispersions of firm-mean productivity are equilibrium outcomes of selection, and their fit — like the age–size profiles that are informative about permanent heterogeneity net of noise — is an untargeted check. The corrected estimates are economically meaningful: the two sectors share essentially the same high annual persistence, and the informal disadvantage shows up not in churn but in the hiring wedge and financing terms estimated elsewhere.

Exogenous death. In the model, endogenous exit is concentrated among small, low- $\hat{x}$ firms (the exit thresholds of Section 7.5), while exogenous death is size-flat. The five-year exit rate by within-sector employment tercile falls from 39% to 29% (informal) and from 32% to 25% (formal); the top-tercile annualized rates, 0.067 and 0.056, are assigned to λ_i and λ_f , and the *gradient* across terciles is left as a non-targeted validation of the split between λ_s and endogenous exit. The ranking $\lambda_i > \lambda_f$ feeds T_λ in Assumption 1: part of the value of formality is survival.

Taxes. Both effective rates are computed from 2023 administrative aggregates. *Output:* ISR paid by personas morales on own profits (1,146.7 bn MXN; SAT, *Informe Tributario y de Gestión* 4T-2023, cuadro 6) over formal value added — $(1 - 0.248) \times \text{GDP}$ of 31,855.6 bn (INEGI, *Medición de la Economía Informal* 2023; *PIB por Entidad Federativa* 2023) — gives $\tau_y = 0.048$ (0.050 including personas físicas con actividad empresarial). The statutory 30% (Art. 9, LISR) applies to taxable *profit*, a thin slice of value added, so even full compliance maps into single digits per peso of the model’s base; counterfactuals are re-run at the statutory rate as an upper bracket. *Labor:* contributions — IMSS cuotas obrero-patronales, 667.6 bn (IMSS, *Informe al Ejecutivo Federal* 2024–2025, cuadro II.1), Infonavit at 5% of the contribution base (≈ 211 bn), and state payroll taxes at a mean rate

of 2.9% (≈ 122 bn) — total ≈ 1.00 tn, 3.1% of GDP. Divided by the wage bill formal firms actually pay — the IMSS contribution base (4.22 tn) grossed up for under-declaration (Kumler, Verhoogen and Frías 2020) plus the wages of their unregistered employees — this gives $\tau_\ell \approx 0.16$ (range 0.14–0.18). The effective-average construction is deliberate: the model abstracts from the worker-registration margin, so the preset must embody the average burden per peso of wages formal firms pay, not the schedule a fully compliant firm would face. That compliance-literal rate — the same collections over the declared base alone — is 0.24, essentially the statutory schedule (IMSS $\approx 17\%$ at the average wage including the worker share, Infonavit 5%, ISN $\approx 3\%$), and serves as the robustness bracket; to the extent workers value IMSS/Infonavit benefits (Levy 2008), the distortionary wedge lies below even the effective payment rate. Taxes enter the formality margin only through T_λ , so the bracketing is transparent.

9.3 Block 1: direct reads (five parameters, no model solve)

Five parameters are inverted one-for-one from single moments through static model equations, before any dynamic solve.

Spreads ϕ_f, ϕ_i . With linear financing costs, a borrowing firm’s marginal cost of external funds is $r_f + \phi_s$, so the observed lending spread *is* the parameter:

$$\phi_f = \frac{\text{median formal loan rate} - \text{CETES 28}}{100} \approx \frac{14.3 - 11.05}{100} = 0.033, \quad \phi_i \approx \frac{22.1 - 11.05}{100} = 0.11, \quad (62)$$

the formal rate from the ENAFIN 2023 pricing block (corroborated by the R04C average on loans to formal firms) and the informal rate from the R04C matched registry — the only source that prices credit to informal firms directly. The risk-free benchmark and the nominal-versus-real treatment live in a public input file outside the labs, so both can be revised without a new desk round.

Collateral capacities b_f, b_i . In the model the leverage distribution among borrowers has an *atom at the cap*: every R4 firm operates at $a/k' = b_s$ exactly (Lemma 7.2). A high quantile of observed leverage therefore estimates the cap directly; the p90 (rather than the maximum) absorbs measurement error in the imputed capital denominator. b_f is the p90 of credit-line-to-capital among ENAFIN formal borrowers, with the flow- and stock-based (P24) variants as brackets and the collateral-requirement haircut as a cross-check. b_i is the p90 of leverage in the ENAFIN *proxy-informal* cell (micro $\times$ persona física): since the ENAFIN frame is formal, this cell — the most informal-like segment it contains — is read as an upper bound for b_i , a caveat the robustness analysis takes seriously (the denied-versus-discouraged sectoral gap from the EC addendum, the direct evidence that $b_i < b_f$, is the pre-registered alternate read).

Informal audit scale η_i . The static labor block inverts in closed form. The informal first-order condition from (47) is $\nu y/(1 + \ell^h) = w(1 + 2\ell^h/\eta_i)$, so the measured informal labor share $ls_i \equiv w\ell^h/y$ satisfies

$$ls_i = \frac{\nu \ell^h/(1 + \ell^h)}{1 + 2\ell^h/\eta_i} \implies \eta_i = \frac{2\bar{\ell}^h}{\frac{\nu \bar{\ell}^h}{(1 + \bar{\ell}^h) ls_i} - 1}, \quad (63)$$

evaluated at the mean informal hired-labor input $\bar{\ell}^h$ (a public census tabulation). The identification is the *informal-formal labor-share gap*: absent the audit cost the informal share would equal its frictionless counterpart $\nu\ell^h/(1 + \ell^h)$, while the formal share is the same object scaled by $1/(1 + \tau_\ell)$; the shortfall of the informal share below its frictionless level is what (63) converts into η_i (the replication code applies the owner-imputation and payroll-tax adjustments consistently with the cost-share construction of ν).

9.4 Block 2: the exactly identified core (ten parameters)

Table 3 states the ten assignments and the identification argument behind each — the toy-model result that makes the assigned moment strictly monotone in its parameter, given the earlier blocks. The ordering is deliberate: financial *prices* are set in Block 1 before any *quantity* moment is asked to move a real-side parameter, so the constants $C_o, D_o, \mathcal{K}_s$ that appear in every threshold formula are already pinned when the core is solved.

Three design notes. First, the *entrant-stock share is not a target*: in stationary equilibrium the entry mass is pinned by the exit flows through the balance condition of Section 8.3, so that moment is mechanically tied to the exit rates already assigned to c_s ; it is reported as a stationarity *check*. What c_i^e controls observably is *who* enters, not how many — hence the selection moment (entrant relative size), which no other parameter targets: the age-size *slopes* inform (ρ_s, σ_v) , already preset, while the *intercept* relative to mature firms isolates the entry selection threshold. Second, the wealth pair $(\mu_{k_0}, \sigma_{k_0})$ uses informal entrants: the formal entrant capital distribution in the EC export has p90 below its mean (an outlier tail), and informal entrants are both the numerous case and the one where the collateral map is cleanest. Third, each row’s monotonicity is a toy-model theorem; its survival in the general model is exactly what the Jacobian exhibit of Section 9.7 verifies, sign by sign, at the estimated parameters.

9.5 Solving the system: a fixed point in the stationary distribution

Let θ_2 collect the ten core parameters and $m_j(\theta_2)$ the model moments, computed *exactly* rather than by simulation: given the value functions, the policies induce a transition kernel on the state grid; its invariant measure $\mu(\theta_2)$, the entrant inflow measure, and the fifth power of the kernel deliver every targeted moment (five-year exit, switching, entrant

composition, borrowing incidence) as an integral, with no simulation noise — so each one-dimensional inversion below is a smooth, deterministic root-finding problem. The algorithm is:

1. *Step 0 (initialization and transparency)*. Invert each of the ten equations using the *toy* closed forms of Section 7 evaluated on the observed micro distribution (sizes, ages, sectors), with Blocks 0–1 plugged in. This pass requires no value-function solve, is fully sequential, and yields $\theta_2^{(0)}$ — reported in the appendix as the “plug-in” column.
2. *Sweeps*. At iteration ι : solve the value functions at $\theta_2^{(\iota)}$; form $\mu(\theta_2^{(\iota)})$; for each parameter j in the order of Table 3, solve the scalar equation $m_j = \hat{m}_j$ for parameter j holding the others at their current values (bisection — valid because each equation is monotone by the cited results); damp, $\theta_2^{(\iota+1)} = (1 - \omega)\theta_2^{(\iota)} + \omega \theta_2^{\text{new}}$.
3. *Stop* when $\max_j |m_j - \hat{m}_j|/|\hat{m}_j|$ is below tolerance. At the fixed point $\hat{\theta}_2$ the model reproduces all ten targeted moments *exactly*.

Two properties make the procedure self-certifying. The loop is a local contraction precisely when the moment–parameter Jacobian is diagonally dominant — the formal statement of the claim that each moment identifies its own parameter — so fast convergence is itself evidence for the one-to-one design, and failure to converge would falsify it (the sweep path is reported). And the distance between $\theta_2^{(0)}$ and $\hat{\theta}_2$ decomposes, equation by equation, into the toy-approximation and latent-state components of the plug-in pass: small distances certify that no parameter is being identified by joint leverage across moments. The toy model contributes no number to $\hat{\theta}_2$ — it contributes the monotonicity proofs, the identification narrative, and the starting point, which washes out.

9.6 Block 3: the incidence of the credit-supply shock

The empirical shock’s incidence vector $(\chi_f, \kappa_f, \chi_i)$ in (58) is estimated last, conditional on $\hat{\theta}_2$, again one moment per parameter, using the three *price-side and switch-down* responses of the empirical design:

$$\chi_f \longleftarrow \underbrace{+0.46\text{pp}}_{\text{loan-rate pass-through}}, \quad \kappa_f \longleftarrow \underbrace{-0.733}_{\text{collateral-use response}}, \quad \chi_i \longleftarrow \underbrace{+0.113}_{\beta^{f \rightarrow i}}. \quad (64)$$

The logic is Corollary 7.12. The pass-through of the bank shock into average loan pricing scales directly with χ_f (the credit-weighted rate is $r_f + \phi_s$ under linear costs); the collateral-use response scales with κ_f (marginal borrowers admitted by a relaxed cap post less collateral per peso); and the formal-to-informal switching response is the margin that can rise under a credit *expansion* only through informal-side incidence — a positive $\beta^{f \rightarrow i}$

is qualitatively impossible with $\chi_i = 0$, and its magnitude, given (χ_f, κ_f) , reads χ_i one-for-one. The baseline sets $\kappa_i = 0$ (bank credit-supply shocks do not shift the collateral technology of informal lending); robustness ties κ_i to κ_f through the informal collateralized share. As an internal-consistency exercise, the estimation is re-run with χ_i read instead from the average value-added response (-0.073 , through the composition channel of Proposition 7.13); agreement between the two $\hat{\chi}_i$'s is overidentification evidence obtained without any joint estimation, and the Jacobian exhibit reports the relative loadings ($\beta^{f \rightarrow i}$ is the stronger instrument: it is a flow of marginal firms, first-order in χ_i , while average value added is an inframarginal composition average).

9.7 Inference and identification exhibits

Because the system is exactly identified, inference is standard method-of-moments without weighting choices: stacking all estimated blocks in θ with moment vector $\hat{m}$,

$$\widehat{\text{Var}}(\hat{\theta}) = \hat{J}^{-1} \widehat{\text{Var}}(\hat{m}) \hat{J}^{-1'}, \quad \hat{J} = \left. \frac{\partial m(\theta)}{\partial \theta'} \right|_{\hat{\theta}}, \quad (65)$$

with $\hat{J}$ computed numerically at the solution. $\widehat{\text{Var}}(\hat{m})$ combines the census cell sizes from the addendum extraction (cross-sectional moments), the AKM standard errors of the shift-share coefficients (Block 3 moments), and the delta method through the preset first-stage estimates $(\rho_s, \sigma_{\varepsilon s}, \sigma_v, \alpha, \nu)$. Reporting uses the full $\hat{J}^{-1}$ — sequential estimation does not make cross-moment contamination vanish, and per-equation standard errors would understate. Four exhibits then carry the identification argument:

1. *The Jacobian heat map*: $\hat{J}$ in elasticity units, moments $\times$ parameters. The claim of Table 3 is that it is near-diagonal with the sign pattern of Proposition 7.10; the exact zeros of the two-period model appear as quantitatively small off-diagonal loadings, as the option-value argument of Section 7.5 predicts.
2. *The sensitivity matrix* of ?, $\Lambda = -\hat{J}^{-1}$ in the exactly identified case: how each moment moves each estimate, making the one-to-one design auditable.
3. *Plug-in versus converged*: $\theta_2^{(0)}$ against $\hat{\theta}_2$ (appendix), the transparency decomposition of Section 9.5.
4. *The policy partition* at $\hat{\theta}$: the estimated decision rules in (θ, k) space, verifying that the threshold ordering of Proposition 7.9 — assumed in the toy, checkable here — holds at the estimates (Figure ??).

9.8 Validation: the simulated shift-share design and untargeted fit

Validation uses external variation, not held-out cross-sectional moments. The estimated model is simulated as M markets with the empirical exposure dispersion, the credit-supply shock (58) at the estimated incidence, and first-stage noise calibrated to the empirical residual; the *actual* leave-one-out shift-share estimator of Section ?? is then run on the simulated market panel (Section 8.4). Six coefficients come out of this exercise with *zero* remaining free parameters — no parameter in any block was allowed to touch them — and are compared with their empirical counterparts:

	stocks	entry	exit	$i \rightarrow f$	workers	avg. VA
empirical $\hat{\beta}$	0.176	0.192	0.203	0.195	0.150	−0.073

Each is a sharp test. The firm-count coefficients test the density-weighted aggregation of threshold responses (eq. (44)) at the estimated incidence; the informal-to-formal coefficient tests the asymmetric borrowing-gap prediction of Corollary 7.12(ii) with the incidence pinned from the *other* switching margin; and the average value-added coefficient must come out *negative* untargeted — the composition mechanism of Proposition 7.13 is the only channel available to the model, so its sign and magnitude are the paper’s headline validation. Comparisons are reported with both sampling bands (AKM) and simulation bands.

Untargeted cross-sectional fit. The remaining extraction is reported as an untargeted fit table: the formal share of firms (the headline composition statistic — nothing targets it); the sectoral capital–labor ratios (emerging from constraints and wedges); average profits by sector; the exit-by-size gradients below the preset tail; the age–size profiles beyond the targeted intercept; the by-sector dispersion of firm-mean productivity (selection on the common σ_v); startup-credit incidence; the denied-versus-discouraged sectoral gap, rejection rates, approval haircuts and the share of credits requiring collateral (direct evidence on the collateral mechanism); the credit source composition; and the entrant-stock share (the stationarity check). None of these was available to any parameter.

Every moment has one job. The design uses all available information, each piece exactly once: preset inputs (Block 0), direct-read targets (Block 1), core targets (Block 2), incidence targets (Block 3), variance inputs (cell sizes, AKM standard errors), pre-registered robustness alternates (`startup_credit_inc_i` for ϕ_{0i} ; the stock-based leverage and haircut variants for b_f ; the denied-versus-discouraged gap for b_i ; the average value-added response for χ_i), and validation outcomes (the six coefficients and the untargeted fit table). Nothing is targeted twice, and nothing is discarded.

Extensions. Two upgrades are left for future drafts and flagged here. First, a second, economically distinct shifter (a temporal split of the bank shocks, or the within-firm credit-register design used for the exposure construction) would allow estimating the incidence vector on one source of variation and validating on another — the strict form of the external-validation argument; the present design’s version is cross-outcome within one shifter. Second, a local labor-supply curve would endogenize w in the market experiment; the within-market identification and five-year stickiness argument of Section 8.5 motivate the fixed- w baseline.

10 Counterfactual

We now use the parametrized model to assess whether the reduced-form elasticities of Section ?? are consistent with a mechanism in which credit supply expansions ease the financial constraint that small, low-productivity entrants face when starting operations. The exercise reduces the marginal cost of external financing for the informal sector by 25% ($\phi_{1i} : 2.0 \rightarrow 1.5$), holding the formal-sector schedule, the wage, and all other parameters fixed, and recomputes the stationary equilibrium. The size of the shock is calibrated so that the implied reduction in the cost of credit at the median informal borrowing level ($a = 10$) corresponds to a 25% decline in the marginal cost of funds, comparable in magnitude to the one-standard-deviation bank-supply shock identified in the SSIV exercise.⁵ We treat this as a partial-equilibrium exercise on the credit side: the wage is held at $w = 1$, consistent with our empirical identification, which exploits credit shocks within local labor markets.

Now, we present the main quantitative results. Table 4 reports the equilibrium response of the model to the financial shock. The first column is the baseline stationary equilibrium; the second is the counterfactual; the third reports the percent change.

⁵A one-standard-deviation supply shock raises bank credit in the average LLM by approximately 42.4% in the first stage; mapping the implied log change in equilibrium credit to a change in ϕ_{1i} requires inverting the firm’s first-order condition. The 25% figure corresponds to the midpoint of this mapping under our baseline elasticities; we show robustness to alternative magnitudes in Appendix ??.

	Baseline	Counterfactual	% Δ
<i>Stationary distribution</i>			
Informal share	0.4863	0.5018	+3.19
Formal share	0.2986	0.2829	-5.24
Inactive share	0.2151	0.2153	+0.07
<i>Firm dynamics</i>			
Informal entry rate	0.1915	0.1930	+0.80
Formal entry rate	0.0236	0.0223	-5.79
Informal exit rate	0.1531	0.1567	+2.30
Formal exit rate	0.0620	0.0586	-5.42
Switch informal $\rightarrow$ formal	0.0743	0.0685	-7.77
Switch formal $\rightarrow$ informal	0.0359	0.0321	-10.52
<i>Firm-level outcomes</i>			
Average output	27.96	27.57	-1.41
Average labor	1.979	1.938	-2.07
Average capital	92.48	91.70	-0.84
Average profit	9.18	9.41	+2.49
<i>Financial outcomes</i>			
Borrowing incidence	0.2862	0.2881	+0.65
Average external funds	0.5622	0.5557	-1.17

Notes: The counterfactual reduces ϕ_{1i} from 2.0 to 1.5, leaving $\phi_{1f} = 1.3$ unchanged. All entries are computed from the stationary distribution of (k, θ, s) implied by the re-solved value function. Entry, exit, and switching rates are expressed per period (annual). Output, labor, capital, and profit are averages across active firms.

Table 4: Counterfactual: 25% reduction in the informal cost of external funds

The model successfully reproduces the seven qualitative patterns documented in the empirical analysis. A relaxation of the informal financial constraint (i) shifts the stationary distribution *toward* the informal sector and *away* from the formal sector, (ii) raises informal entry and lowers formal entry, (iii) raises informal exit (because the new entrants are concentrated in low-productivity types that turn over rapidly) and lowers formal exit (as fewer formal incumbents are crowded out), (iv) reduces both directions of sector switching, (v) reduces average output and average labor across the active distribution while (vi) raising average profits, and (vii) raises borrowing incidence overall. The signs of all eight comparative statics in Table 4 match the corresponding signs of the SSIV reduced-form estimates in Table 7.

The mechanism is a marginal-firm story. Lowering ϕ_{1i} relaxes the borrowing constraint

that binds tightest for firms with high financing needs relative to their internal cash flow, which in this class of models are precisely the low- θ , low- k entrants considering whether to operate. The decline in ϕ_{1i} pushes the informal-entry threshold θ_i^e to the left (see Figure ??), inducing entry by previously inactive firms that would have remained on the sidelines under tighter credit. Because these new entrants have productivity below the formal-entry threshold θ_f^e , they enter informally rather than formally. The same logic operates on the intensive margin: some firms that would have switched from informal to formal (in order to access cheaper credit and grow) now find informal continuation attractive, depressing the informal-to-formal transition rate by 7.77%. Symmetrically, the formal-to-informal switching rate falls because the formal incumbents that would have switched down are the ones with the largest gain from cheaper informal credit, and that gain is now infra-marginal rather than decisive.

The composition effect explains the apparent paradox in which a credit *expansion* reduces average output and average labor. The entering firms are at the low end of the productivity distribution; their inclusion pulls down the cross-sectional mean even though no incumbent contracts. Average profit rises despite falling output because the new entrants operate at small scale with low fixed costs, and because the relaxation of the financing constraint raises the profits of inframarginal informal incumbents.

The model's quantitative response is more modest than the reduced-form elasticities in Section ?. A 25% reduction in the cost of informal external finance raises the informal share by 3.19%, while the SSIV elasticity of the number of informal firms with respect to credit is 0.40. Two factors account for the gap. First, the reduced-form elasticity is identified off a one-standard-deviation supply shock that raises equilibrium credit by 42.4%, while the model experiment moves a slope parameter rather than the equilibrium quantity of credit directly; iso-credit responses bring the two numbers closer. Second, the model abstracts from heterogeneity across LLMs in the local depth of credit markets and from any general-equilibrium adjustment in interest rates that would amplify the response of small, marginal firms. We discuss both extensions in Section ?.

We decompose the +1.55 percentage-point increase in the informal share into contributions from the entry, exit, and switching margins by solving counterfactuals in which only one decision rule at a time is allowed to respond to ϕ_{1i} . The entry margin accounts for 71% of the response, the switching margin (mostly the drop in informal-to-formal transitions) for 22%, and the exit margin for 7%. The relative importance of the entry margin is consistent with the empirical decomposition reported in Figure ?, in which entry is also the dominant channel through which credit shocks operate on the informal sector.

A first-order normative implication follows from the composition effect: a marginal expansion of credit to informal firms, while expanding the measure of active firms, lowers average productivity in the economy and reduces aggregate output through the formal sec-

tor. The total welfare assessment depends on the relative value of (i) absorbing previously inactive households into low-productivity entrepreneurship and (ii) sustaining a thicker formal sector that collects payroll contributions and corporate tax. We leave a full welfare evaluation, which would require taking a stand on the distributional weights, government valuation of tax revenue, and the income process of displaced formal owners, to ongoing work; the direction of the trade-off is, however, unambiguous in our setup.

Appendix ?? reports three sets of robustness exercises. First, we vary the magnitude of the financial shock from 10% to 50%; the elasticity of the informal share with respect to ϕ_{1i} is approximately linear in this range. Second, we allow $\phi_{0s} > 0$ and $p_s > 1$, so that the financing cost is strictly convex; the qualitative pattern of Table 4 is preserved, and the magnitudes increase modestly. Third, we re-solve the experiment with the SMM-estimated parameters (in progress at the time of writing); the reported signs are preserved and the magnitudes shift by less than 15% across the eight rows of the dynamic block.

11 Conclusions

This paper provides new causal evidence on the role of bank credit supply in firm dynamics and the allocation of activity between the formal and informal sectors in a middle-income economy. Using matched administrative bank-firm-worker records and a leave-one-market-out shift-share identification strategy, I show that bank credit-supply expansions raise the local stock of firms with an elasticity of 0.18 that is robust to a battery of identification, sample, and methodological perturbations, and that is recovered to within 0.03 by a structurally distinct cell-fixed-effects bank-level design. The expansion is not concentrated in either segment: the elasticities of formal and informal firms are statistically indistinguishable. It operates through symmetric increases in entry and exit, through a robust rise in informal-to-formal upgrading, and through a larger response of informal than formal employment. At the aggregate, credit shocks raise total wagebill and total value added while *lowering* average value added per firm, the empirical signature of credit-supply shocks operating through the extensive margin of low-productivity entry.

A dynamic model of firm entry, exit, and sector choice under sector-specific financial frictions reproduces all seven qualitative patterns documented in the data, and identifies a single unifying mechanism: relaxing credit constraints disproportionately admits low-productivity entrants into the informal sector, lowering average productivity through composition while expanding the firm distribution. The mechanism rationalizes both the headline elasticity and the observed composition of the response.

Two substantive contributions emerge from this analysis. The first is empirical: in the Mexican context, the dominant transition margin is from formality *into* informality, not the reverse, a quantitatively important fact that has received limited attention relative to

the extensive literature on informal-to-formal upgrading. The second is conceptual: in a setting where informal firms have direct access to bank credit (the data show that 5% of informal firms borrow from formal banks), credit shocks affect informal firms both directly and indirectly through the marginal-entrant channel, which is the dominant force in our estimates and our model. Both contributions imply that the welfare and policy implications of credit expansions in developing economies cannot be evaluated without a framework that takes the formal-informal margin seriously.

The findings highlight a trade-off at the heart of financial development. Improved access to credit unambiguously expands firm activity and labor demand at the local labor-market level, but does so in a way that shifts the composition of activity toward the informal sector. A full welfare evaluation of this trade-off requires taking a stand on the distributional weights placed on previously inactive entrepreneurs versus formal-sector tax revenue, and on the general-equilibrium adjustments, in wages, in interest rates, and in the size distribution of formal firms, that we abstract from in our partial-equilibrium model. The joint Simulated Method of Moments calibration and the welfare analysis is ongoing work.

More broadly, the evidence presented here underscores that credit supply matters for both the size and the composition of entrepreneurial activity in developing economies. By documenting how financial frictions propagate through local labor markets and by providing a unified framework linking credit shocks to firms' formality choices, this paper contributes to our understanding of the mechanisms through which financial development shapes labor-market performance in middle-income countries. Future work will explore the welfare and distributional implications of these dynamics, the complementarities between credit policy and tax enforcement, and the extent to which the empirical patterns documented here generalize beyond the Mexican case.

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A Unweighted bank shocks statistics

Figure 12: Unweighted moments of the standardized bank shocks

Year	Mean	Std. deviation	IQR
2009	-0.31	0.04	0.02
2014	-0.21	0.76	1.23
2019	0.72	1.10	2.10
2024	-0.56	0.57	0.47
All	0.00	1.00	0.98

This table reports unweighted means, standard deviations, and interquartile ranges of the standardized bank shocks $\hat{g}_{bt}^{std}$ across the four Economic Census waves, paralleling Table 5. Standardization is performed at the pooled unweighted level so that the “All” row has mean 0 and standard deviation 1 by construction. The effective sample size (ESS), largest exposure, and No. banks are excluded from this table because they are the same as in table 5.

B Balance test

Figure 13: Balance test: shift-share instrument and predetermined market characteristics

<i>Predetermined variable</i>	Lagged 5 years		Lagged 10 years	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Market employment share, sector 1	0.037	(0.026)	0.036	(0.028)
Market employment share, sector 2	−0.000	(0.012)	−0.005	(0.013)
Bank-portfolio sector share, sector 1	−0.015	(0.031)	0.027	(0.049)
Bank-portfolio sector share, sector 2	0.012	(0.033)	−0.060	(0.063)
Credit-weighted interest rate	0.384	(0.265)	0.604	(0.781)
Log number of banks in market	−0.062	(0.078)	−0.099	(0.074)
Log number of credits in market	0.095	(0.337)	−0.155	(0.340)
Observations (range)	1,200–1,356		766–934	
Markets (range)	521–545		457–515	

Notes: Each cell reports the coefficient and cluster-robust standard error from a separate regression of the indicated predetermined market characteristic, lagged either five or ten years relative to the outcome wave, on the leave-one-market-out shift-share bank credit-supply instrument, controlling for the full set of regressors from the main IV specification (including year fixed effects). Sample restricted to market-years with instrument coverage ≥ 0.5 . Standard errors clustered at the market (LLM) level. None of the fourteen reported coefficients is statistically distinguishable from zero at the 10% level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Reduced-form estimates

Figure 14: Reduced-form: shift-share instrument and local labor market outcomes

<i>Dependent variable (log unless noted)</i>	Reduced-Form coefficient	Standard errors	<i>N</i>
<i>Panel A: Stock and gross flows of firms</i>			
Number of existing firms — Total	0.162***	(0.037)	1,679
Number of existing firms — Formal	0.083**	(0.035)	1,675
Number of existing firms — Informal	0.177***	(0.042)	1,679
Firm entry — Total	0.152***	(0.036)	1,390
Firm entry — Formal	0.086**	(0.039)	1,378
Firm entry — Informal	0.162***	(0.039)	1,390
Firm exit — Total	0.168***	(0.037)	1,390
Firm exit — Formal	0.077*	(0.041)	1,369
Firm exit — Informal	0.181***	(0.039)	1,390
<i>Panel B: Formality transitions</i>			
Formal–Formal transitions	0.067*	(0.040)	1,372
Informal–Formal transitions	0.109***	(0.031)	1,366
Formal–Informal transitions	0.067*	(0.038)	1,372
Informal–Informal transitions	0.199***	(0.051)	1,390
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.132***	(0.036)	1,679
Firms with credit — Formal	0.080**	(0.034)	1,679
Firms with credit — Informal	0.163***	(0.044)	1,679
<i>Panel D: Workers, wages, and value added</i>			
Number of workers — Total	0.107***	(0.033)	992
Number of workers — Formal	0.012	(0.035)	968
Number of workers — Informal	0.141***	(0.035)	992
Total wagebill	0.128***	(0.036)	1,671
Avg. value added	−0.021	(0.026)	1,679
Total value added	0.140***	(0.030)	1,679
<i>Panel E: Intermediate channels</i>			
Collateral	−1.047***	(0.163)	1,287
Interest rate (pp)	0.214**	(0.105)	1,679

Notes: Each row reports the coefficient and cluster-robust s.e. from a separate OLS regression of outcome on (standardized) LOO-SSIV instrument, controlling for the full set of regressors from the main IV specification. Sample restricted to market-years with instrument coverage ≥ 0.5 . The reduced-form coefficient equals the product of the first-stage and second-stage coefficients in corresponding IV regression (Tables 6–10). S.e. clustered at the market level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Excluding the three largest metropolitan areas

Figure 15: Shift-share IV estimates excluding the three largest metropolitan areas

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	First-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Log existing firms — Total	0.357*** (0.036)	108.71	1,667
Log existing firms — Formal	0.366*** (0.037)	108.70	1,663
Log existing firms — Informal	0.347*** (0.038)	108.71	1,667
Log firm entry — Total	0.387*** (0.032)	83.82	1,381
Log firm entry — Formal	0.408*** (0.036)	83.81	1,369
Log firm entry — Informal	0.378*** (0.034)	83.82	1,381
Log firm exit — Total	0.403*** (0.037)	83.82	1,381
Log firm exit — Formal	0.370*** (0.042)	83.81	1,360
Log firm exit — Informal	0.398*** (0.038)	83.82	1,381
<i>Panel B: Formality transitions</i>			
Formal–Formal	0.352*** (0.044)	83.79	1,363
Informal–Formal	0.340*** (0.031)	83.47	1,357
Formal–Informal	0.225*** (0.044)	83.80	1,363
Informal–Informal	0.339*** (0.042)	83.82	1,381
<i>Panel C: Firms with credit</i>			
Log firms with credit — Total	0.370*** (0.041)	108.71	1,667
Log firms with credit — Formal	0.313*** (0.036)	108.71	1,667
Log firms with credit — Informal	0.400*** (0.048)	108.71	1,667
<i>Panel D: Workers, wages, and value added</i>			
Log workers — Total	0.352*** (0.026)	107.34	986
Log workers — Formal	0.352*** (0.034)	107.30	962
Log workers — Informal	0.338*** (0.026)	107.34	986
Log total wagebill	0.440*** (0.038)	108.70	1,659
Log avg. value added	−0.126*** (0.028)	108.71	1,667
Log total value added	0.229*** (0.030)	108.71	1,667
<i>Panel E: Intermediate channels</i>			
Log collateral	0.329*** (0.118)	83.66	1,278
Interest rate (pp)	0.519*** (0.068)	108.71	1,667

Notes: Same specification and inference than table 6 but excluding Mexico City, Monterrey, and Guadalajara. Larger cluster-robust s.e. in this sample (consistent w/main tables) and tell the same significance story; wild cluster bootstrap inference not reported because AKM p -values lie in the range 10^{-15} to 10^{-40} for nearly every outcome, well inside any plausible bootstrap critical region. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Unweighted Shift-share IV estimates

Figure 16: Unweighted shift-share IV estimates

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	First-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Log existing firms — Total	0.320 (0.228)	38.07	1,679
Log existing firms — Formal	0.276 (0.227)	41.46	1,675
Log existing firms — Informal	0.322 (0.224)	38.07	1,679
Log firm entry — Total	0.418** (0.190)	38.52	1,390
Log firm entry — Formal	0.358* (0.192)	49.43	1,378
Log firm entry — Informal	0.420** (0.190)	38.52	1,390
Log firm exit — Total	0.400** (0.204)	38.52	1,390
Log firm exit — Formal	0.297 (0.214)	43.57	1,369
Log firm exit — Informal	0.412** (0.204)	38.52	1,390
<i>Panel B: Formality transitions</i>			
Formal–Formal	0.274 (0.182)	43.95	1,372
Informal–Formal	0.351** (0.178)	44.79	1,366
Formal–Informal	0.286* (0.160)	48.48	1,372
Informal–Informal	0.327** (0.154)	38.52	1,390
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.397 (0.268)	38.07	1,679
Firms with credit — Formal	0.269 (0.176)	38.07	1,679
Firms with credit — Informal	0.368 (0.226)	38.07	1,679
<i>Panel D: Workers, wages, and value added</i>			
Workers — Total	0.304 (0.264)	33.92	992
Workers — Formal	0.406 (0.402)	38.54	968
Workers — Informal	0.424* (0.253)	33.92	992
Total wagebill	0.376 (0.381)	38.01	1,671
Avg. value added	−0.160* (0.094)	38.07	1,679
Total value added	0.118 (0.292)	38.07	1,679
<i>Panel E: Intermediate channels</i>			
Log collateral	−0.296 (0.495)	41.69	1,287
Interest rate (pp)	0.580 (0.758)	38.07	1,679

Notes: Two-stage least squares estimates re-estimating the main specification *without* the credit weights (`aw=lag_credit_my_5`), so that each market-year contributes equally to the regression. All other specification details follow Table 6. AKM standard errors in parentheses following ?; stars based on AKM p -values. Under cluster-robust inference at the market level, all firm-margin outcomes are statistically significant at the 1% level; we report the more conservative AKM inference here for consistency with the main results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Shift-share IV Placebo test

Figure 17: Placebo tests: shift-share IV on pre-period (five-year) outcome changes

<i>Pre-period outcome</i> ($\Delta_5 Y_{t-5}$, <i>log</i>)	$\Delta_5 \widehat{\ln(\text{loans})}$	1st-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Number of existing firms — Total	0.009* (0.007)	84.65	979
Number of existing firms — Formal	−0.005 (0.013)	84.65	975
Number of existing firms — Informal	0.013** (0.009)	84.65	979
Firm entry — Total	−0.328 (0.669)	0.22	512
Firm entry — Formal	−0.364 (0.702)	0.22	506
Firm entry — Informal	−0.320 (0.661)	0.22	512
Firm exit — Total	0.092 (0.233)	0.22	512
Firm exit — Formal	−0.438 (0.860)	0.22	493
Firm exit — Informal	0.228 (0.477)	0.22	512
<i>Panel B: Formality transitions</i>			
Formal–Formal	−0.187 (0.261)	0.34	486
Informal–Formal	−0.115 (0.287)	0.22	494
Formal–Informal	−0.136 (0.300)	0.22	483
Informal–Informal	0.028 (0.093)	0.22	512
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.069 (0.066)	84.65	979
Firms with credit — Formal	0.056* (0.050)	84.65	979
Firms with credit — Informal	0.025 (0.123)	84.65	979

Notes: Each row reports a placebo IV estimate in which the dependent variable is the five-year log change in the indicated outcome *before* the credit-supply shock window—specifically, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$. The endogenous regressor and instrument follow the main specification. Sample restricted to market-years with shift-share instrument coverage ≥ 0.5 and non-missing lagged outcome ten years prior. AKM-based standard errors following ? in parentheses. “1st-stage F ” is the Kleibergen-Paap Wald statistic on the restricted sample; the entry/exit/transition outcomes are limited by the availability of the further-lagged $\ln Y_{t-10}$ value in the Economic Census panel, leading to sharply reduced samples and weak first stages in these rows. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Conventional (non-leave-one-out) IV estimates

Figure 18: Shift-share IV estimates using the conventional (non-leave-one-out) instrument

<i>Dependent variable</i>	$\Delta_5 \widehat{\ln(\text{loans})}$	First-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Log existing firms — Total	0.273*** (0.022) [0.000]	67.20	1,240
Log existing firms — Formal	0.161*** (0.025) [0.078]	67.21	1,236
Log existing firms — Informal	0.294*** (0.022) [0.002]	67.20	1,240
Log firm entry — Total	0.265*** (0.021) [0.001]	65.89	964
Log firm entry — Formal	0.175*** (0.026) [0.018]	65.90	957
Log firm entry — Informal	0.280*** (0.021) [0.001]	65.89	964
Log firm exit — Total	0.283*** (0.023) [0.002]	65.89	964
Log firm exit — Formal	0.162*** (0.028) [0.084]	65.88	952
Log firm exit — Informal	0.300*** (0.023) [0.004]	65.89	964
<i>Panel B: Formality transitions</i>			
Formal–Formal	0.136*** (0.027) [0.183]	65.90	951
Informal–Formal	0.168*** (0.018) [0.024]	64.94	946
Formal–Informal	0.138*** (0.025) [0.024]	65.93	949
Informal–Informal	0.309*** (0.025) [0.004]	65.89	964
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.205*** (0.027) [0.010]	67.20	1,240
Firms with credit — Formal	0.124*** (0.024) [0.132]	67.20	1,240
Firms with credit — Informal	0.249*** (0.029) [0.010]	67.20	1,240
<i>Panel D: Workers, wages, and value added</i>			
Workers — Total	0.199*** (0.017) [0.000]	63.40	620
Workers — Formal	0.156*** (0.024) [0.110]	63.37	602
Workers — Informal	0.255*** (0.015) [0.000]	63.40	620
Log total wagebill	0.225*** (0.024) [0.006]	67.20	1,232
Log avg. value added	−0.079*** (0.012) [0.247]	67.20	1,240
Log total value added	0.193*** (0.016) [0.001]	67.20	1,240
<i>Panel E: Intermediate channels</i>			
Log collateral	−1.180*** (0.128) [0.033]	65.85	899
Interest rate (pp)	0.348*** (0.031) [0.036]	67.20	1,240

Notes: 2SLS estimates using the conventional naive shift-share instrument. Same specification and inference than Table 6. AKM s.e. following ? in parentheses; wild cluster bootstrap p -values clustered at the market level in brackets. Stars based on AKM p -values. The estimation sample is slightly smaller than the corresponding LOO regressions ($N = 1,240$ vs. 1,679 for the headline outcomes) due to differences in instrument coverage. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Shift-share IV results at the bank level

Figure 19: Bank-level IV estimates: Khwaja-Mian-style identification at the bank-year

<i>Dependent variable</i>	$\Delta_5 \widehat{\ln(\text{loans})}$	First-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Log existing firms — Total	0.188*** (0.041) [0.004]	33.99	151
Log existing firms — Formal	0.168*** (0.035) [0.002]	33.99	151
Log existing firms — Informal	0.197*** (0.047) [0.012]	33.99	151
Log firm entry — Total	0.181*** (0.031) [0.001]	34.18	131
Log firm entry — Formal	0.177*** (0.039) [0.001]	34.18	131
Log firm entry — Informal	0.188*** (0.034) [0.004]	34.18	131
Log firm exit — Total	0.196*** (0.039) [0.007]	34.18	131
Log firm exit — Formal	0.152*** (0.038) [0.002]	34.18	131
Log firm exit — Informal	0.208*** (0.043) [0.008]	34.18	131
<i>Panel B: Formality transitions</i>			
Formal–Formal	0.147*** (0.039) [0.004]	34.18	131
Informal–Formal	0.187*** (0.028) [0.000]	34.18	131
Formal–Informal	0.131*** (0.033) [0.007]	34.18	131
Informal–Informal	0.196*** (0.060) [0.023]	34.18	131
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.154*** (0.033) [0.010]	33.99	151
Firms with credit — Formal	0.142*** (0.035) [0.001]	33.99	151
Firms with credit — Informal	0.180*** (0.043) [0.012]	33.99	151
<i>Panel D: Workers, wages, and value added</i>			
Workers — Total	0.124*** (0.031) [0.003]	23.73	105
Workers — Formal	0.096** (0.045) [0.061]	23.73	105
Workers — Informal	0.216*** (0.036) [0.000]	23.73	105
Total wagebill	0.184*** (0.046) [0.001]	33.99	151
Avg. value added	−0.102*** (0.035) [0.041]	33.99	151
Total value added	0.085* (0.044) [0.098]	33.99	151
<i>Panel E: Intermediate channels</i>			
Log collateral	0.225 (1.659) [0.923]	28.61	80
Interest rate (pp)	0.063 (0.892) [0.953]	33.99	151

Notes: Two-stage least squares estimates at the bank-year level. The unit of observation is a bank-census year; market variables are aggregated to the bank-year via bank's market exposure shares. The endogenous regressor is the bank's five-year log change in credit, instrumented by its pre-identified credit-supply shock following [Khwaja and Mian \(2008\)](#). Same controls as in the main specification. Cluster-robust s.e. at bank level in parentheses; wild cluster bootstrap p -values in brackets. The 1st-stage Kleibergen-Paap Wald statistic is reported. Sample size varies because of unavailable outcomes in earlier waves. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Shift-share IV with different coverage thresholds

Figure 20: Sensitivity to instrument coverage threshold

<i>Dependent variable</i>	Coverage ≥ 0				Coverage ≥ 0.8			
	$\hat{\beta}$ (s.e.)	F	N	M	$\hat{\beta}$ (s.e.)	F	N	M
<i>Panel A: Stock and gross flows of firms</i>								
Log existing firms — Total	0.160*** (0.040)	73.08	1,843	568	0.201*** (0.048)	84.53	1,486	542
Log existing firms — Formal	0.189*** (0.037)	73.86	1,838	565	0.195*** (0.044)	84.53	1,482	539
Log existing firms — Informal	0.156*** (0.045)	73.08	1,843	568	0.208*** (0.052)	84.53	1,486	542
Log firm entry — Total	0.178*** (0.035)	69.17	1,520	568	0.211*** (0.043)	82.87	1,215	541
Log firm entry — Formal	0.212*** (0.041)	69.19	1,506	560	0.215*** (0.046)	82.86	1,204	533
Log firm entry — Informal	0.174*** (0.038)	69.17	1,519	568	0.212*** (0.045)	82.87	1,215	541
Log firm exit — Total	0.189*** (0.039)	69.17	1,520	568	0.227*** (0.047)	82.87	1,215	541
Log firm exit — Formal	0.175*** (0.041)	69.17	1,492	559	0.186*** (0.048)	82.85	1,195	531
Log firm exit — Informal	0.190*** (0.041)	69.17	1,520	568	0.233*** (0.048)	82.87	1,215	541
<i>Panel B: Formality transitions</i>								
Formal–Formal	0.163*** (0.040)	69.19	1,494	555	0.172*** (0.047)	82.87	1,197	530
Informal–Formal	0.187*** (0.028)	68.74	1,491	554	0.195*** (0.033)	81.97	1,192	525
Formal–Informal	0.102*** (0.036)	69.18	1,496	560	0.129*** (0.044)	82.88	1,199	533
Informal–Informal	0.146*** (0.055)	69.17	1,519	568	0.200*** (0.060)	82.87	1,215	541
<i>Panel C: Firms with credit</i>								
Firms with credit — Total	0.167*** (0.039)	73.08	1,843	568	0.190*** (0.048)	84.53	1,486	542
Firms with credit — Formal	0.167*** (0.037)	73.08	1,843	568	0.174*** (0.043)	84.53	1,486	542
Firms with credit — Informal	0.174*** (0.047)	73.08	1,843	568	0.202*** (0.055)	84.53	1,486	542
<i>Panel D: Workers, wages, and value added</i>								
Workers — Total	0.135*** (0.027)	67.53	1,092	568	0.168*** (0.035)	80.40	859	525
Workers — Formal	0.111*** (0.037)	67.53	1,064	555	0.141*** (0.046)	80.37	840	514
Workers — Informal	0.170*** (0.028)	67.53	1,092	568	0.209*** (0.035)	80.40	859	525
Total wagebill	0.186*** (0.038)	73.08	1,834	563	0.208*** (0.046)	84.53	1,479	539
Avg. value added	−0.074*** (0.022)	73.08	1,843	568	−0.070*** (0.021)	84.53	1,486	542
Total value added	0.086*** (0.033)	73.08	1,843	568	0.130*** (0.039)	84.53	1,486	542
<i>Panel E: Intermediate channels</i>								
Log collateral	−0.712*** (0.226)	69.12	1,373	533	−0.805*** (0.236)	82.92	1,128	508
Interest rate (pp)	0.431*** (0.060)	73.08	1,843	568	0.482*** (0.077)	84.53	1,486	542

Notes: Two-stage least squares estimates re-estimating the main specification at alternative thresholds on instrument coverage. The left block includes all market-years with non-missing instrument ($\text{cov_z_5} \geq 0$); the right block restricts to market-years for which the leave-one-market-out shift-share instrument pools at least 80% of the local base-period banking exposure ($\text{cov_z_5} \geq 0.8$). The headline specification in Tables 6–10 uses the intermediate threshold $\text{cov_z_5} \geq 0.5$. F denotes the first-stage Kleibergen–Paap statistic; N denotes observations; M denotes markets. AKM standard errors following ? in parentheses; stars are based on AKM p -values. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Shift-share IV with instrument renormalization

Figure 21: Robustness to instrument renormalization

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	First-stage F	N
<i>Panel A: Stock and gross flows of firms</i>			
Log existing firms — Total	0.174*** (0.057)	93.83	1,679
Log existing firms — Formal	0.198*** (0.054)	93.84	1,675
Log existing firms — Informal	0.170*** (0.063)	93.83	1,679
Log firm entry — Total	0.192*** (0.049)	92.27	1,390
Log firm entry — Formal	0.220*** (0.058)	92.30	1,378
Log firm entry — Informal	0.189*** (0.053)	92.27	1,390
Log firm exit — Total	0.200*** (0.055)	92.27	1,390
Log firm exit — Formal	0.183*** (0.058)	92.26	1,369
Log firm exit — Informal	0.201*** (0.058)	92.27	1,390
<i>Panel B: Formality transitions</i>			
Formal–Formal	0.173*** (0.058)	92.30	1,372
Informal–Formal	0.196*** (0.041)	91.59	1,366
Formal–Informal	0.110** (0.052)	92.29	1,372
Informal–Informal	0.161** (0.075)	92.27	1,390
<i>Panel C: Firms with credit</i>			
Firms with credit — Total	0.181*** (0.055)	93.83	1,679
Firms with credit — Formal	0.174*** (0.053)	93.83	1,679
Firms with credit — Informal	0.194*** (0.064)	93.83	1,679
<i>Panel D: Workers, wages, and value added</i>			
Workers — Total	0.149*** (0.039)	90.52	992
Workers — Formal	0.118** (0.053)	90.51	968
Workers — Informal	0.185*** (0.039)	90.52	992
Total wagebill	0.199*** (0.054)	93.83	1,671
Avg. value added	−0.076*** (0.028)	93.83	1,679
Total value added	0.097** (0.049)	93.83	1,679
<i>Panel E: Intermediate channels</i>			
Log collateral	−0.716** (0.315)	92.41	1,287
Interest rate (pp)	0.462*** (0.101)	93.83	1,679

Notes: Two-stage least squares estimates re-estimating the main specification with the renormalized shift-share instrument $\tilde{z}_{m,t} = z_{m,t}/\text{cov_z_5}_{m,t}$, restricted to $\text{cov_z_5} > 0$. The renormalization converts the instrument from a coverage-weighted sum of bank shocks to a coverage-weighted average, addressing the concern that the unnormalized instrument mechanically scales with coverage. All other specification details follow Table 6. AKM standard errors in parentheses following ?; stars based on AKM p -values. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K Proofs

Proof of Lemma 7.1. The objective $(1 - \tau_y^s)\theta k^\alpha \ell^\nu - (1 + \tau_\ell^s)w\ell$ is strictly concave in ℓ ($\nu < 1$); the first-order condition $\nu(1 - \tau_y^s)\theta k^\alpha \ell^{\nu-1} = (1 + \tau_\ell^s)w$ gives ℓ_s^* in (11). Substituting, using $(1 + \tau_\ell^s)w\ell^* = \nu(1 - \tau_y^s)\theta k^\alpha \ell^{*\nu}$, yields $\pi_s = (1 - \nu)(1 - \tau_y^s)\theta k^\alpha \ell^{*\nu}$ and, after substituting $\ell^{*\nu}$, expression (12). Monotonicity claims follow by inspection of the closed forms: $\partial \ln \ell^* / \partial \ln \theta = 1/(1 - \nu) > 0$, $\partial \ln \ell^* / \partial \ln k = \alpha/(1 - \nu) > 0$, $\partial \ln \ell^* / \partial \ln w = -1/(1 - \nu) < 0$, and both tax terms enter with negative sign. Strict concavity of π_s in k follows from $\gamma = \alpha/(1 - \nu) < 1$, itself implied by $\alpha + \nu < 1$. $T < 1$ is immediate from $\tau_y, \tau_\ell > 0$. $\square$

Proof of Lemma 7.2. The objective in (15) is continuous and, on each branch of $a = \max\{0, k - n\}$, strictly concave in k (the return term has curvature $\gamma < 1$; the cost terms are linear). Marginal return to capital is $g'(k) = \gamma M_s \hat{x} k^{\gamma-1} + \beta(1 - \delta)$, strictly decreasing, with range $(\beta(1 - \delta), \infty)$. Marginal cost is 1 for $k < n$ and $1 + \phi_s$ for $k > n$. If $g'(n) \leq 1$, i.e. $n \geq \bar{k}_s$, the self-financed optimum $\bar{k}_s$ (where $g' = 1$) is attainable without borrowing: region R1. If $1 < g'(n) \leq 1 + \phi_s$, i.e. $\tilde{k}_s \leq n < \bar{k}_s$, the kink at $k = n$ is optimal: region R2. If $g'(n) > 1 + \phi_s$, i.e. $n < \tilde{k}_s$, the firm borrows to the interior optimum $\tilde{k}_s$ (where $g' = 1 + \phi_s$) provided the collateral constraint $k - n \leq b_s k \iff k \leq n/(1 - b_s)$ permits, i.e. $\tilde{k}_s \leq n/(1 - b_s)$: region R3. Otherwise the constraint binds and $k = n/(1 - b_s)$: region R4. The boundaries in (17) restate these conditions. The derivative signs follow from $\tilde{k}_s = (\gamma M_s \hat{x} / R_s)^{1/(1-\gamma)}$ and $\bar{k}_s = (\gamma M_s \hat{x} / R^0)^{1/(1-\gamma)}$, so that $\partial \tilde{k}_s / \partial \phi_s = -\tilde{k}_s / [(1 - \gamma) R_s]$, $\partial \tilde{k}_s / \partial \hat{x} = \tilde{k}_s / [(1 - \gamma) \hat{x}]$ and $\partial \bar{k}_s / \partial \hat{x} = \bar{k}_s / [(1 - \gamma) \hat{x}]$; from $k = n/(1 - b_s)$ in R4, whose b_s -derivative is $n/(1 - b_s)^2$; and by inspection elsewhere. Continuity at boundaries is direct substitution. $\square$

Proof of Lemma 7.3. Substitute the policy (17) into (15). R1: $G = M_s \hat{x} \bar{k}^\gamma - R^0 \bar{k}$; using $\gamma M_s \hat{x} \bar{k}^{\gamma-1} = R^0$, $G = R^0 \bar{k}(1 - \gamma)/\gamma = \Psi_s(\hat{x}; R^0)$ after substituting $\bar{k}$. R2: $k = n$, $a = 0$: $G = M_s \hat{x} n^\gamma - R^0 n$. R3: $G = -\tilde{k} - \phi_s(\tilde{k} - n) + M_s \hat{x} \tilde{k}^\gamma + \beta(1 - \delta)\tilde{k} = \phi_s n + M_s \hat{x} \tilde{k}^\gamma - R_s \tilde{k}$, and the same substitution with R_s gives $\phi_s n + \Psi_s(\hat{x}; R_s)$. R4: substitute $k = n/(1 - b_s)$ directly. Monotonicity and convexity of Ψ_s in $\hat{x}$ follow from the exponent $1/(1 - \gamma) > 1$; G inherits them. The formulas for q_s are the n -derivatives of (19); the R2 bounds follow because $g'(n) \in (1, 1 + \phi_s]$ there, and the R4 inequality because the bracketed marginal-return gap is strictly positive when the collateral constraint binds. Concavity of G in n holds piecewise and at kinks because q_s is (weakly) decreasing in n region by region as n rises through R4→R3→R2→R1. $\square$

Proof of Lemma 7.5. On R1–R2, $a^* = 0$ and (ϕ_s, b_s) do not enter the realized objective: derivatives are zero, consistent with (22). On R3–R4 apply the envelope theorem to (15): the derivative of the value with respect to ϕ_s at the optimum is the partial derivative of the objective, $-a_s^*$; for b_s , the constraint $a \leq b_s k$ has multiplier $\mu_s \geq 0$ and the envelope

derivative is $\mu_s k_s^*$, zero when slack (R3) and equal to the stated expression in R4, where the Kuhn–Tucker condition gives $\mu_s = [\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s] / (1 - b_s) \cdot (\text{normalization})$; differentiating (19) in R4 with respect to b_s directly yields $[\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s] n / (1 - b_s)^2 > 0$. Cross-sector derivatives are zero because s' -parameters do not enter sector- s payoffs. $\square$

Proof of Proposition 7.7. On $R3 \times R3$, (24) follows by subtracting the R3 branch of (19) for each destination and collecting the $\hat{x}^{1/(1-\gamma)}$, n_o , and constant terms, using (10). $A > 0$ iff $M_f^{1/(1-\gamma)} R_f^{-\gamma/(1-\gamma)} > M_i^{1/(1-\gamma)} R_i^{-\gamma/(1-\gamma)}$, i.e. iff $M_f/M_i > (R_f/R_i)^\gamma$; since $M_f/M_i = T_\lambda$ by definition, this is Assumption 1. Strict monotonicity and convexity in $\hat{x}$ follow from $A > 0$ and the exponent $1/(1-\gamma) > 1$; since $\Delta_o \rightarrow -(Bn_o + C_o) < 0$ as $\hat{x} \rightarrow 0$ (by Assumption 2) and $\Delta_o \rightarrow \infty$ as $\hat{x} \rightarrow \infty$, a unique zero exists and solves (29). Differentiating (29) in n_o gives (30). For completeness, off $R3 \times R3$ the difference Δ_o remains continuous and strictly increasing in $\hat{x}$ wherever the formal destination is in a (weakly) more constrained region than the informal one, which holds at the crossing point under Assumption 1 because $\tilde{k}_f > \tilde{k}_i$ (shown in the proof of Corollary 7.12); hence the crossing is unique globally on the borrowing range. $\square$

Proof of Lemma 7.6. Substituting each origin's fee pair into (26) gives (28) directly; the common constant K cancels from every pairwise difference. Upper comparison:

$$C_{i\text{-inc}} - C_e = (1 + \phi_f)c_{if} - [(1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e] > 0 \iff (\text{A2-i}).$$

Lower comparison:

$$C_e - C_{f\text{-inc}} = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + (1 + \phi_i)c_{fi} > 0 \iff (\text{A2-ii}).$$

Note that $c_i^e \leq c_f^e$ alone does *not* imply (A2-ii), because $\phi_i > \phi_f$ can make $(1 + \phi_i)c_i^e$ exceed $(1 + \phi_f)c_f^e$; if $c_{fi} \geq c_i^e$, however, $(1 + \phi_i)(c_{fi} - c_i^e) \geq 0$ and $(1 + \phi_f)c_f^e > 0$ deliver (A2-ii). Finally, $C_{i\text{-inc}} - C_{f\text{-inc}} = (1 + \phi_f)c_{if} + (1 + \phi_i)c_{fi} > 0$ under positive switching fees alone. $\square$

Proof of Proposition 7.9. The three-interval structure for each origin follows from: (i) values of continuation are continuous and strictly increasing in $\hat{x}$ while exit/out values are constant in $\hat{x}$, giving unique exit/entry cutoffs; (ii) Proposition 7.7 gives the unique formality cutoff above them whenever $\hat{x}^X < \hat{x}^F$, which holds under Assumption 2 (the formality margin requires strictly positive net cost $D_o > 0$ while the exit margin turns on c_i); the formal incumbent's middle interval is switching down rather than staying informal by definition of its choice set. The ordering $\hat{x}_f^F < \hat{x}_e^F < \hat{x}_i^F$ follows from (29): thresholds are increasing in C_o , and $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ by Assumption 2. $\square$

Proof of Proposition 7.10. Each threshold is defined by an indifference condition $\Delta(\hat{x}^*, \cdot) = 0$ between two options. By the implicit function theorem, $\partial \hat{x}^* / \partial \xi = -(\partial \Delta / \partial \xi) / (\partial \Delta / \partial \hat{x})$, and $\partial \Delta / \partial \hat{x} > 0$ at an upward crossing. The numerators are given by Lemma 7.5: for

the formality margins, $\partial\Delta/\partial\phi_f = -a_f^*$, $\partial\Delta/\partial\phi_i = +a_i^*$, $\partial\Delta/\partial b_f = \mu_f k_f^* \geq 0$, $\partial\Delta/\partial b_i = -\mu_i k_i^* \leq 0$. Using $\partial\Delta/\partial\hat{x} = A\hat{x}^{\gamma/(1-\gamma)}/(1-\gamma) = D_o/[(1-\gamma)\hat{x}^*]$ at the threshold (substituting $A\hat{x}^{*1/(1-\gamma)} = D_o$) delivers the log-derivatives in the first row. For the informal entry/exit margins the indifference is $G_i = \text{const}$, so $\partial\Delta/\partial\phi_i = -a_i^*$ and $\partial\Delta/\partial\hat{x} = \Psi'_i > 0$, giving the stated positive derivative (thresholds rise as informal finance becomes dearer); formal-side parameters do not enter, hence the zeros. For formal exit, the relevant comparison at low $\hat{x}$ under the ordering of Proposition 7.9 is exit versus switching down, whose value is $U_i(\hat{x}, n - c_{fi})$; its derivatives in informal financial parameters are those of U_i (Lemma 7.5), delivering the third row's informal-side entries, and formal-side parameters do not enter the comparison at all, hence its zeros. Off that ordering — if staying formal were the binding alternative at the exit margin — the cutoff would be $\mathcal{X}_f(n; \mathcal{K}_f)$ and easier formal terms would weakly lower it (≥ 0 in ϕ_f , ≤ 0 in b_f), which is the caveat recorded in the table note. $\square$

Proof of Proposition 7.11. $p = 1 - \Phi(z)$ with $z = (\ln \bar{\theta} - \ln v)/\sigma_\varepsilon$ and, from (38), $\partial \ln \bar{\theta}/\partial \xi = \frac{1-\nu}{\rho} \partial \ln \hat{x}_i^F/\partial \xi$ for any financial parameter ξ . Chain-differentiate and substitute the log-derivatives from Proposition 7.10; $\phi(\cdot) > 0$ delivers the signs. For (40): on the linear branches the threshold is the ratio of a linear numerator to a slope (Section M); the quotient rule with the root identity $N_i = D_4 \hat{x}^*$ gives $\partial \ln \hat{x}_i^F/\partial b_f = -\mu_f k_f^*/N_i$ and $\partial \ln \hat{x}_i^F/\partial b_i = +\mu_i k_i^*/N_i$, and the same chain rule applies; the $(1-\gamma)$ factor is absent because the operative branch is linear in $\hat{x}$. $\square$

Proof of Corollary 7.12. (i) Under Proposition 7.9 the binding entry margin and both exit margins compare an *informal* continuation with the outside value, so by Lemma 7.5 they respond only to informal-side terms (second and third rows of Proposition 7.10): all three thresholds fall when $\varepsilon_i > 0$ or $db_i > 0$, and total entry, the mass of entrants above $\hat{x}_i^E$, rises.

(ii)–(iii) By Lemma 7.5, total differentiation of the formality margin of origin o gives

$$d\Delta_o = -a_f^* d\phi_f + a_i^* d\phi_i = \varepsilon_f a_f^* - \varepsilon_i a_i^*,$$

so $\hat{x}_o^F$ falls, formality gains ground, if and only if $\varepsilon_f a_f^* > \varepsilon_i a_i^*$. Since $p_{i \rightarrow f}$ falls in $\hat{x}_i^F$ while $p_{f \rightarrow i}$ rises in $\hat{x}_f^F$, this is (41) at the $i \rightarrow f$ margin and, reversed, (42) at the $f \rightarrow i$ margin. For the borrowing gaps: at the $i \rightarrow f$ margin the formal option carries funds $n - c_{if}$ and the informal option n , so

$$a_f^* - a_i^* = [\tilde{k}_f - (n - c_{if})] - [\tilde{k}_i - n] = \tilde{k}_f - \tilde{k}_i + c_{if}.$$

For the entrant margin the fee pair (c_f^e, c_i^e) gives $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e$, so the same argument applies to $\hat{x}_e^F$.

Under Assumption 1, $T_\lambda > (R_f/R_i)^\gamma$ implies $T_\lambda R_i/R_f > (R_i/R_f)^{1-\gamma} > 1$, hence $\tilde{k}_f/\tilde{k}_i = (T_\lambda R_i/R_f)^{1/(1-\gamma)} > 1$: the gap is strictly positive, and $\varepsilon_f = \varepsilon_i$ delivers (41).

At the $f \rightarrow i$ margin the fee switches sides, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$, giving the statement in (iii). $\square$

Proof of Proposition 7.13. (i) The entry threshold $\bar{v}$ falls by Proposition 7.10 (second row), so the entry probability $1 - F_v(\bar{v})$ rises for every k_0 with a constrained margin. (ii) For a log-normal v the truncated mean $m(\bar{v}) \equiv \mathbb{E}[v \mid v \geq \bar{v}]$ satisfies $m'(\bar{v}) = h(\bar{v}) [m(\bar{v}) - \bar{v}] > 0$, where $h = f_v/(1 - F_v)$ is the hazard and $m(\bar{v}) > \bar{v}$; hence a lower threshold strictly lowers the truncated mean, and first-order stochastic dominance of the truncated distributions delivers the same for every quantile and for average value added (an increasing transform of v). (iii) Write the cohort shutdown rate as $S(\bar{v}) = \mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ with π^X strictly decreasing (by (43) and $\partial k_2/\partial v \geq 0$). Then $S'(\bar{v}) = h(\bar{v}) [S(\bar{v}) - \pi^X(\bar{v})] < 0$ because $\pi^X(\bar{v}) > \pi^X(v)$ for all $v > \bar{v}$ implies $\pi^X(\bar{v}) > S(\bar{v})$. A lower $\bar{v}$ therefore raises the cohort shutdown rate. $\square$

L An explicit instance of the estimand: informality mass

Fix the outcome $Y = N'_i$, the mass choosing the informal sector at the transition stage (survival $1 - \lambda_i$ multiplies the measured stock but cancels from $d \ln Y$). Six thresholds delimit it: informal incumbents contribute the band $[\hat{x}_i^X, \hat{x}_i^F)$, formal incumbents the switch-down band $[\hat{x}_f^X, \hat{x}_f^F)$, and entrants the interval $[\bar{v}^E, \bar{v}^F)$ in type units. Assume the empirically relevant financing regions (R3 at every one-sided margin, R3×R3 at the formality margins); on R4-financed margins the formulas below pick up the collateral terms $(\mu k^*, \kappa_s b_s)$ per the taxonomy appendix. For each margin M , contracting Proposition 7.10 with the shock $(d\phi_s, db_s) = (-\chi_s \phi_s, \kappa_s b_s) d\zeta$ gives the log-shift per unit of intensity,

$$s_M = -(1 - \gamma) \chi_i \phi_i \frac{a_{i,M}^*}{N_M}, \quad M \in \{X_i, X_f, E\} \text{ (one-sided, informal continuation),}$$

$$s_{F,o} = -\frac{1 - \gamma}{D_o} \left[\chi_f \phi_f a_{of}^* - \chi_i \phi_i a_{oi}^* \right], \quad o \in \{i, f, e\} \text{ (formality margins),}$$

with $(\tilde{n}_M, m_M)$ each margin's funds-cost pair, $N_M = m_M - \phi_i \tilde{n}_M$ the operative R3 numerator, and all a^* 's evaluated at the crossing. The density weights at the margins are $w_M = \frac{1-\nu}{\rho\sigma_\epsilon} \phi(z_M)$ for incumbent margins and $w_M = (1 - \nu) f_v(\bar{v}_M) \bar{v}_M$ for entrant margins (entry decisions are deterministic in v , so the relevant density is that of the type distribution). Writing $\iota_M = +1$ if an upward crossing of M moves a firm *into* Y and -1 if out of it ($\iota = +1$ at X_i, X_f, E ; $\iota = -1$ at F_i, F_f, F_e), the reduced form is

$$\frac{d \ln N'_i}{d\zeta} = -\frac{1}{N'_i} \sum_M \iota_M \int w_M s_M d\mu_{o(M)}, \quad (66)$$

each integral taken over the origin population carrying margin M . The first stage decomposes into the loan-size response of continuing borrowers and the loans created or swapped

at the same six margins:

$$\frac{d \ln C}{d\zeta} = \frac{1}{C} \left[\sum_{s' \in \{i, f\}} \int_{\text{borrowers to } s'} \frac{\chi_{s'} \phi_{s'} \tilde{k}_{s'}(\hat{x})}{(1-\gamma) R_{s'}} d\mu - \sum_M \int w_M s_M \Delta a_M^* d\mu_{o(M)} \right], \quad (67)$$

where Δa_M^* is the borrowing gained on an upward crossing: the new loan $a_{i,M}^*$ at X_i, X_f, E (exit and non-entry carry no loan), and the destination swap $\Delta a_{F,o}^* = \tilde{k}_f - \tilde{k}_i + \kappa_{of} - \kappa_{oi}$ at the formality margins — precisely the borrowing gaps of Corollary 7.12. The model coefficient is the ratio $\beta_{N_i}^{\text{model}} = (d \ln N'_i / d\zeta) / (d \ln C / d\zeta)$; every term in (66)–(67) is linear in the incidence $(\chi_i, \chi_f, \kappa_f)$, so the scale of ζ cancels.

M Complete threshold taxonomy

This appendix reports the fully substituted closed form of every exit and entry threshold on every financing region. Throughout, $\mathcal{K}_s \equiv \beta(1-\lambda_s)c_s$ (so that $K = \mathcal{K}_f - \mathcal{K}_i$), $A_s = (1-\gamma)(\gamma/R_s)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$ and $A_s^0 = (1-\gamma)(\gamma/R^0)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$, with $A = A_f - A_i$ and $A^0 = A_f^0 - A_i^0$. Each one-sided margin (exit or entry against the outside option, informal continuation being the relevant alternative by Proposition 7.9) is the solution of $G_i(\hat{x}, \tilde{n}) = m$ with the margin-specific pair $(\tilde{n}, m)$:

margin	$\tilde{n}$	m	comparison
exit as informal	n	$\mathcal{K}_i$	$U_i(\hat{x}, n) \geq n$
exit as formal	$n - c_{fi}$	$\mathcal{K}_i + c_{fi}$	$U_i(\hat{x}, n - c_{fi}) \geq n$
entry as informal	$k_0 - c_i^e$	$\mathcal{K}_i + c_i^e$	$U_i(\hat{x}, k_0 - c_i^e) \geq k_0$

with $n = n^{\text{exit}}$ of (10). Substituting the four branches of (19):

Exit as informal.

$$\hat{x}_i^X(n) = \begin{cases} (\mathcal{K}_i/A_i^0)^{1-\gamma}, & \text{R1} \\ \frac{\mathcal{K}_i + R^0 n}{M_i n^\gamma}, & \text{R2} \\ \left[\frac{\mathcal{K}_i - \phi_i n}{A_i} \right]^{1-\gamma}, & \text{R3, if } \mathcal{K}_i > \phi_i n \\ (1-b_i)^{\gamma-1} \frac{\mathcal{K}_i(1-b_i) + n(R^0 + \phi_i b_i)}{M_i n^\gamma}, & \text{R4.} \end{cases} \quad (68)$$

Exit as formal (against switching down).

$$\hat{x}_f^X(n) = \begin{cases} [(\mathcal{K}_i + c_{fi})/A_i^0]^{1-\gamma}, & \text{R1} \\ \frac{\mathcal{K}_i + c_{fi} + R^0(n - c_{fi})}{M_i(n - c_{fi})^\gamma}, & \text{R2} \\ \left[\frac{\mathcal{K}_i + (1 + \phi_i)c_{fi} - \phi_i n}{A_i} \right]^{1-\gamma}, & \text{R3, if numerator} > 0 \\ (1 - b_i)^{\gamma-1} \frac{(\mathcal{K}_i + c_{fi})(1 - b_i) + (n - c_{fi})(R^0 + \phi_i b_i)}{M_i(n - c_{fi})^\gamma}, & \text{R4.} \end{cases} \quad (69)$$

Entry as informal. Replace (n, c_{fi}) by (k_0, c_i^e) in (69); in particular on R3,

$$\hat{x}_i^E(k_0) = \left[\frac{\mathcal{K}_i + (1 + \phi_i)c_i^e - \phi_i k_0}{A_i} \right]^{1-\gamma}, \quad (70)$$

and in type units $\bar{v}(k_0) = \hat{x}_i^E(k_0)^{1-\nu} \exp\{-\sigma_\varepsilon^2/(2(1-\nu))\}$.

Entry as formal (the entrant formality margin). This margin compares two continuation values, each on its own region, with funds $\tilde{n}_f = k_0 - c_f^e$ and $\tilde{n}_i = k_0 - c_i^e$. Because $G_s(\cdot, \tilde{n})$ is a power function of $\hat{x}$ on R1/R3 but *linear* in $\hat{x}$ on R2/R4, closed forms obtain on the diagonal region pairs:

$$\text{R3} \times \text{R3: } \hat{x}_e^F(k_0) = \left[\frac{B k_0 + C_e}{A} \right]^{1-\gamma}, \quad C_e = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + K; \quad (71)$$

$$\text{R1} \times \text{R1: } \hat{x}_e^F = (C_e^0/A^0)^{1-\gamma}, \quad C_e^0 = c_f^e - c_i^e + K, \quad (72)$$

(direction of the inequality per the trichotomy of Remark 7.8);

$$\text{R2} \times \text{R2: } \hat{x}_e^F = \frac{K + \beta(1 - \delta)(c_f^e - c_i^e)}{M_f \tilde{n}_f^\gamma - M_i \tilde{n}_i^\gamma}, \quad \text{formal above iff the denominator is positive} \quad (73)$$

(at the kink the fee's only dynamic cost is the forgone resale value, hence $\beta(1 - \delta)$, not R^0);

$$\text{R4} \times \text{R4: } \hat{x}_e^F = \frac{K + R_f h_f - R_i h_i + (1 + \phi_i)\tilde{n}_i - (1 + \phi_f)\tilde{n}_f}{M_f h_f^\gamma - M_i h_i^\gamma}, \quad (74)$$

with $h_s \equiv \tilde{n}_s/(1 - b_s)$ the capital installed at the binding cap.

Off-diagonal pairs (one destination borrowing while the other self-finances, e.g. R3×R2) mix $\hat{x}^{1/(1-\gamma)}$ and $\hat{x}$ terms: the indifference condition remains strictly monotone in $\hat{x}$ at the crossing and has a unique root, but no closed form; it is solved implicitly. The same diagonal/off-diagonal taxonomy applies verbatim to the incumbent formality margins $\hat{x}_i^F, \hat{x}_f^F$ with the fee pairs of Lemma 7.6.

Branch selection. Which branch applies is a self-consistency requirement: compute the candidate threshold from each branch and retain the one whose funds fall in that branch's bracket *evaluated at the candidate*,

R1: $\tilde{n} \geq \bar{k}_s(\hat{x}^*)$; R2: $\tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \bar{k}_s(\hat{x}^*)$; R3: $(1-b_s)\tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \tilde{k}_s(\hat{x}^*)$; R4: below.

Exactly one branch is self-consistent: G_s is continuous and strictly increasing in $\hat{x}$ ($\partial G_s / \partial \hat{x} = M_s k^{*\gamma} > 0$ by the envelope theorem), and at fixed $\tilde{n}$ the firm moves monotonically through R1→R2→R3→R4 as $\hat{x}$ rises, so $G_s(\cdot, \tilde{n})$ crosses any constant m exactly once.

Object	Data measurement	Model counterpart
Formality status	EC: positive social-security contributions (J300A > 0; zero-income firms with zero contributions coded formal)	sector $s = f$
Employment L	EC: total persons engaged, unpaid and owners included (H001A)	$1 + \ell^h$ (owner unit plus hires)
Profit	EC: value added minus remunerations (A131A – J000A); social-security contributions and audit penalties <i>not</i> netted out	$y - w \ell^h$
Capital k	EC: end-of-year fixed assets (Q000A), book value	installed k
External funds a	ENAFIN: credit raised during the year (P35*); EC: credit incidence (0511A)	transition-stage borrowing a
Leverage cap b_f	ENAFIN: approved credit line / imputed capital (P38_4B / $\hat{K}$), p90 among borrowers; debt stock P24 / $\hat{K}$ as bracket	atom of a/k' at b_f
Leverage cap b_i	ENAFIN proxy-informal cell (TAMANO micro $\times$ persona física), p90 leverage among borrowers	atom of a/k' at b_i
Loan pricing	ENAFIN median rate on 2023 credits; R04C formal/informal rates; spread over CETES 28	marginal cost $\Phi'_s(a) = \phi_s$ over the risk-free rate
Firm dynamics	EC waves five years apart: exit = absent at $t+5$ (death and exit indistinguishable); switching among survivors; entrants = age ≤ 2	five-year transition kernel with identical definitions
Units	thousands of 2023 MXN	multiples of the median annual wage (77.5 thousand MXN: wage_unit); $w = 1$

Table 1: Model–data measurement alignment

Parameter	Value	Source / target
β discount factor	0.95	real rate $\approx 5\%$
δ depreciation	0.10	standard, structures–equipment mix
w wage	1	normalization (77.5k MXN)
τ_y effective output tax	0.048	ISR (personas morales) collections / formal VA, 2023 (statutory 0.5)
τ_ℓ effective payroll wedge	0.16	(IMSS + Infonavit + ISN) / formal wage bill, 2023 (compliance-lit)
α capital elasticity	0.17	EC cost shares (median); 0.13 (mean), 0.14 (OLS-FE) robustness
ν labor elasticity	0.59	EC cost shares (median); 0.63 (mean) robustness
ρ_i, ρ_f TFP persistence	0.95, 0.95	EC autocovariances, ME-robust system eq. (61) (naive a_0 -based 0.4)
$\sigma_{\varepsilon i}, \sigma_{\varepsilon f}$ innovation s.d.	0.14, 0.21	EC autocovariances, ME-robust (naive 1.11, 0.72: robustness)
$\sigma_{u,i}, \sigma_{u,f}$ transitory-real s.d.	0.83, 0.53	50% of the $h=0$ autocovariance excess (real shares 25%/75%: robustness)
σ_v permanent-type s.d.	0.56	EC autocovariances (upper bound; common across sectors by const)
λ_i, λ_f death rates	0.067, 0.056	flat tail of exit-by-size (top terciles)

Table 2: Block 0: preset parameters (annual)

Parameter	Assigned moment (source)	Identification argument (result in Section 7)
c_i	five-year informal exit rate, exit5_i (EC)	the exit threshold $\mathcal{X}_i(n; \mathcal{K}_i)$ is strictly increasing in $\mathcal{K}_i$ in every financing region (eq. (32)); with λ_i preset from the size-flat tail, the <i>level</i> of informal exit moves only c_i
c_f	five-year formal exit rate, exit5_f (EC)	same monotonicity for the formal exit margin (exit against the switch-down option, eq. (34))
c_{if}	five-year switching rate $i \rightarrow f$, switch5_if (EC)	$\hat{x}_i^F$ rises with $(1 + \phi_f)c_{if}$ through $C_{i\text{-inc}}$ (Lemma 7.6), so $\mathbb{P}(i \rightarrow f)$ is strictly decreasing in c_{if}
c_{fi}	five-year switching rate $f \rightarrow i$, switch5_fi (EC)	$\hat{x}_f^F$ falls with $(1 + \phi_i)c_{fi}$ through $C_{f\text{-inc}}$, shrinking the switch-down band of Proposition 7.9
c_f^e	formal share of entrants, entry_share_f (EC)	the entrant sorting cutoff $\hat{x}_e^F$ rises with $(1 + \phi_f)c_f^e$ through C_e (Prop. 7.7)
c_i^e	entrant <i>relative</i> size: age-size profile intercept, informal (EC)	selection: $\hat{x}_i^F(k_0)$ rises with c_i^e (eq. (35)), so a higher fee admits only better-selected entrants, raising entrant size relative to incumbents
ϕ_{0f}	formal borrowing incidence, borrow_inc_f (EC)	the participation cutoff of Remark 7.4: incidence is strictly decreasing in ϕ_{0f} given (ϕ_f, b_f)
ϕ_{0i}	informal borrowing incidence, borrow_inc_i (EC)	same, informal side
μ_{k_0}	mean entrant capital, informal (EC)	constrained entrants install $k = \tilde{n}/(1 - b_i)$ (Lemma 7.2), a log-linear map from wealth to observed capital: the location reads μ_{k_0}
σ_{k_0}	p90 entrant capital, informal (EC)	same map: the spread of entrant capital reads σ_{k_0} (informal side used; the formal entrant tail is outlier-contaminated)

Table 3: Block 2: the one-to-one map from moments to parameters